HIPPO Emergency Medicine High-Yield Review

Comprehensive Deduplicated Study Book

18 Chapters | 1,927 Source Questions
Exam Tricks, Algorithms, Pitfalls, and Brief Arabic Synthesis

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Educational disclaimer: This book is a revision resource, not a substitute for local protocols, specialist advice, or independent clinical judgment. Verify time-sensitive recommendations against current local guidance before clinical use.

CHAPTER 01: ENVIRONMENTAL DISORDERS

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1. Clinical overview and pathophysiology

Environmental Disorders — high-yield revision notes (deduplicated HIPPO items)

Core concepts and pathophysiology
- Heat illnesses
- Continuum: heat cramps → heat exhaustion (volume/electrolyte depletion, T often elevated but <≈40–40.5°C) → heat stroke (core hyperthermia with CNS dysfunction, often >40–40.5°C). Exertional heat stroke often in young athletes; non-exertional in elderly.
- Pathophysiology: failure of heat dissipation → direct thermal injury + inflammatory cascade → multiorgan dysfunction, rhabdomyolysis. Rapid cooling reduces morbidity/mortality. (Rosen)
- Hypothermia and frostbite
- Hypothermia produces bradyarrhythmias, Osborn (J) waves on ECG; severity correlates with core temp.
- Frostbite: ice crystal formation → endothelial injury, microthrombosis, reperfusion injury. Superficial (1st/2nd) → clear blisters; deep (3rd/4th) → hemorrhagic blisters, tissue loss. Thrombolysis (tPA) may be considered for severe deep frostbite in selected patients.
- Drowning / submersion
- Hypoxic/anoxic injury → lactic acidosis is common. Prognosis worsened by prolonged submersion (>~5 min), delayed CPR, age <3 y, warm water drowning, GCS 3, prolonged resuscitation. (Szpilman / NEJM)
- Envenomations & bites
- Snake (Crotalids): venom = cytotoxins + hemotoxins → local swelling, tissue destruction, coagulopathy, thrombocytopenia, systemic toxicity. Pulmonary/airway compromise or progressive coagulopathy → antivenom indicated.
- Spiders: Latrodectus (black widow) → presynaptic latrotoxin causing massive neurotransmitter release → diffuse muscle cramps, autonomic features. Loxosceles (brown recluse) → sphingomyelinase → local necrosis sometimes systemic hemolysis.
- Scorpion: severity graded I–IV (local → regional → cranial nerve/skeletal muscle

dysfunction → severe cardiopulmonary). Most US stings are mild; severe require antivenom/critical care.
- Reptile (e.g., iguana) bites: often colonized with Salmonella spp.; risk of wound infection.
- Mammalian (dog/cat) bites: Pasteurella multocida common; polymicrobial including Staph/Strep/Capnocytophaga.
- Marine hazards
- Jellyfish: venom delivered by nematocysts. Treatment focuses on preventing further nematocyst firing (no fresh water/rubbing), tentacle removal, and hot-water immersion or species-specific measures (vinegar for some species).
- Vibrio vulnificus: gram-negative marine organism → fulminant cellulitis/sepsis in hours; risk factors: chronic liver disease, hemochromatosis, immunocompromise.
- Altitude & decompression
- Acute Mountain Sickness (AMS): headache, nausea at >2000 m; acetazolamide speeds acclimatization (causes mild metabolic acidosis → hyperventilation).
- HACE (high-altitude cerebral edema): ataxia, AMS, altered mental status — definitive therapy = descent; adjuncts: O2, dexamethasone, hyperbaric bag if descent impossible.
- HAPE: high-altitude pulmonary edema due to hypoxic pulmonary vasoconstriction → pulmonary hypertension → non-cardiogenic edema; therapy: descent, O2, nifedipine or PDE-5 inhibitors if needed.
- Decompression sickness (DCS): nitrogen bubble formation on rapid ascent → Type I (skin / musculoskeletal) and Type II (CNS). Definit

Core concepts and pathophysiology
- Thermoregulatory extremes
- Hypothermia: core temperature <35°C. Progressive physiologic depression → bradycardia, dysrhythmias (VF/VT more likely at low temps), coagulopathy, insulin resistance; myocardium irritable (guidewire/defib can provoke arrhythmia). Rewarming reverses. (Ref: NBK545239)
- Frostbite: freezing injury → intracellular/extracellular ice crystals → endothelial injury, thrombosis, ischemia/necrosis. Damage evolves over days. Rewarming required; avoid dry heat. (Ref: NBK536914)
- Chilblains (pernio): superficial inflammatory response to nonfreezing cold; burning/pruritic erythematous plaques; usually self-limited. (Ref: NBK549842)
- Hyperthermia / heat-related illness
- Heat cramps: exercise-associated muscle cramps from electrolyte losses (Na+, K+,

Mg2+, Cl–) — treat with oral/IV rehydration and stretching. (Ref: NBK553117)
- Exertional heat stroke: core temp >40°C with CNS dysfunction; multisystem injury (rhabdomyolysis, hepatic injury, DIC). Rapid cooling is lifesaving. (Ref: NBK553117)
- Classic (nonexertional) heat stroke: elderly/chronically ill with impaired heat dissipation.
- Diving and barotrauma
- Arterial gas embolism (AGE): gas bubbles entering systemic circulation during ascent → sudden focal neuro/cardiac deficits, often within minutes of surfacing. Hyperbaric oxygen definitive. (Ref: NBK482321)
- Decompression sickness: delayed multisystem symptoms from inert gas bubbles.
- Oxygen toxicity: at depth, high PO2 → CNS toxicity (seizures, twitching). Occurs while at depth. (Ref: NBK430743)
- Radiation and thyroid protection
- Radioactive iodine uptake by thyroid → increased long-term thyroid cancer risk; potassium iodide (KI) saturates thyroid to block uptake in relevant exposures. (Ref: NBK542320)
- Animal and envenomation syndromes (high-yield patterns)
- Elapidae (sea snakes, coral snakes, cobras): potent neurotoxins → early tetany then flaccid paralysis and respiratory failure (sea snakes may cause rhabdomyolysis). Sea-snake antivenom (Elapidae) where available. Coral-snake bites may have delayed neurotoxicity → observe/ICU. (Refs: NBK532973; Rosen's)
- Crotalinae (rattlesnakes, pit vipers): hemotoxic — local tissue necrosis, coagulopathy; Crotalidae antivenom for systemic/ progressive envenomation.
- Scorpion (Centruroides): neurotoxin → autonomic storm (tachy, HTN), cranial nerve dysfunction, opsoclonus/hemiballismus; Centruroides antivenom for severe neurotoxicity. (Ref: NBK430928)
- Stingray: venom is heat-labile → hot water immersion + wound care; leave wound open and irrigate. (Ref: Rosen's)
- Brown recluse (Loxosceles): cytotoxic venom → painless initial bite with possible necrotic ulcer later; supportive care, tetanus update; avoid early debridement. (Ref: NBK537045)
- Honeybee/Africanized bees: toxicity from large numbers of stings (swarming), not inherently more toxic venom per sting; treat anaphylaxis/systemic toxicity supportively. (Ref: Rosen's)
- Electrical injury
- Low-voltage (<600 V, e.g., household 120 V): if asymptomatic and normal ECG and

urinalysis (no myoglobinuria), may be discharged with wound care/tetanus. High-volta

Core concepts and pathophysiology
- Heat illness spectrum: heat cramps → heat exhaustion → heat stroke. Heat stroke = life-threatening failure of thermoregulation with core hyperthermia and central nervous system dysfunction. Two types:
- Exertional (young, sweaty, hyperdynamic): profuse sweating, rhabdomyolysis, coagulopathy.
- Classic/classical (elderly, comorbid, impaired cardiovascular response/meds): less tachycardia, higher mortality risk.
- Cooling physiology: evaporative and conductive heat loss are the fastest; peripheral vasoconstriction (e.g., from early vasopressors) impairs cooling.
- Hypothermia: degree ranges from mild to severe; severe cases may have bradycardia, hypotension, arrhythmias. Rewarming strategies range from passive to active internal (lavage, ECMO). ECMO provides fastest core rewarming and oxygenation (can raise core temperature rapidly) (NCBI; Rosen's).
- Immersion/drowning: primary problem is hypoxemia — ventilation is priority. Submersion time is strongest predictor of outcome (shorter is better).
- Barotrauma and decompression/diving injuries:
- Arterial gas embolism (AGE): ascent → gas bubble entry/systemic emboli → neuro/cardiac ischemia → treat with hyperbaric oxygen.
- Pulmonary over-pressurization (POPS): alveolar rupture → pneumomediastinum/pneumothorax (radiographic continuous diaphragm sign).
- Cold-injury types:
- Frostbite staging similar to burns: 1st (superficial), 2nd (clear blisters), 3rd (hemorrhagic blisters/full thickness → hypodermis), 4th (deep tissue involvement → muscle/bone).
- Trench/immersion foot: prolonged exposure to wet, nonfreezing cold → ischemia; prevention = keep feet dry.
- Envenomations and marine exposures:
- Stingray: venom heat-labile → hot water immersion 40–45°C for analgesia.
- Black widow: neurotoxin → severe abdominal wall muscle spasm can mimic peritonitis.
- Brown recluse: local loxoscelism → necrotic ulceration; dapsone not supported; surgical review if ulceration.
- Toxic chemical: hydrofluoric acid (HF) penetrates tissue, binds calcium →

hypocalcemia, hypomagnesemia, hyperkalemia; treat with calcium (topical gel, IV calcium gluconate).
- Wound infections from marine/animal exposures:
- Vibrio vulnificus (salt/brackish water, oysters/wound exposure): fulminant wound infection and sepsis; high mortality; risk amplified by liver disease, hemochromatosis, diabetes.
- Mycobacterium marinum (aquarium/fish tanks): chronic skin lesions (“fish tank granuloma”).
- Animal bites: Pasteurella multocida common in cat bites; human bites polymicrobial (Eikenella, Strep, Staph).

(References: Rosen’s Emergency Medicine; NCBI Bookshelf: NBK537135, NBK553117, NBK537045, NBK536914, NBK482364, NBK441829, NBK482321, NBK482348, NBK441920.)

> ملخص بالعربية: المفاهيم الأساسية تشمل فشل تنظيم الحرارة (ضربة حرارية)، إعادة تسخين المفرط في انخفاض الحرارة (ECMO عند الضرورة)، أولوية التهوية في الغرق، وخطورة التعرض البحري (Vibrio) وسموم مثل HF. المصادر: Rosen’s وكتابات NCBI.

Recognition and differential
- Heat stroke: core temp often >40°C with CNS dysfunction (confusion, combativeness, seizures). Differentiate from sepsis, neuroleptic malignant syndrome, serotonin syndrom

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Environmental Disorders.

CHAPTER 02: ABDOMINAL AND GASTROINTESTINAL DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Acute pancreatitis: premature activation of pancreatic enzymes → autodigestion; may be interstitial (sterile) or necrotizing. Necrosis can become secondarily infected days later. Prophylactic antibiotics are not recommended; treat infection only when proven/clinically evident (fever, sepsis, gas in collections, positive cultures, or CT-guided aspiration). (Dellinger et al., Ann Surg 2007; UpToDate/NIH reviews)
- Biliary obstruction/cholangitis: biliary stasis + infection → Charcot's triad (fever, RUQ pain, jaundice); Reynolds pentad (add hypotension + altered mental status) signals severe disease—biliary decompression (usually ERCP) + antibiotics required.
- Mesenteric ischemia: embolic (AF) or thrombotic occlusion → sudden severe pain often "out of proportion" to exam; lactate may be elevated; can progress to bowel necrosis—diagnose with CTA; urgent revascularization and surgery if necrotic bowel.
- Small vs large bowel obstruction: SBO commonly from adhesions; radiograph/CT: SBO shows dilated small bowel loops with valvulae conniventes across lumen; LBO often from carcinoma, shows haustra that do not cross entire lumen, large-caliber colon dilation.
- Volvulus: sigmoid presents with "coffee-bean" sign; initial nonoperative reduction via sigmoidoscopic decompression unless peritonitis or perforation.
- Upper GI bleeding: risk stratify with Glasgow-Blatchford Score (GBS); GBS = 0 low risk → outpatient possible; melena, tachycardia, elevated BUN are higher-risk features. Witnessed melena correlates strongly with significant UGIB (Srygley et al., JAMA 2012).
- Esophageal/gastric pathology: Mallory-Weiss tears follow retching/vomiting and often need urgent endoscopy if higher risk; gastric adenocarcinoma risk factors include H. pylori infection.
- Perianal disease: fissure = midline linear posterior (painful defecation, bright

blood on paper); perianal abscess = fluctuant, tender collection → require incision & drainage (antibiotics not routine unless systemic/immune compromise).
- Bezoars: trichobezoar (hair) resists chemical dissolution; attempt endoscopic removal first; surgical if endoscopy fails.
- Neonatal jaundice: physiologic unconjugated hyperbilirubinemia peaks in first days of life (usually benign unless very high or Coombs-positive, hemolysis, conjugated bilirubin elevation).
- Hepatic/autoimmune markers: Primary biliary cholangitis → anti-mitochondrial antibody (AMA) positive; alcoholic hepatitis commonly shows AST>ALT (often ~2:1) and elevated GGT.

> Arabic (موجز علمي)
> - الملخص: التهابات وموانع التدفق المعوية غالبًا تتطلب تمييز بين العمليات المعقمة والمصابة، تقييم الاستقرار الهيموديناميكي، واستخدام التصوير التشخيصي الملائم (CTA للـنقص التروية، ERCP لانسداد القنوات الصفراوية). التدابير العلاجية تتدرج من الإفراغ البطني أو التنظير إلى الجراحة الطارئة حسب علامة الخطر.

Recognition and differential (key presentations and discriminators)
- Severe epigastric pain radiating to back + lipase elevation: acute pancreatitis; if necrosis on CT → watch for secondary infection (fever, rising WBC, gas in necrosis).
- Retching + sudden hematemesis after forceful emesis: Mallory-Weiss (consider urgent endoscopy i

Core concepts and pathophysiology
- Infections
- Clostridioides (Clostridium) difficile: antibiotic-associated colitis from dysbiosis; toxin-mediated pseudomembranous colitis; spores resistant to many disinfectants. First-line therapy for initial non-severe disease: oral fidaxomicin or oral vancomycin (IDSA 2021). IV vancomycin does not treat luminal C. difficile (not secreted into gut). Rectal vancomycin considered if ileus.
- Campylobacter jejuni: curved Gram-negative rod causing inflammatory diarrhea; postinfectious complication = Guillain–Barré syndrome via molecular mimicry.
- Entamoeba histolytica: trophozoites can invade bowel and seed liver → amebic liver abscess (travel-associated).
- Spontaneous bacterial peritonitis (SBP): common precipitator of hepatorenal syndrome; high morbidity in cirrhotics.
- Obstructive / ischemic processes

- Gallbladder disease: acute calculous cholecystitis vs acalculous (critically ill, TPN, stasis → ischemia/necrosis). Acalculous cholecystitis often in ICU; percutaneous cholecystostomy for those too unstable for surgery.
- Ascending cholangitis: biliary obstruction (stones) + infection → Charcot triad (RUQ pain, fever, jaundice); Reynolds pentad adds hypotension + confusion (severe/septic).
- Hernia incarceration → strangulation: vascular compromise → ischemia, elevated lactate, peritonitis; urgent surgical management.
- Perforation / barotrauma
- Boerhaave syndrome: full-thickness distal esophageal rupture after forceful vomiting → pneumomediastinum (Hamman's crunch), rapid deterioration; needs emergent surgical management.
- Pediatric-specific emergencies
- Button battery in esophagus: alkali current causes liquefactive necrosis and rapid perforation—medical emergency requiring immediate removal.
- Hypertrophic pyloric stenosis: projectile nonbilious vomiting, palpable "olive" mass, hypochloremic hypokalemic metabolic alkalosis; male predominance; surgery definitive (pyloromyotomy).
- Meckel diverticulum: true diverticulum from vitelline duct; "rule of 2s"; often asymptomatic; when bleeding, painless lower GI bleed in young child; definitive treatment surgical resection.
- Functional / postoperative
- Postoperative ileus: bowel dysfunction with gas throughout colon and rectum on plain film; management supportive (NPO/IV fluids, electrolyte correction, early mobilization).

Arabic summary (pathophysiology)

ملخص: التهابات معوية (C. difficile، Campylobacter، Entamoeba) مضبوطة بالعلاج المعياري؛ حالات انسداديةٍ وإقفارية (التهاب المرارة الحاد، التهاب الأقنية الصفراوية) قد تتطوّر إلى صدمة أو احتياج جراحي؛ حالات طوارئ طفلية مميزة (بطارية زر في المريء، تضيُّق البواب عند الرضع، رتج ميكل) تتطلب تصرفًا عاجلًا.

Recognition and differential (presentation cues and red flags)
- C. difficile: recent antibiotics, watery → sometimes bloody diarrhea, leukocytosis; severe disease = fever, leukocytosis, hypotension, ileus, abdominal distention.
- Red flags: toxic megacolon, lactic acidosis, hypotension.
- Button-battery ingestion: sudden coughing, drooling, dysphagia, inability to

tolerate liquids; x-ray AP shows “double ring/halo” and lateral may show “step-off.”
- Red flags: esophageal locatio

Abdominal & Gastrointestinal Disorders — High-Yield Emergency Revision Notes

Core concepts and pathophysiology
- Structural vs motility vs inflammatory vs infectious vs ischemic causes produce overlapping presentations; use pattern recognition (onset, progression, systemic signs, stool characteristics, prior surgery, immunosuppression).
- Mechanical obstruction: small-bowel obstruction (SBO) most commonly from postoperative adhesions (developed countries); hernia, neoplasm, volvulus less common. Obstruction → distension, vomiting, pain, absent flatus; risk of strangulation/ischemia.
- Dysphagia: oropharyngeal (transfer/neurologic) vs esophageal (mechanical vs motor). Progressive solids→liquids = mechanical (stricture, malignancy); intermittent solids = motility. Prior radiation/caustic injuries → complex, proximal, often fibrotic strictures.
- Upper GI alarm features requiring urgent endoscopy: new dysphagia/odynophagia, weight loss, GI bleeding, refractory GERD with red flags, age-related new dyspepsia (guideline thresholds such as age >50 commonly used for referral).
- Infectious diarrheas: secretory vs inflammatory vs malabsorptive (steatorrhea). Foul, greasy, voluminous stools → consider Giardia (camping, water exposure); stool antigen or nucleic acid tests preferred over routine culture or serology for acute diagnosis.
- Liver infections: pyogenic liver abscesses are usually polymicrobial (enteric facultative + anaerobes), often associated with biliary obstruction; Klebsiella important in primary liver abscess in Asia; S. aureus more after instrumentation.
- Pancreatitis: most necrotizing pancreatitis is initially sterile; infected necrosis presents later—surgical debridement reserved for infected necrosis failing antibiotics or hemodynamic instability.
- Cirrhosis complications: ascites → risk of spontaneous bacterial peritonitis (SBP); diagnostic ascitic PMN count ≥250 cells/mm3 indicates SBP and triggers empiric antibiotics.
- Hepatic encephalopathy: ammonia from gut bacteria; lactulose lowers colonic pH and traps NH_4^+ and increases transit—first-line therapy.
- Post-bariatric/Roux-en-Y: long-term malabsorption and micronutrient deficiencies (notably B12) — require lifetime supplementation.

- Pediatric specifics: pyloric stenosis (infant projectile nonbilious vomiting, palpable olive) — ultrasound first; foreign body ingestion → chest/neck x-rays initially (AP + lateral) for radioopaque items.

> Arabic summary (مفاهيم أساسية): التمييز بين أسباب ميكانيكية، التهابية، ومعدية يوجّه التشخيص العاجل؛ علامات الإنذار تستلزم منظار عاجل؛ عدوى الكبد غالبًا متعددة الميكروبات؛ عدوى البنكرياس المتحللة تحتاج جراحة فقط إذا ملوّثة وفشلت المعالجة المحافظَة.

Recognition and differential (presentations → top differentials and discriminators)
- Progressive dysphagia (solids → liquids over wks): mechanical stricture +/- malignancy (risk factors: radiation, caustic ingestion, smoking, weight loss). First-line imaging before endoscopy if prior radiation or caustic ingestion: barium contrast swallow to delineate complex/proximal strictures and reduce endoscopy perforation risk.
- Odynophagia in immunocompromised: consider esophagitis (Candida, HSV, CMV) → end

Core concepts and pathophysiology
- Acute pancreatitis: autodigestion of pancreas by activated pancreatic enzymes. Common causes: gallstones (obstruction of biliary/pancreatic ducts) and alcohol. Classic metabolic association: hypocalcemia (calcium saponification with fat necrosis). Complications: pancreatic pseudocyst (late, walled-off fluid collection), pancreatic necrosis, infection, chronic pancreatitis → insulin deficiency/diabetes, pancreatic calcifications. Diagnosis = 2 of 3: characteristic epigastric pain (often radiates to back), serum lipase/amylase ≥3× upper limit OR imaging consistent with pancreatitis. (Ref: NCBI Bookshelf, Rosen’s)
- Biliary disease spectrum:
- Biliary colic: transient cystic duct obstruction.
- Acute cholecystitis: persistent cystic duct obstruction → GB wall inflammation, Murphy sign, US shows stones, GB wall thickening, pericholecystic fluid.
- Choledocholithiasis → obstruction of common bile duct; can cause cholangitis.
- Ascending cholangitis: Charcot triad = fever + RUQ pain + jaundice; Reynolds pentad adds hypotension/shock and altered mental status (severe).
- Management distinctions: cholecystectomy definitive for cholecystitis; ERCP for obstructing CBD stone in cholangitis or retained choledocholithiasis. (Ref: NCBI Bookshelf)
- Small-bowel obstruction (SBO): most commonly from adhesions (prior surgery).

Presents with crampy abdominal pain, vomiting, distention, high-pitched bowel sounds early.
- Spontaneous bacterial peritonitis (SBP): infection of ascitic fluid in cirrhosis; common organisms = enteric gram-negatives (E. coli, Klebsiella). Diagnostic threshold: PMNs in ascitic fluid >250 cells/μL triggers empiric therapy. (Ref: NCBI Bookshelf)
- Infectious diarrheas and extraintestinal complications:
- Giardia lamblia: foul-smelling, greasy, steatorrhea; stool O&P (multiple samples) or antigen testing.
- Shigella dysenteriae: bloody/mucoid diarrhea, high fever; can cause seizures/ encephalopathy in children.
- Salmonella Typhi (typhoid): fever, abdominal symptoms, rose spots, relative bradycardia (pulse–temperature dissociation / “Faget sign”); treat empirically with fluoroquinolones where susceptible.
- Entamoeba histolytica: amebic liver abscess after diarrheal illness → treat with metronidazole.
- Special syndromes:
- Dumping syndrome after gastric bypass (Roux-en-Y): postprandial cramping, diarrhea, vasomotor sx (flushing, dizziness), syncope; managed with dietary measures.
- Cystic fibrosis: exocrine pancreatic insufficiency → steatorrhea, recurrent pancreatitis and pulmonary infections; diagnose with sweat chloride test.
- Necrotizing enterocolitis (NEC) in preterm neonates: ischemic necrosis, pneumatosis intestinalis on AXR, can progress to perforation/sepsis.
- Rectal foreign body: glass/sharp objects require surgical consultation; risk of perforation.
- Zollinger–Ellison (gastrinoma): refractory PUD, elevated gastrin; consider MEN1 associations (parathyroid, pituitary adenoma, pancreatic NET). (Refs: Rosen’s, Tintinalli, NCBI)

> ملخّص عربي — المفاهيم الأساسية والفيزيولوجيا
> - التهاب البنكرياس: تنشأ من تفعيل الإنزيمات البنكرياسية ذاتياً؛ أسباب شائعة: الحصيات الصفراوية

Abdominal and Gastrointestinal Disorders — compact high-yield emergency revision notes
(Integrated, deduplicated synthesis of common HIPPO QBank concepts)

Core concepts and pathophysiology
- Acute abdomen etiologies to prioritize: perforated viscus, mesenteric ischemia, complicated biliary disease (acute cholecystitis, emphysematous cholecystitis, choledocholithiasis, ascending cholangitis), acute appendicitis, bowel obstruction (volvulus, incarcerated/strangulated hernia), ischemic colitis, pancreatitis (acute vs chronic), intussusception (pediatrics), acute radiation proctitis, GI infection ± complications (HUS), hepatobiliary disease (viral hepatitis, hepatic encephalopathy).
- Biliary specifics:
- Emphysematous cholecystitis = gas-forming infection of gallbladder wall (Clostridia and gram-negatives); strong association with diabetes → high risk of perforation and mortality. (See NCBI Bookshelf)
- Choledocholithiasis → CBD obstruction; ultrasound “double-barrel sign” (CBD ≈ portal vein) and CBD dilation suggest obstruction. CBD considered enlarged by usual sonographic criteria (commonly >6 mm; note age and post-cholecystectomy modifiers in radiology references).
- Porcelain gallbladder = calcified gallbladder wall; associated with increased gallbladder malignancy risk → consider cholecystectomy.
- Vascular:
- Acute mesenteric ischemia: pain out of proportion to exam, rapidly rising lactate; often arterial embolus/thrombosis in patients with vascular disease. Vasoconstrictors may worsen splanchnic perfusion.
- Pancreas:
- Chronic pancreatitis may present with calcifications on XR, chronic pain, and normal/low lipase; management is analgesia/supportive, not acute pancreatitis protocols.
- Pediatric:
- Intussusception commonly follows viral illness (Peyer patch hypertrophy) — classical intermittent colicky pain with vomiting; treat with pneumatic/contrast enema unless signs of perforation/peritonitis.
- Infection/toxic:
- HUS (STEC): triad microangiopathic hemolytic anemia, thrombocytopenia, acute renal failure after bloody diarrheal illness; antibiotics may worsen outcome in STEC-HUS — supportive care is primary.
- Liver:
- Hepatic encephalopathy associated with elevated ammonia; TIPS increases risk

of encephalopathy post-procedure. Spontaneous bacterial peritonitis (SBP) commonly precipitates hepatorenal syndrome.

> Arabic summary (مختصر مفهوم رئيسي):
> التهابات المرارة الغازية غالباً مع مرضى السكري؛ انسداد القناة الصفراوية يبين بتوسّع القناة في السونار؛ آلام شديدة مع فحص بسيط → فكر في نقص التروية المساريقية؛ التِهابات الأمعاء عند الأطفال قد تتبع عدوى فيروسية وتسبب انغلاف الأمعاء؛ متلازمة HUS بعد إسهال دموي تتطلب علاجاً داعماً وليس مضادات حيوية.

Recognition and differential (clinical discriminators)
- Pain features:
- Pain out of proportion to exam → mesenteric ischemia.
- Sudden severe epigastric pain with free air under diaphragm on upright CXR → perforated viscus (peptic ulcer perforation) → surgical emergency.
- RUQ pain + fever + Murphy sign → acute cholecystitis; crepitus in RUQ → consider emphysematous cholecystitis.
- RUQ pain + jaundice + fevers (Charcot triad) → ascending cholangitis (urgent ERCP).
- Coli

Core concepts and pathophysiology
- Inflammatory bowel disease
- Crohn’s disease: transmural, granulomatous inflammation; “skip lesions”; can affect any part mouth→anus; predisposes to fistulae, strictures, malabsorption. (Rosen’s/Tintinalli, NCBI Bookshelf)
- Ulcerative colitis: mucosal inflammation limited to colon/rectum, continuous, always involves rectum; risk of toxic megacolon, colorectal cancer.
- Perforation syndromes
- Perforated viscus (e.g., peptic ulcer): full-thickness GI wall defect → free intraperitoneal air → peritonitis.
- Boerhaave’s syndrome: spontaneous transmural esophageal rupture after violent retching → mediastinitis, pneumomediastinum, pleural effusion (often left-sided).
- Hepatobiliary disease
- Acute cholecystitis: usually gallstone obstructing cystic duct → inflammation ± secondary bacterial infection (E. coli common; also Klebsiella, Enterococcus, Enterobacter).
- Ascending cholangitis: biliary obstruction + infection → Reynolds pentad: RUQ

pain, fever, jaundice ± hypotension, altered mental status → emergency decompression (ERCP).
- Gallstone pancreatitis: gallstone obstructing ampulla → elevated lipase; CBD dilation increases suspicion of retained/obstructing stone.
- Liver failure and complications
- Cirrhosis → portal hypertension → ascites; bacterial translocation → spontaneous bacterial peritonitis (SBP).
- Hepatic encephalopathy: precipitated by SBP, GI bleed, hypovolemia, etc.; treat precipitant + lactulose.
- Infectious gastroenteritis and parasites
- Norovirus: rapid-onset (24–48 h), short duration (48–72 h), vomiting + watery diarrhea; common food/wedding outbreaks.
- Giardia lamblia: greasy, foul-smelling, malabsorptive diarrhea after contaminated water; treat metronidazole.
- Hepatitis A: fecal–oral, incubation 2–6 weeks; prodrome → jaundice.
- Pediatric surgical causes
- Intussusception: telescoping bowel → intermittent severe colicky pain, “currant jelly” stool; ultrasound “target sign.” Air-contrast enema first-line if stable; surgery if peritonitis/perforation.
- Hirschsprung disease: failure to pass meconium in neonate; diagnosis requires rectal biopsy (aganglionosis).
- Postoperative physiology
- Dumping syndrome after gastric bypass: early (osmotic/vasomotor) and late (hypoglycemic) presentations; treat with dietary modification.

> Arabic (موجز فسيولوجي):
> أمراضٌ التهابيةٌ: كرون نزعةٌ عبر الجدار وتظهر بقفزات؛ التهاب قولون غير مستمر ويشمل المستقيم. ثقب الجهاز الهضمي يسبب هواء حرّاً ببطن أو وسط الصدر؛ متلازمة بويرهافه ثقب مريئي بعد التقيؤ. التهاب المرارة غالباً ناتج عن حصاة ثم عدوى بكتيرية (E. coli شائع). التهاب الأقنية المرارية صعب ويتطلب فك الانسداد سريعاً. التهاب الصفاق التلقائي يحدث عند مرضى تليف والكشف بواسطة بزل البطن.

Recognition and differential (key presentations and discriminators)
- Acute RUQ pain
- Cholecystitis: RUQ pain, fever, Murphy’s sign; RUQ ultrasound first-line.
- Choledocholithiasis/cholangitis: RUQ pain + jaundice ± fever; consider Reynolds pentad (add hypotension, AMS) → urgent ERCP.

- Epigastric pain
- Peptic ulcer / perforation: sudden severe generalized pain and peritonitis; upright CXR → free air under diaphragm.
- Boerhaave's:

Abdominal and Gastrointestinal Disorders — compact high-yield ED revision notes

Core concepts and pathophysiology

- Broad categories: surgical emergencies (perforation, obstruction, ischemia, volvulus), hepatobiliary/pancreatic disease, infectious enteritis, inflammatory colitis, esophageal foreign bodies/impactions, neutropenic enterocolitis.
- Important mechanisms:
  - Acute mesenteric ischemia (AMI): arterial embolus (cardiac emboli, e.g., AF), arterial thrombosis on atherosclerotic plaque, venous thrombosis, non-occlusive hypoperfusion. Ischemia → mucosal necrosis → transmural infarction/perforation if untreated.
  - Neutropenic enterocolitis (typhlitis): mucosal injury from cytotoxic chemo + profound neutropenia → bacterial translocation, necrosis of cecum/terminal ileum.
  - Pancreatic head adenocarcinoma: obstructive painless jaundice from common bile duct compression; Courvoisier sign = palpable, nontender gallbladder.
  - Infectious diarrheas: protozoal (Giardia, Cryptosporidium) vs bacterial (Campylobacter, Salmonella, E. coli) — different incubation, exposures, stool findings.
  - Esophageal food impaction: impacted bolus commonly over an underlying stricture, ring, eosinophilic esophagitis, or malignancy.

Arabic: المفاهيم الأساسية تتضمن أسباب إقفارية، التهابات معوية طفيلية/بكتيرية، أورام neutropenic. كل حالة لها فيزيولوجيا البنكرياس، انسداد المريء بالطعام، والتهاب الأمعاء مختلفة تؤدي إلى ألم أو انسداد أو يرقان.

Recognition and differential (red flags, discriminators)

- Acute mesenteric ischemia
  - Red flags: sudden severe abdominal pain, "pain out of proportion" to exam, soft abdomen early, AF/paroxysmal AF or recent embolic source, advanced age.
  - Pulse, BP changes, tachycardia; lactate may be normal early.

- Differential: bowel obstruction, pancreatitis, perforated viscus, ischemic colitis.
- Neutropenic enterocolitis
- Red flags: recent cytotoxic chemo, neutropenia, fever + new RLQ/ diffuse abdominal pain; risk of rapid progression to perforation/sepsis.
- Painless obstructive jaundice (pancreatic head mass)
- Discriminator: painless jaundice + palpable nontender RUQ mass → (Courvoisier) suspect pancreatic carcinoma until proven otherwise.
- Infectious diarrheas
- Crypto: exposure to contaminated swimming pools, incubation ≈5–10 days, watery diarrhea; more severe in immunocompromised.
- Giardia: exposure to untreated fresh water (lakes/streams), incubation 1–3 weeks, foul-smelling, greasy stools, weight loss; cysts/trophozoites on O&P.
- Campylobacter/Salmonella: bloody or inflammatory diarrhea, shorter incubation, associated food exposures.
- Esophageal food impaction
- Red flags: dyspnea, drooling, inability to handle secretions → airway compromise → urgent endoscopy; if breathing OK and obstruction resolves, still refer for endoscopy.

Arabic: تحتاج الحالات إلى تمييز سريري دقيق: ألم شديد مفاجئ مع الرجفان → شكّ بالإقفار المعوي؛ مريض تحت كيماويّ مصحوب بحمى → اشتبه في typhlitis؛ يرقان بلا ألم مع كتلة RUQ → فكر في ورم البنكرياس؛ مياه حمامات السباحة → cryptosporidium؛ فضلات رائحتها نتنة ودهنية → Giardia.

Diagnostics — targeted tests and imaging

- Initial ED tests common to many acute abd presentations: CBC, ele

7. Arabic rapid-review summary

تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Abdominal and Gastrointestinal Disorders.

CHAPTER 03: TRAUMATIC DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Primary survey and priorities: Airway (with C-spine protection) > Breathing > Circulation > Disability (neuro) > Exposure. Always assume c-spine injury until cleared.
- Glasgow Coma Scale (GCS): eye (4), verbal (5), motor (6). Motor score is the single best predictor of outcome. GCS ≤ 8 generally indicates need for airway protection/intubation.
- Intracranial mass effect/herniation: expanding epidural hematoma (brief LOC → lucid interval → rapid deterioration) and uncal (temporal) herniation produce ipsilateral CN III palsy (dilated, “down-and-out” pupil) and contralateral hemiparesis; Cushing’s reflex = hypertension + bradycardia + irregular respirations → suggests markedly elevated ICP.
- Epidural vs subdural vs subarachnoid vs diffuse axonal CT patterns:
- Epidural: lentiform (biconvex), extra-axial, does not cross suture lines; classically arterial (middle meningeal) → rapid decline.
- Subdural: crescent-shaped, can cross sutures, usually venous → slower.
- Subarachnoid: hyperdensity outlining sulci/ventricles.
- Diffuse axonal: petechial hemorrhages at gray-white junction, corpus callosum, brainstem.
- Secondary brain injury: raised ICP, hypoxia, hypotension — prevent these to improve outcome.
- Blast injuries: primary (pressure wave — TM rupture, pulmonary contusion, bTBI), secondary (projectiles), tertiary (body displacement, blunt trauma), quaternary (burns, inhalation).
- Compartment syndrome: increased interstitial pressure after crush/fracture → ischemia; early sign = severe pain out of proportion + pain on passive stretch; late signs = paresthesia, pallor, pulselessness, paralysis.
- Penetrating vs blunt abdominal trauma triage: unstable + positive eFAST → OR. Stable + positive eFAST → CT and trauma surgery consult.

Arabic (مختصر فسيولوجي):
الأساس: ABCDE مع حماية العمود الفقري. درجة غلاسكو ≥8 عادةً تتطلب تأمين مجرى الهواء. النزف داخل الجمجمة يسبب تأثير كتلي وهرسلة؛ ثلاثية كوشينغ تدل على ارتفاع ضغط داخل القحف خطير. حالات الطوارئ: فشل الطرف المصاب (متلازمة الحيز) ونزف بطني إيجابي في الفحص بالسونار يتطلبان تدخل جراحي عاجل.

Recognition and differential
- Altered mental status after head trauma: differentiate intoxication, seizure/postictal state, hypoglycemia, stroke, intracranial hemorrhage (look for Cushing's reflex), and metabolic causes.
- Signs of basilar skull fracture: Battle's sign, periorbital ecchymosis (raccoon eyes), hemotympanum, CSF rhinorrhea/otorrhea; CT may miss nondisplaced basilar fractures — consult neurosurgery if clinical suspicion.
- Ocular trauma: Seidel's sign (fluorescein streaming away) = globe rupture/aqueous leak. Rust ring suggests retained metallic foreign body.
- Urogenital: pelvic fracture → suspect posterior urethral injury (blood at meatus, high-riding prostate, perineal hematoma); straddle injury → suspect anterior urethral injury.
- Extremity: pain out of proportion after crush/injury, tense compartments → compartment syndrome. Thumb ulnar collateral ligament (UCL) injury = "gamekeeper's/skier's thumb": MCP ulnar-side tenderness and laxity with valgus stress.
- Pediatric minor head injury: use validated clinical decision rules (eg, PECARN) to identify v

Core concepts and pathophysiology
- Blunt and penetrating trauma produce predictable injury patterns by mechanism (axial load, hyperflexion/extension, shear, compression).
- C1 (Jefferson) — burst fracture from axial loading/vertical compression; lateral mass displacement on odontoid (open-mouth) view. (NBK534091)
- C2 (Hangman) — bilateral pars interarticularis fracture of C2 from hyperextension with distraction.
- Bilateral facet dislocation — anterior spondylolisthesis/perching and marked prevertebral swelling from severe flexion-distraction; often unstable.
- Central cord syndrome — hyperextension injury in older patients with preexisting cervical spondylosis causing greater motor/sensory loss in the upper extremities, possible urinary retention. (NBK441932)

- SCIWORA (spinal cord injury without radiographic abnormality) — common in children due to ligamentous laxity; neurologic deficit despite normal x-ray/CT; MRI for diagnosis.
- Intracranial hemorrhages:
- Epidural hematoma — lens-shaped, often MMA tear, lucid interval, best surgical prognosis if evacuated. (Rosen’s Head Trauma)
- Subdural/subarachnoid — diffuse injury or cortical bridging vein rupture; often worse outcome because of underlying brain injury.
- Torso trauma:
- FAST detects intraperitoneal free fluid rapidly but is operator dependent; positive FAST + hemodynamic instability → OR. Stable + positive FAST → CT to localize source. (NBK431087)
- Bladder: extraperitoneal (commonly with pelvic fracture) → conservative management with catheter; intraperitoneal → surgical repair. (NBK557875)
- Diaphragmatic rupture/hernia — may present with bowel in hemithorax on CXR; definitive therapy is operative repair (often laparoscopic). (Gaine FA et al.)
- Facial/orbital/ocular trauma:
- Traumatic iritis — anterior chamber cells/flare, photophobia, small/sluggish pupil; treat with cycloplegic (e.g., cyclopentolate) and ophthalmology follow-up. (Tintinalli / Walker & Adhikari)
- Hyphema grading I–IV (I <33%; II 33–50%; III >50%; IV = total). Grade I–II: bed rest, HOB elevation, ophthalmology f/u. Sickle cell = urgent ophtho.
- Wound/animal-bite infections: polymicrobial including Pasteurella; first line amoxicillin–clavulanate; in true beta-lactam allergy use agents with Pasteurella + anaerobic coverage (e.g., doxycycline, TMP-SMX, or a fluoroquinolone with anaerobic activity such as moxifloxacin). (Rosen’s, literature on bite prophylaxis)

> Arabic (مختصر الفيزيولوجيا المرضية)
> - الإصابات الرضّية لها أنماط متوقعة حسب آلية القوة. كسور مثل جيفرسون من ضغط محوري، وجراحات الوجه والحوض لها تبعات وظيفية ومعدية خاصة.

Recognition and differential
- Cervical spine:
- Red flags: midline cervical tenderness, neurological deficit, distracting injury, altered mental status, intoxication. Use NEXUS/Canadian C-spine rules to decide imaging.
- Patterns: Jefferson (C1 burst) vs odontoid vs Hangman vs facet dislocation —

distinguish on open-mouth, lateral, and CT.
- Head injury:
- Red flags for CT in adults: GCS <15, focal neuro deficit, deterioration, suspected skull fracture, signs of basilar skull fracture (Battle sign, hemotympanum, CSF otorrhea/rhinorrhea), severe mechanism,

Traumatic Disorders — condensed high-yield revision notes (HIPPO-style synthesis)

Core concepts and pathophysiology
- Trauma triage/approach: primary survey (Airway with C-spine control → Breathing → Circulation → Disability → Exposure). Life threats must be recognized and treated immediately. (Rosen's; ATLS principles)
- Compartments of occult bleeding: intrathoracic, intraperitoneal, retroperitoneal, pelvic/soft-tissue, long-bone. Some spaces (eg retroperitoneum, diaphragm) are poorly evaluated by FAST.
- Blunt deceleration injuries: traumatic aortic isthmus tears (isthmus most common), diaphragmatic rupture (left side common), proximal clavicle posterior displacement → mediastinal/vascular injury.
- Intracranial hemorrhage patterns:
- Epidural = arterial (middle meningeal), lens/convex, classic lucid interval, does NOT cross suture lines.
- Subdural = bridging veins, crescent, can cross sutures, worse prognosis.
- Uncal (transtentorial) herniation = ipsilateral CN III compression → dilated, nonreactive pupil; uncus (temporal lobe) pushes through tentorial notch.
- Spinal cord syndromes:
- Anterior cord: loss of motor + pain/temp below lesion; preserved light touch/proprioception (dorsal columns spared).
- Brown-Séquard, central cord, cauda equina have distinct patterns.
- Crush syndrome/rhabdomyolysis: myocyte lysis → hyperkalemia, hyperphosphatemia, hypocalcemia (early), myoglobinuria → AKI. ECG changes from hyperkalemia can be life-threatening.
- Penetrating abdominal wounds with evisceration or positive intraperitoneal fluid in unstable patients indicate operative management.

> Arabic summary (Core concepts)
> - المبادئ الأساسية: مسح وتشخيص الإصابات الحادة بترتيب ABCDE. بعض إصابات النزف

> تكون مخفية (خاصة التجويف خلف البريتوني والحجاب الحاجز). إصابات الدماغ والحبل الشوكي لها أنماط مميزة تساعد في التمييز.

Recognition and differential (red flags, diagnostic discriminators)
- Grey Turner sign / flank/ecchymosis → suspect retroperitoneal injury (eg pancreatic, renal, retroperitoneal bleed) — FAST may be negative.
- Massive hemothorax indicators → chest tube initial output >1.5 L OR ongoing >200–250 mL/hr (texts vary) → operative exploration/thoracotomy.
- Diaphragmatic rupture clues → respiratory distress, orthopnea, bowel sounds in chest, CXR with bowel/air-fluid level in thorax; CT ~50% sensitive; maintain high clinical suspicion; diagnostic/therapeutic laparoscopy/laparotomy is gold standard.
- Penetrating abdominal injury with evisceration → immediate surgical consult/OR.
- Proximal (medial) clavicle fracture with posterior displacement → high risk mediastinal/vascular injury (subclavian vessels, trachea, esophagus) → urgent imaging/vascular/thoracic consult.
- Testicular rupture after blunt scrotal trauma → heterogeneous testicular parenchyma, loss of contour, hematocele on ultrasound → urgent urology/surgical exploration.
- Nail puncture through shoe (sole) → high risk Pseudomonas and S. aureus; delayed/infected puncture → IV therapy with anti-pseudomonal coverage.
- Cat bite → Pasteurella multocida (treat with amoxicillin-clavulanate).
- Oculovestibular reflex (cold irrigation) = COWS mnemonic: Cold → Opposite (fast component

Core concepts and pathophysiology
- ATLS/primary survey mindset: airway with cervical spine control, breathing, circulation, disability, exposure. Mechanism guides suspicion for specific injuries (axial load → C1 burst; steering-wheel blow → diaphragmatic rupture; high-energy pelvic crush → major pelvic hemorrhage).
- Energy transfer and pediatric differences: children have more pliable ribs, large intra-abdominal organs, thick periosteum → greenstick/buckle fractures and higher risk of solid-organ injury despite seemingly minor external signs.
- Compartment syndrome: rising intracompartmental pressure → ischemia. Diagnostic thresholds used clinically: absolute compartment pressure > 30 mmHg or delta pressure (diastolic BP − compartment pressure) < 30 mmHg.
- Penetrating vs blunt chest: penetrating may cause pericardial effusion/ tamponade; blunt chest more likely to cause tension pneumothorax, pulmonary

contusion, diaphragmatic rupture.
- Hand/soft-tissue special cases: high-pressure injection injuries produce extensive deep tissue necrosis despite tiny entry wounds; fingertip amputations require zone-based decision making (Allen zones).
Sources: Rosen’s Emergency Medicine; multiple NCBI/NCBI Bookshelf trauma chapters (see refs).

> مفاهيم أساسية: اتبع مسار ATLS، عدّل الشك السريري حسب آلية الإصابة (تحميل محوري، صدمات حزام الأمان، طلق ناري). لدى الأطفال اختلافات تشريحية مهمة (عظام قادرة على الانثناء، أعضاء كبيرة نسبياً). ضغط الحجرة >30 مم زئبق أو Δ<30 يشير إلى متلازمة الحجرة.

Recognition and differential
- Chest:
- Tension pneumothorax: severe dyspnea, hypotension, JVD, decreased/absent breath sounds, tracheal shift (late). Immediate decompression required.
- Cardiac tamponade: hypotension, JVD, muffled heart sounds (Beck’s triad) + FAST pericardial fluid; electrical alternans and low QRS voltage on ECG.
- Diaphragmatic rupture: mechanism (blunt steering wheel), chest pain worse when supine, vomiting, NG tube coils in thorax, bowel/solid viscera in hemithorax on CXR; left > right more common.
- Abdomen/pelvis:
- Positive FAST in unstable → OR; positive FAST in stable → CT abdomen/pelvis.
- Hollow viscus/mesenteric/duodenal injuries may be occult on initial CT — persistent/worsening pain after negative CT → admit, serial exams, early surgical consultation.
- Pelvic fractures → major retroperitoneal hemorrhage; instability, pelvic tenderness, open book on radiograph.
- Extremities:
- Compartment syndrome: severe pain out of proportion, pain on passive stretch, tense swelling, paresthesia; distal pulses may remain early.
- High-pressure injection: tiny punctum, severe underlying contamination — high risk of necrosis/amputation.
- Pediatric fractures: greenstick (one cortex break), torus/buckle (compression buckling) — stable.
- Face/neck:
- Le Fort fractures: Le Fort I (maxillary alveolar), II (pyramidal, midface mobility), III (craniofacial dissociation — whole midface moves).

- Penetrating neck: violation of platysma → consider deeper injury; zone 2 (cricoid → angle mandible) most common.
- Medial canthal lacerations → canalicular injury; need ophthalmic/oral-maxillofacial specialist.
- Joints/soft tissue:
- Patellar di

Core concepts and pathophysiology
- Blunt vs penetrating trauma produce distinct injury patterns: direct compression/axial load → burst/pars fractures (C1 Jefferson, C2 Hangman), deceleration → aortic, cardiac contusion, visceral shear, diaphragmatic rupture; focal impact (handlebar, fencepost, ball) → pancreas/duodenum hematoma, urethral/penile/perineal injury, orbital/subconjunctival hemorrhage.
- Spine stability: C1 (atlas) burst with lateral mass displacement (Jefferson) vs C2 pedicle/pars fracture (Hangman) vs odontoid (dens) types I–III; unstable fractures need immobilization and neurosurgical input. (See NBK534091, NBK519496, NBK441956)
- Vascular risk after joint dislocations: knee dislocation risks popliteal artery injury even after spontaneous reduction; diminished distal pulses or high-energy mechanism mandates vascular imaging. (NBK470595)
- Diaphragm injuries: left > right; NG tube coiling in thorax, delayed GI/respiratory symptoms; CT/CXR have low sensitivity — diagnostic laparoscopy/laparotomy is gold standard. (NBK557647)
- Pediatric differences: open sutures allow greater accommodation of expanding hematoma; head CT sensitivity varies; fluid/blood resuscitation volumes differ (children: 20 mL/kg crystalloid × up to 2, then PRBC 10 mL/kg if persistent hypotension). (Rosen’s; see Puskarich/Jones)
- Penetrating neck trauma: historically zone-based risk; current practice relies on presence of hard signs to proceed to OR, otherwise CTA is rapid and sensitive in stable patients. (EAST guidance; NBK470422)

Arabic summary (مختصر مفاهيمي)
- الإصابات الناتجة عن الصدمات يمكن أن تكون كدمات عميقة أو اختراقية؛ بعض الكسور العنقية (C1, C2, dens) غير مستقرة وتحتاج تثبيت واستشارة جراحة أعصاب. التشخيص المحدد (مثل تمزق الحجاب الحاجز أو إصابة الشريان خلف الركبة) قد يتطلب تصوير متقدم أو تنظير جراحي.

Recognition and differential
- Cervical spine: midline cervical tenderness, mechanism (diving, high-speed deceleration), neurological signs. DDX: ligamentous injury vs atlas burst vs odontoid vs teardrop.
- Thoracoabdominal: seatbelt sign, Kehr's sign (left shoulder pain = diaphragmatic irritation → splenic rupture), abdominal bruising, persistent vomiting after blunt impact → duodenal or pancreatic injury (can be occult on early CT).
- Penetrating wounds: hard signs (active hemorrhage, expanding hematoma, airway compromise, pulsatile bleeding, shock, bruit/thrill, absent distal perfusion) → immediate OR. Soft signs → CTA and specialist consult.
- GU trauma: blood at meatus, scrotal/perineal hematoma, inability to void → urethral injury (straddle → anterior; pelvic fracture → posterior). Gross hematuria + pelvic fracture → bladder rupture (retrograde/CT cystogram to distinguish intra- vs extraperitoneal).
- Extremity: knee with high-energy mechanism or spontaneously reduced dislocation → suspect popliteal artery injury even if pulses initially present; fibular head fracture + foot drop → common peroneal nerve injury.
- Ocular: subconjunctival hemorrhage after blunt eye trauma with normal vision/IOP and no corneal uptake → conservative management.

> Arabic summary (التعرف والتمييز)
> - العلامات الحمراء: نزيف نابض، غرابة عقلية عنق الرحم، ألم كتفي يسار بعد

Traumatic Disorders — condensed high-yield revision notes
(Integrated, de-duplicated from multiple HIPPO QBank items; sources: Rosen's Emergency Medicine; Tintinalli's Emergency Medicine; relevant NCBI/UpToDate reviews cited in source materials.)

1) Core concepts and pathophysiology
- Mechanisms matter: axial load → burst (C1 Jefferson) or compression injuries; hyperflexion → bilateral facet dislocation, flexion teardrop (unstable); hyperextension → Hangman's (C2 pars) fractures; axial + rotation → meniscal tears; FOOSH (fall on outstretched hand) → scaphoid, lunate/perilunate injuries, radial head fracture.
- Pediatric growth-plate injuries (Salter-Harris): types I–V; I and V may be radiographically occult (type V = compression; risk of growth disturbance).
- Blast injuries: primary = blast overpressure (TM, lungs, bowel), secondary =

projectiles, tertiary = body displacement causing crush/fractures/traumatic amputations, quaternary = burns/toxic inhalation.
- High-pressure injection injuries (grease/paint guns): small skin wound but extensive deep tissue damage — surgical emergency.
- Chest wall: flail chest = ≥2 fractures of ≥3 consecutive ribs → paradoxical segment; often associated pulmonary contusion → V/Q mismatch, respiratory fatigue.
- Intracranial herniation patterns: uncal (ipsilateral CN III palsy/blown pupil ± contralateral hemiparesis), subfalcine (cingulate; may be silent), tonsillar (brainstem compression → coma/arrest).
- Urologic trauma: pelvic fracture → high risk for posterior urethral injury; avoid blind Foley until urethra excluded.

> Arabic summary (مختصر فسيولوجيا الإصابة):
> - آليات الإصابة تحدد النوع: السقوط/التحميل المحوري/الانثناء/الانقلاب تؤدي لإصابات عظمية أو رباطية مميزة. صفائح نمو الأطفال قد تبدو طبيعية بالأشعة (النوع I وV). إصابات الطوارق عالية الضغط تحتاج تدخل جراحي عاجل.

2) Recognition and differential (key clinical signs)
- Wrist: anatomic snuffbox tenderness → suspected scaphoid fracture (occult on initial x-ray). Lunate dislocation → “spilled teacup” on lateral, “piece-of-pie” on AP; perilunate spectrum distinguished by alignment of capitate with radius.
- Elbow: posterior fat pad (on true lateral) → occult intra-articular fracture (radial head in adults, supracondylar in children).
- Finger injuries: mallet = terminal extensor avulsion; boutonniere = central slip injury (PIP flexion, DIP hyperextension).
- Pelvis: blood at meatus, high-riding prostate, perineal/scrotal hematoma → suspect urethral injury.
- Abdomen: “seatbelt sign” (lap belt ecchymosis) → increased risk hollow-viscus/mesenteric shear injuries.
- Blast: displacement of victim, evidence of penetrating fragments, burns, inhalation.
- Spinal radiographs: flexion teardrop → unstable; bilateral facet dislocation → acute neurologic deficit possible.
- Child abuse clues: classic metaphyseal/“bucket-handle” lesions, bilateral symmetric corner fractures in infants.

Arabic summary (العلامات السريرية المميزة):
- تيبس أو ألم في صندوق المنخول = اشتباه كسر العظم الزورقي. شارة الحزام المقعدي تدل على خطر إصابات أحشاء داخلية خفية. وجود دم في مجرى البول مع كسر حوضي → فكر في إصابة الإحليل.

3) Diagnostics — discriminators and modalities
- Plain radiographs:

Core concepts and pathophysiology
- Spinal cord syndromes (mechanism = focal cord compression/hemisection/ischemia)
- Brown–Séquard (hemisection, classically from penetrating trauma): ipsilateral UMN weakness and loss of vibration/proprioception below lesion; contralateral loss of pain and temperature beginning a few levels below the lesion (spinothalamic decussation).
- Anterior cord syndrome: anterior 2/3 of cord → motor and pain/temperature loss with preserved dorsal column (vibration/proprioception).
- Central cord: greater motor weakness in the upper>lower limbs (cervical hyperextension in elderly).
- Posterior cord: predominant dorsal column signs (ataxia, loss of vibration/proprioception).
- SCIWORA (spinal cord injury without radiographic abnormality): pediatric ligamentous elasticity allows cord injury without bony fracture; neuro deficits may be delayed. MRI is diagnostic for cord/ligamentous injury.
- Hyphema: blood in anterior chamber. Major early complication is delayed/recurrent bleeding after clot retraction; risk of raised IOP and optic compromise.
- Traumatic amputation: viability depends on ischemia time and tissue handling; replantation candidates require clean preservation of the part and urgent microsurgical consultation.
- Cardiac/pericardial injury in trauma: any pericardial fluid after trauma is abnormal and may represent hemopericardium → tamponade physiology; penetrating thoracic trauma is high risk.
- Commotio cordis: blunt precordial blow at a vulnerable phase of ventricular repolarization → primary ventricular fibrillation in otherwise structurally normal heart (typically young athletes).
- Pediatric distal-radius injuries: torus (buckle) fracture = compression of cortex, stable, requires immobilization only; greenstick and physeal injuries are less stable

and need ortho follow-up.
- Mandibular fractures: many (angle, ramus, symphysis) communicate with oral cavity → treated as “open” and receive prophylactic antibiotics. Condylar fractures typically do not require prophylactic antibiotics.
- Hollow viscus injury after blunt abdominal trauma: may be occult; CT with IV contrast is best test in hemodynamically stable patients, but sensitivity for bowel injury is imperfect; manage stable, equivocal CT with observation unless peritonitis/instability.
References: hyphema (NBK507802); amputations (NBK538153); pericardial/trauma (NBK535444); commotio cordis (NBK526014); pediatric buckle fracture (CMAJ); mandibular fractures (NBK507705); abdominal trauma (Drexel et al.).

> Arabic (باختصار العِلمِي): التصنيفات الفقارية تحدد أنماط الفقد الحسي/الحركي؛ إصابة الحبل الشوكي قد تكون دون كسر واضح خاصة عند الأطفال (SCIWORA)؛ أي سائل حول التامور بعد الصدمة يجب اعتباره مَرضياً؛ إصابات العينّ قد تُعاد بالنزف؛ إصبع مقطوع يجب حفظه رطبًا ومبردًا دون وضعه مباشرة في الماء/ثلج.

Recognition and differential
- Spinal cord injury patterns — key discriminators
- Brown–Séquard: ipsilateral motor + vibration loss; contralateral pain/temp loss.
- Anterior cord: motor + pain/temp deficits with preserved dorsal columns.
- Central cord: older patient, greater upper-limb motor deficit.
- Posterior cord: isolated proprioceptive/vib

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Traumatic Disorders.

CHAPTER 04: CARDIOVASCULAR DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Acute coronary syndromes (ACS): plaque rupture → thrombosis → STEMI/NSTEMI. Reperfusion priority: PCI if timely; fibrinolysis when PCI delayed/unavailable [ACC/AHA STEMI guidelines].
- Arrhythmias:
- SVT/AF/AFlutter: re-entrant or focal automaticity; hemodynamic stability guides therapy (rate control vs rhythm control vs cardioversion).
- WPW: accessory pathway → pre-excitation (short PR, delta wave); AV-nodal blockers can paradoxically accelerate ventricular rate during atrial fibrillation/flutter by promoting conduction via accessory pathway.
- AV blocks: Mobitz I = progressive PR prolongation then drop; Mobitz II = fixed PR with sudden dropped beats; Mobitz II → high risk progression to complete heart block.
- Ventricular arrhythmias: torsades de pointes is polymorphic VT associated with prolonged QT (drug-induced, electrolyte disturbance) and can degenerate to VF.
- Structural/cardiac mechanical:
- Aortic stenosis: fixed outflow obstruction, preload dependent; severe AS → syncope/exertional angina/heart failure — urgent evaluation and valve replacement if symptomatic.
- Hypertrophic obstructive cardiomyopathy (HOCM): asymmetric septal hypertrophy and systolic anterior motion → outflow obstruction; syncope in young athletes; echo first-line.
- Myocarditis: viral etiologies (parvovirus B19, enteroviruses, HHV-6); can mimic ACS — troponin may be elevated, CXR cardiomegaly.
- Pericarditis: pleuritic chest pain, friction rub, diffuse ST elevation with PR depression; colchicine + NSAID/aspirin therapy.
- Vascular emergencies:
- DVT/PE in pregnancy: physiologic hypercoagulability; LMWH is preferred anticoagulant in pregnancy.
- Phlegmasia cerulea dolens: massive iliofemoral DVT causing venous occlusion,

cyanosis, absent pulses — limb-threatening → urgent thrombectomy/thrombolysis.
- Carotid artery dissection: headache, neck pain, ipsilateral Horner syndrome, transient monocular blindness; antithrombotic therapy to prevent stroke.
- Abdominal aortic aneurysm (AAA): rupture → hypotension, abdominal/back pain; bedside ultrasound rapid diagnostic in unstable patient; immediate vascular/OR mobilization.

> Arabic summary (مختصر فيزيولوجيا مرضية): نظم القلب، انسداد الشرايين، وأمراض تركيبية للقلب قد تؤدي إلى نقص تروية مفاجئ، اختلال توصيل كهربائي، أو فشل وظيفي. التمييز المبكر بين أذية عضلية، اضطراب موضع القناة المساعدة، وسبل الإنعاش الحاد يحدد العلاج الطارئ.

Recognition and differential (key presentations and red flags)
- Chest pain with ST-elevation or equivalent (posterior MI: ST depression V1–V3 with ST elevation in lateral leads) → STEMI.
- Hemodynamic instability with narrow-complex rapid tachycardia → synchronized cardioversion.
- Hypotension + brady/AV block or syncope → consider high-degree AV block (Mobitz II, complete block).
- Exertional or exercise-related syncope, family history of sudden death, or syncope with abnormal ECG → high-risk cardiac syncope; admit for monitoring and workup.
- New fever, weight loss, recent dental procedure, valvular disease, positive blood cultures → suspect infective endocarditis (TTE → TEE if needed) [Duke criteria guidance].
- Pleuritic chest pain + di

Cardiovascular Disorders — condensed high-yield ED revision notes
(Integrated and deduplicated from HIPPO QBank material; selected sources: Rosen’s Emergency Medicine, Tintinalli’s Emergency Medicine, ACC guidance, NIH/NLM topic reviews)

1) Core concepts and pathophysiology
- Ischemic coronary syndromes: spectrum from unstable angina → NSTEMI → STEMI. Acute complications (early post-MI, days) include papillary muscle rupture → acute MR; interventricular septal rupture → VSD/left→right shunt; free wall rupture → tamponade. Mechanical complications often present with new murmur, shock, pulmonary edema (depending on lesion) and need urgent surgical support.

(Rosen’s; ACC)
- Pericardial disease vs myocarditis: pericarditis = pericardial inflammation → pleuritic positional chest pain, pericardial rub, evolving diffuse ST elevations with PR depressions; myocarditis often causes troponin leak and arrhythmia risk. Chagas (Trypanosoma cruzi) is a major global cause of myocarditis/cardiomyopathy. (Rosen’s; NLM)
- Cardiac tamponade: pericardial fluid compresses all chambers → preload dependence, Beck’s triad (hypotension, JVD, muffled heart sounds), pulsus paradoxus, electrical alternans on ECG. Initial resuscitation aims to maintain preload until definitive pericardiocentesis. (Rosen’s)
- Aortic dissection: intimal tear producing true/false lumen. Stanford A (ascending) → surgical emergency. Hypertension common trigger; aortic regurgitation murmur and pulse deficits may be present. CTA chest is the emergent diagnostic study if patient stable. (Tintinalli; NLM)
- Arrhythmias: SVT (re-entrant) → abrupt tachycardia; WPW (ventricular pre-excitation) shows short PR and delta wave and is a cause of syncope/SCD in young athletes. Ventricular tachycardia (monomorphic/polymorphic) may be unstable → cardioversion. Multifocal atrial tachycardia (MAT) = >3 P-wave morphologies, irregular rhythm; strongly associated with COPD/exacerbation. Pediatric differences: calcium-channel blockers contraindicated in infants <1 year; ductal-dependent congenital lesions present after ductus closure and require PGE1 to maintain ductal patency. (Rosen’s)
- Hypertensive emergencies: organ-threatening BP elevations (e.g., ICH, pulmonary edema). Drug choice depends on target organ (e.g., nicardipine/labetalol for ICH to SBP 140–160; nitroglycerin preferred for hypertensive acute pulmonary edema). (Guideline summaries; NLM)

> الشرح المختصر: هذه النقاط الأساسية تشرح الآليات التي تؤدي إلى المضاعفات الطارئة القلبية الشائعة: مضاعفات ما بعد الاحتشاء، التامبوناد، تمزق الشريان الأورطي، واختلافات علاجية في اضطرابات النظم والحمل الشرياني الدموي.

2) Recognition and differential (diagnostic discriminators / red flags)
- Sudden collapse during exertion in a young athlete + palpitations → think WPW/HOCM/primary arrhythmia. EKG: short PR + delta wave → WPW. HOCM often has deep narrow Q waves, LVH; echo may show asymmetric septal hypertrophy.
- New holosystolic murmur after recent MI:
- Harsh holosystolic at LLSB → VSD (interventricular septal rupture) → hypotension,

left→right shunt.
- Holosystolic apex radiating to axilla → acute MR (papillary muscle dysfunction/rupture).
Both can cause pul

Core concepts and pathophysiology
- Syncope: transient loss of consciousness from global cerebral hypoperfusion. Major categories — reflex (vasovagal), orthostatic, and cardiac (arrhythmic or structural). Cardiac syncope carries highest mortality risk.
- Arrhythmogenic ECG syndromes:
- Brugada syndrome: coved ST-elevation in V1–V2 (type 1); risk of ventricular arrhythmia and sudden death.
- Hypertrophic cardiomyopathy (HCM): asymmetric septal hypertrophy → outflow obstruction; ECG: high QRS voltages, “dagger-like” deep narrow Q waves in lateral/inferior leads.
- Acute aortic syndromes: intimal tear → true + false lumen; ascending (type A) surgical emergency. Wall shear stress reduction (HR then BP) reduces propagation risk.
- Acute coronary ischemia: plaque rupture → STEMI → transmural ischemia (ECG ST-elevation); posterior MI classically shows ST depression and tall R waves in V1–V3.
- Coronary vasospasm (Prinzmetal): transient coronary constriction → transient ST-elevation relieved by nitrates; diagnosis with coronary angiography + provocative testing.
- Pericardial effusion/tamponade: fluid → impaired diastolic filling → hypotension, JVD, muffled heart sounds; electrical alternans on ECG may be seen.
- Congenital cyanotic lesions (pediatrics):
- Tetralogy of Fallot (TOF): VSD + pulmonary stenosis + overriding aorta + RVH → “tet spells” (hypercyanotic episodes) from increased RVOT obstruction and right-to-left shunt.
- Eisenmenger physiology: long-standing L→R shunt → pulmonary vascular remodeling → reversal to R→L with cyanosis.
- Patent ductus arteriosus (PDA): left-to-right shunt causing volume overload; ductal-dependent lesions require prostaglandin E1 (PGE1) to maintain patency.
- Abdominal aortic aneurysm (AAA): dilation of aorta; rupture most commonly into the retroperitoneum.

Arabic (مفاهيم أساسية)
- الإغماء نتيجة انخفاض مؤقت بتروية الدماغ: أهم الأنواع انعكاسي، وضعيفي، وقلبي. الانغماس القلبي الأخطر.
- متلازمة برجادا، اعتلال عضلة القلب التضخمي، تمزق الشريان الأورطي الحاد، واعتلالات قلبية خلقية مثل تترالوجي أوف فالات تشكل كيانات علاجية طارئة.

Recognition and differential (red flags and discriminators)
- Syncope red flags (admit / urgent cardiology):
- Syncope during exertion or supine, syncope with palpitations, family history sudden cardiac death, known structural heart disease, abnormal ECG (Brugada pattern, WPW, high-grade AV block, ischemia).
- Distinguishing etiologies:
- Vasovagal: prodrome (nausea, diaphoresis, emotional trigger), transient, normal post-event exam/ECG.
- Orthostatic syncope: BP drop ≥20 mmHg systolic or ≥10 mmHg diastolic within 3 minutes of standing.
- Cardiac: sudden onset without prodrome, exertional or supine, abnormal ECG, structural disease.
- Pediatric cyanotic spells:
- TOF “tet spell”: sudden cyanosis, irritability/agitation, harsh systolic ejection murmur, boot-shaped heart on CXR.
- PDA/ductal dependency: poor feeding, respiratory distress, continuous “machine” murmur; collapse when ductus closes.
- Aortic dissection clues: tearing chest/back pain, asymmetric pulses/weak femoral pulse, widened mediastinum on CXR, new aortic or aortic-valve findings.
- Carotid dissection: ips

Core concepts and pathophysiology
- Valvular disease
- Rheumatic heart disease — classically affects the mitral valve (mitral stenosis in adults → low-pitched mid-diastolic rumble with an opening snap). Mitral stenosis → left atrial hypertension → pulmonary venous congestion, hemoptysis, atrial fibrillation and embolic stroke [NCBI / Harrison’s/Rosen’s].
- Pericardial disease
- Acute pericarditis presents with pleuritic chest pain relieved by leaning forward, diffuse ST-elevation and PR depression; pericardial effusion → cardiomegaly on CXR when >≈250 mL; tamponade physiology → JVD, hypotension, pulsus

paradoxus. Late/post-MI autoimmune pericarditis = Dressler (1–6 weeks post-MI). Uremic pericarditis occurs in missed/advanced dialysis patients; dialysis is primary therapy.
- Aortic disease
- Aortic dissection: intimal tear → flap separating true/false lumen. Presents with severe chest/back pain, pulse/radial-femoral delay, HTN. Rate control (beta-blocker) MUST precede vasodilation to avoid reflex shear stress.
- Arrhythmias & conduction disease
- Atrial fibrillation (AF): irregularly irregular; management distinguishes stable vs unstable; rate control with beta-blockers or nondihydropyridine CCBs is first-line for most stable patients. Adenosine is for re-entrant SVT only.
- Lyme carditis: can cause high-grade AV block (including 3rd-degree) in young patients — treat severe conduction disease with IV ceftriaxone; milder PR prolongation (PR <300 ms) can be treated with doxycycline.
- Long QT → polymorphic VT (torsades) and syncope; congenital and drug-induced causes; QTc > 480–500 ms increases risk.
- Brugada syndrome: type I (coved ST elevation V1–V2), syncope or ventricular arrhythmia → ICD for secondary prevention.
- Wolff-Parkinson-White (WPW): short PR, delta wave; if SVT — consider AVRT; if AF with WPW avoid AV-nodal blockers (beta-blockers, CCBs, adenosine, digoxin) — use procainamide or electrical cardioversion.
- Pacemaker-mediated tachycardia (PMT): re-entry involving device — acute therapy: magnet over device to convert to asynchronous pacing and stop PMT; cardiology/device team follow-up.
- Right heart / pulmonary hypertension (cor pulmonale)
- Chronic lung disease or recurrent pulmonary emboli → pulmonary hypertension → right ventricular hypertrophy/failure. Gold standard diagnostic test: right heart catheterization; echo is screening tool [Rosen’s].

> Arabic summary (concise)
> - الأمراض الصِمامية والالتهاب التاموري وتمزق الأبهري والاضطرابات النَظْمية كلٌ لها علامات وأسباب مميزة. فحص الصدر بالموجات فوق الصوتية والإكو مهمان، وتعطى الأولوية للسيطرة على الضَربات وضغط الدم عند الإصابة بتمزق الأبهر أو السَّكَّان القَلبيّ.

Recognition and differential (red flags and discriminators)
- Aortic dissection red flags: severe tearing chest/back pain, unequal pulses or BP (radial–femoral delay), new aortic regurgitation murmur, hypotension/shock.

(Immediate BP/HR control + emergent imaging/surgical consult.)
- Pericarditis vs MI: pleuritic, positional pain + diffuse ST elevation/PR depression → pericarditis; focal ST elevation with reciprocal changes → MI. Post-MI within 1–6 weeks with pericarditic picture → Dressler.
- Cardi

Core concepts and pathophysiology
- Ischemia spectrum: STEMI = transmural infarction from acute full-thickness coronary occlusion → immediate reperfusion (PCI ± fibrinolysis if no PCI). NSTEMI = subendocardial/non-transmural ischemia from partial/temporary occlusion → medical therapy ± delayed PCI.
- Valvular disease: severe aortic stenosis → fixed outflow, preload dependence, exertional syncope, narrow pulse pressure; prosthetic valves risk early prosthetic valve endocarditis (PVE), early PVE commonly Staphylococcus aureus.
- Aortic disease: aortic dissection = intimal tear creating false lumen. Stanford A (ascending) = surgical emergency; Stanford B (descending) initially medical (BP control) unless complications.
- Pericardial disease: pericarditis often viral; classic pleuritic, better leaning forward, pericardial friction rub, diffuse concave ST elevation ± PR depression. Large/rapid effusion → tamponade (Beck's triad: hypotension + JVD + muffled heart sounds) with pulsus paradoxus.
- Myocarditis: viral prodrome, chest pain, fever, tachycardia, elevated troponin, variable ECG and echocardiogram; can present like ACS or cause cardiogenic shock/arrhythmia.
- Arrhythmias:
- Ventricular tachycardia: classify as pulseless vs perfusing; monomorphic vs polymorphic. Stable monomorphic VT → antiarrhythmic (procainamide commonly used in stable perfusing VT). Unstable → synchronized cardioversion. Pulseless VT/VF → immediate defibrillation + ACLS.
- Torsades de pointes (polymorphic VT associated with prolonged QT): precipitated by QT-prolonging drugs, electrolyte disturbances (↓K, ↓Mg), congenital LQTS.
- PEA/asystole: treat per ACLS (epinephrine) and search for reversible causes (Hs and Ts).
- Venous vs arterial thromboembolism: DVT predicted by Virchow's triad: stasis, endothelial injury, hypercoagulability. Acute arterial occlusion presents with 6 P's and requires immediate anticoagulation (heparin) and vascular surgery consult.
- Congenital ductal-dependent lesions (neonate): ductus arteriosus closure →

cardiogenic shock or differential pulses (bounding upper, weak lower) → prostaglandin E1 to reopen ductus.

> Arabic summary (مفاهيم أساسية):
> النوبات القلبية: STEMI → انصمام كامل → إعطاء ترميم فوري. التهاب التامور: ألم صدري يتزايد بالاستلقاء ويخف بالانحناء؛ علاج أولي مضادات الالتهاب. تمزق الشريان الأورطي/نادراً: تصوير CTA وتهيئة جراحي. اضطرابات النظم: torsades → ماغنيسيوم؛ الرجفان البطيني/VT غير مستقرة → صدمة كهربائية.

Recognition and differential
- Syncope:
- Cardiac red flags: syncope during exertion, sudden syncope, syncope with structural heart disease, abnormal ECG (new BBB, high-degree AV block, VT) → high risk for adverse events and admission.
- Vasovagal features (prodrome of nausea, diaphoresis, warm, prolonged standing, situational trigger) → low risk.
- Chest pain differential (high-yield discriminators):
- STEMI: focal ST-elevation in contiguous leads, reciprocal changes, high troponin → reperfusion.
- Pericarditis: diffuse concave ST elevation ± PR depression, pleuritic pain improved leaning forward; troponin may be mildly elevated if myocardial involvement.
- Myocarditis: viral prodrome, persistent tachycardia, fever

Cardiovascular Disorders — condensed high-yield emergency revision notes (Consolidated from the supplied QBank explanations; includes referenced guidance where noted)

> **EXAM TRICK / تريكة امتحانية**
> Headings: Core concepts and pathophysiology; Recognition and differential; Diagnostics; Emergency management (algorithms & red flags); Procedures/ drugs/disposition; Exam tricks and pitfalls; Arabic take-home.

1) Core concepts and pathophysiology
- Aortic dissection: intimal tear → propagation of blood within media. Type A (ascending) is surgical emergency. Rate and pressure reduction are key to reduce shear stress (first reduce heart rate, then BP). Beta-blockade before vasodilators to avoid reflex tachycardia. (See AHA/ESC guidance; AHJ reference in Q text.)
- Pericardial tamponade: pericardial fluid under pressure → impaired diastolic

filling; reduced cardiac output. Beck’s triad (JVD, muffled heart sounds, hypotension); pulsus paradoxus (most sensitive physical sign).
- Post-MI mechanical complications: interventricular septal rupture → acute holosystolic murmur, left→right shunt, acute heart failure; definitive management is surgical; temporize with afterload reduction.
- Coronary vasospasm (Prinzmetal): transient coronary artery constriction → transient ST elevation; requires angiography to exclude fixed lesion.
- Hypertensive emergency: end-organ damage (e.g., renal injury, pulmonary edema) — initial controlled BP reduction: reduce MAP/SBP by about ≤25% in first hour then gradual reduction (per Q-explanation).
- Cocaine/sympathomimetic hypertension: alpha-mediated vasoconstriction; pure β-blockers are contraindicated because of risk of unopposed α stimulation; alpha-blockade (phentolamine) and benzodiazepines are appropriate.
- DVT: thrombus in deep venous system; use Wells score to risk-stratify, combine D-dimer and compression ultrasound as appropriate.
- Long QT syndrome: prolonged ventricular repolarization → torsades de pointes; congenital or acquired; QTc thresholds noted: adolescents normal <0.42 s, QTc >0.55 s associated with high sudden-death risk.
- Brugada syndrome: sodium-channelopathy (autosomal dominant) → Type I ECG pattern in V1–V2 and high risk of ventricular fibrillation / sudden death; syncope in Brugada implies malignant ventricular arrhythmia.
- Neonatal ductal-dependent lesions (e.g., coarctation presenting when ductus closes): prostaglandin E1 maintains ductal patency and restores systemic perfusion.
- Cor pulmonale: right-sided failure from pulmonary hypertension; acute cause commonly pulmonary embolus; chronic causes include COPD.

> Arabic (concise): المفاهيم الأساسية: تمزق الأورتا → تقليل سرعة القلب ثم الضغط؛ الانصباب التاموري يسبب فشل تعبئة أثناء الانبساط؛ صدمات ميكانيكية بعد احتشاء ممكنة وتحتاج إلى تخفيف المقاومة الطرفية ثم جراحة؛ في حالات فرط ضغط من منشأ سمبثاوي (مثل الكوكايين) تجنّب حاصرات بيتا النقية، استخدم حاصرات ألفا.

2) Recognition and differential (red flags and discriminators)
Red flags suggesting aortic dissection
- Sudden severe “tearing” chest/back pain
- Marked hypertension, pulse/blood pressure asymmetry, focal neuro signs, mediastinal widening

- New ischemic ECG changes (can mimic STEMI if coronary ostia involved)
Discriminator:

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Cardiovascular Disorders.

CHAPTER 05: CUTANEOUS DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology

- Infectious vs inflammatory vs autoimmune vs toxic/drug-mediated skin /disease. Recognition hinges on lesion morphology (macule/papule/vesicle ,bulla/ulcer), distribution, timing, systemic features, and exposures (meds .(travel, pets, seafood, close contacts
- :Blistering spectrum
  - → Intraepidermal acantholytic (flaccid bullae, mucosal involvement) .pemphigus vulgaris
  - .Subepidermal (tense bullae, older adults) → bullous pemphigoid
  - /Epidermal necrosis with mucosal involvement ± widespread sloughing → SJS .TEN (usually drug-triggered; spectrum by %BSA)
  - Exfoliative toxin-mediated in neonates → Staphylococcal Scalded Skin .Syndrome (SSSS) (toxin cleaves desmoglein-1; mucosa spared)
- :Common infectious mechanisms
  - Superficial bacterial (impetigo: S. aureus ± GAS) → pustules → honey-colored .crusts
  - Deep dermal/lymphatic (erysipelas: GAS) → sharply demarcated, raised, bright .erythema
  - Dermal/subcutaneous infections and necrotising infections (GAS, Vibrio .(vulnificus with raw seafood/liver disease
- Dermatophytes infect keratinized tissues; hair involvement (tinea capitis) .requires systemic therapy because topical agents do not penetrate hair shafts
- /Contact allergic/irritant dermatitis: Type IV hypersensitivity (poison ivy/oak .(sumac → linear vesicular rash
- Immune-mediated target lesions: Erythema multiforme (commonly post-HSV .(or Mycoplasma
- ;Viral pox/umbilicated papules: Molluscum contagiosum — self-limited .consider immunosuppression if adult/widespread

> Arabic (موجز): المفاهيم الأساسية: فرْقٌ بين الأسباب العَدْوِيَّة، الالتهابية، المناعية، ودوائية؛ البنية النسيجية للفقاعات تحدد التشخيص؛ بعض الفطريات تحتاج علاجاً فموياً عند إصابة الشعر.

Recognition and differential (practical discriminators / “red flags”)
- Impetigo: children 2–5 y, vesicle/pustule → honey-colored crusts; highly contagious. (Differential: contact cheilitis, perioral dermatitis, seborrheic dermatitis.)
- Tinea corporis: annular lesion with central clearing and raised scaly border (“ringworm”). Avoid steroids (tinea incognito).
- Tinea capitis / kerion: scalp hair loss + inflammatory boggy plaque (kerion) → oral antifungal (griseofulvin/terbinafine).
- Scabies: intense nocturnal pruritus, interdigital web spaces, wrist creases, linear burrows; household contacts affected.
- Head lice: pruritus, nits on hair shafts—diagnose by wet combing with fine-toothed comb (most sensitive).
- Contact dermatitis (poison ivy/oak/sumac): linear streaks, vesicles after outdoors exposure → prolonged steroid taper for severe cases.
- Erythema multiforme: target/iris lesions on acral surfaces; check for recent HSV infection.
- SJS/TEN vs SSSS vs bullous diseases:
- SJS/TEN: mucosal involvement, painful skin, positive Nikolsky, prodrome of fever/myalgias, often drug-related; SJS ≤10% BSA, TEN >30% BSA.
- SSSS: neonates/young children, generalized erythema and desquamation, positive Nikolsky, mucosa spared; systemic staph infection/toxin.
- Bullous pemphigoid: older adults, tense subepidermal bullae, pruritus, Nikolsky negative.
- Pemphigus vulgaris: flaccid bullae, prominent mucosal lesions, Nikolsky positive.
- Erysipelas v

Cutaneous Disorders — high-yield emergency revision notes (deduplicated)

Core concepts and pathophysiology
- Classification (useful ED framework)
- Infectious: bacterial (impetigo, erysipelas/cellulitis), viral (herpes zoster), fungal (sporotrichosis, tinea).
- Infestations: pediculosis capitis (head lice), scabies (Sarcoptes scabiei).
- Hypersensitivity / contact: allergic contact dermatitis (urushiol/poison ivy), irritant contact, dyshidrotic eczema, pityriasis rosea.

- Immune-mediated bullous diseases: pemphigus vulgaris (autoantibodies to desmogleins → acantholysis, flaccid intraepidermal blisters, Nikolsky +), bullous pemphigoid (antibodies to hemidesmosomes → subepidermal tense bullae, Nikolsky −).
- Severe drug reactions / mucocutaneous syndromes: erythema multiforme, Stevens–Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), DRESS (drug reaction with eosinophilia and systemic symptoms).
- Mechanisms to remember
- Type IV (delayed) hypersensitivity mediates contact dermatitis and poison ivy (urushiol).
- Immune complex/autoantibody mechanisms produce bullous disorders (direct immunofluorescence diagnostic).
- SJS/TEN = widespread keratinocyte death → epidermal detachment; mucosal involvement common and life-threatening.
- Sources: NCBI Bookshelf reviews cited in stems (eg, pediculosis, scabies, pemphigus, sporotrichosis, erysipelas, herpes zoster).

> Arabic summary (مختصر بالعربية)
> - الأمراض الجلدية الطارئة تُصنَّف إلى معدية، طفيلية، تحسسية، مناعية ومضاعفات دوائية شديدة. بعض الحالات (SJS/TEN, pemphigus) تهدد الحياة وتتطلب رعاية مُكثفة.

Recognition and differential (key presentations + discriminators)
- Targetoid lesions → erythema multiforme (EM). EM minor = target lesions ± limited; EM major / SJS = mucosal involvement, systemic features.
- Flaccid bullae + oral erosions + positive Nikolsky → pemphigus vulgaris (autoimmune). Admit; high-dose steroids.
- Tense bullae, elderly, pruritic → bullous pemphigoid (subepidermal).
- Rapidly spreading, warm, tender erythema with sharply demarcated raised border on face/leg → erysipelas (Group A Strep). Diffuse poorly demarcated cellulitis more often S. aureus/ polymicrobial.
- Vesiculopustular lesions on tympanic membrane + target lesions elsewhere (or preceding URI) → erythema multiforme; triggers commonly HSV, Mycoplasma.
- Linear, streaky vesicles after outdoor exposure, intensely pruritic/burning → plant contact dermatitis (poison ivy/oak; urushiol).
- Small vesicles in web spaces, severe nocturnal pruritus → scabies (treat patient + close contacts).
- Asymmetric vesicular eruption on face with lesions on nose (Hutchinson’s sign) →

herpes zoster ophthalmicus → high risk ocular involvement.
- Greasy yellow scaling on nasolabial folds, forehead → seborrheic dermatitis.
- Herald patch followed by Christmas-tree distribution of scaly ovoid lesions → pityriasis rosea (self-limited; consider RPR if sexual risk).
- Single enlarging keratotic/ulcerated scalp lesion → consider squamous cell carcinoma (biopsy definitive).
- Papulonodular lesions tracking along lymphatics in gardeners → lymphocutaneous sporotrichosis (Sporothrix schenckii).
- Painless, recurrent round reddish-br

Cutaneous Disorders — condensed emergency revision notes (HIPPO-style)

Core concepts and pathophysiology
- Basic lesion types: macule, papule, pustule, vesicle (<5–10 mm), bulla (>5–10 mm), erosion (epidermis loss), ulcer (dermis/in deeper loss), crust, scale.
- Blister mechanisms:
- Intraepidermal/acantholytic: keratinocyte desmosomal disruption → flaccid bullae; classically pemphigus vulgaris (autoantibodies to desmoglein-3 ±1). Histology: suprabasal acantholysis.
- Subepidermal: separation at basement membrane → tense bullae; bullous pemphigoid (linear IgG/C3 at basement membrane on IF).
- Toxin-mediated superficial cleavage: Staphylococcal Scalded Skin Syndrome (SSSS) — exfoliative toxin targets desmoglein-1 → superficial epidermal splitting; mucous membranes usually spared.
- Full-thickness epidermal necrosis: Stevens–Johnson syndrome (SJS) / Toxic epidermal necrolysis (TEN) — drug-mediated cytotoxicity, mucosal involvement common; SJS/TEN differ by %BSA.
- Hypersensitivity types:
- Type I: urticaria, angioedema (IgE mediated)
- Type IV: contact dermatitis (e.g., poison ivy — T-cell mediated macrophage activation)
- Infectious processes: primary bacterial (impetigo, erysipelas, cellulitis), viral (herpes → herpetic whitlow), fungal (tinea capitis/kerion), ectoparasites (pediculosis), parasitic (cutaneous larva migrans).
- Drug reactions: morbilliform exanthem, DRESS (drug reaction with eosinophilia and systemic symptoms), SJS/TEN (severe mucocutaneous necrosis).

Arabic summary — المفاهيم الأساسية:
الجلد يتأثر بأنماط آلية مختلفة: طفرات بين الخلايا (acantholysis) تسبب بثور رخوة (pemphigus)، فصل تحت البشرة يسبب بثور مشدودة (pemphigoid)، سموم المكورات تسبّب تقشير سطحي (SSSS)، واعتلالات دوائية قد تسبب نخر بطانة بشرة كامل (SJS/TEN).

Recognition and differential (practical discriminators)
- Nikolsky sign (epidermal shearing with lateral pressure): positive in pemphigus vulgaris, SSSS, SJS/TEN; negative in bullous pemphigoid.
- Mucous membrane involvement:
- Present → think pemphigus vulgaris, SJS/TEN; usually absent in SSSS.
- Bullae character:
- Flaccid, easily ruptured, superficial → pemphigus vulgaris.
- Tense, intact, deeper → bullous pemphigoid.
- Age and context:
- Neonate/young child with diffuse sloughing after remote staph infection → SSSS.
- New antiepileptic or allopurinol exposed adult with fever, diffuse rash, mucosal erosions → SJS/TEN or DRESS (DRESS typically 2–8 weeks after drug start, systemic organ involvement, lymphadenopathy, eosinophilia).
- Focal infections:
- Erysipelas: raised, well-demarcated, often facial, typical Group A strep.
- Cellulitis: ill-defined, deeper; strep or Staph aureus.
- Impetigo: honey-coloured crusts or bullous lesions — Staph aureus (and strep).
- Herpetic whitlow vs bacterial felon/paronychia:
- Herpetic: grouped vesicles, tender, sterile fluid; DO NOT I&D. Consider oral acyclovir.
- Pediculosis:
- Nits attached to proximal hair shafts — treat with permethrin 1%.
- Tinea capitis / kerion:
- Scalp boggy inflammatory mass (kerion) — treat systemically (oral antifungal).
- Cutaneous larva migrans:
- Intensely pruritic serpiginous tracks on lower extremities afte

7. Arabic rapid-review summary

تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Cutaneous Disorders.

CHAPTER 06: HEAD, EAR, EYE, NOSE, THROAT DISORDERS

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1. Clinical overview and pathophysiology

Head, Ear, Eye, Nose, Throat — high-yield synthesis (HIPPO style)

Core concepts and pathophysiology
- Upper airway/infective spaces:
- Peritonsillar abscess (quinsy): pus between tonsillar capsule and pharyngeal muscles. Common organisms: Streptococcus spp., Staphylococcus. Presents unilateral bulge, trismus, uvular deviation.
- Retropharyngeal abscess: deep neck space infection anterior to prevertebral fascia; can rapidly compromise airway and spread to mediastinum. Common: Group A Strep, Staph aureus.
- Epiglottitis: inflammation/edema of epiglottis/supraglottic tissues (HIB historically). Rapid airway obstruction risk.
- Ear & skull base:
- Malignant (necrotizing) otitis externa: Pseudomonas in diabetics/immune suppression → skull base osteomyelitis.
- Auricular (pinna) hematoma: subperichondrial bleeding → cartilage ischemia → cauliflower ear if not evacuated.
- Ramsay Hunt (herpes zoster oticus): VZV reactivation in geniculate ganglion → CN VII palsy + vesicles in ear.
- Salivary/parotid disease:
- Parotitis (viral—mumps—or bacterial) → painful parotid swelling; serum amylase often ↑.
- Sialadenitis (submandibular/Wharton’s duct): purulent duct discharge; common Staph aureus.
- Eye:
- Conjunctivitis: viral (adenovirus most common), bacterial, allergic.
- Orbital vs preseptal cellulitis: orbital = postseptal involvement with pain on EOM, proptosis, ↓ vision; requires CT ± IV antibiotics ± surgery.
- Hyphema: blood in anterior chamber graded 0–4 by % fill.

- Central retinal artery occlusion (CRAO): acute painless monocular vision loss; pale retina with cherry-red spot; usually embolic (cardiac or carotid).
- Vertigo syndromes:
- Peripheral (vestibular neuritis, BPPV, Meniere): positive head impulse, horizontal/torsional nystagmus, severe spinning, hearing ± tinnitus.
- Central (cerebellar/brainstem ischemia, vertebrobasilar insufficiency): vertical nystagmus, non-fatigable, additional brainstem signs, normal head impulse.
References: NCBI Bookshelf (Rosen's EM chapters cited across items) and specific reviews (see NCBI links).

> Arabic concise: المفاهيم الأساسية — الخَراجات العنقية/التهاب الأذن القاعدي/التهاب الغدد اللعابية/أمراض العين والدوار مع أسباب بكتيرية/فيروسية ومضاعفات تهدد مجرى الهواء والرؤية.

Recognition and differential — key discriminators & red flags
- Sore throat with trismus, muffled "hot potato" voice, unilateral tonsillar bulge/uvula deviation → peritonsillar abscess. Red flags: airway compromise, drooling, severe trismus.
- Fever, drooling, tripod or sniffing posture, muffled voice, rapid progression → epiglottitis. Red flags: stridor, respiratory distress → prepare airway.
- Neck stiffness, increased prevertebral soft tissue on lateral neck film, prolonged fever after sinusitis → retropharyngeal abscess. Red flags: mediastinitis risk, airway compromise.
- Periorbital swelling:
- No pain on EOM, no proptosis, normal vision → preseptal (can be outpatient with oral antibiotics if non-toxic).
- Pain on EOM, proptosis, ↓ vision, systemic toxicity → orbital cellulitis (urgent CT, IV antibiotics, ophtho).
- Unilateral foul nasal discharge in a child (especially chronic/foul, intermittent epistax

Head / Ear / Eye / Nose / Throat — High-Yield Emergency Revision Notes
(Integrated, de-duplicated from multiple QBank items. Key references cited.)

1) Core concepts and pathophysiology
- Otitis externa (OE, "swimmer's ear"): infection/inflammation of external auditory canal — commonly Pseudomonas aeruginosa and Staphylococcus aureus; risk factors = swimming, trauma, hearing aids, occlusion. Malignant (necrotizing) OE = deep extension into skull base (diabetics, immunocompromised).

Reference: Hajioff & MacKeith; NCBI review on OE. (PMC4466798; NBK556055)
- Otitis media / bullous myringitis: middle ear infection; common pathogens: Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis. Bullous myringitis often same pathogens as AOM.
- Mastoiditis: extension of middle ear infection into mastoid air cells; can progress to subperiosteal abscess, coalescent mastoiditis, intracranial complications.
- Epiglottitis: rapid supraglottic inflammation → airway obstruction (H. influenzae type b historically; now rarer with vaccination). Agitation risks precipitating obstruction.
Reference: NCBI Epiglottitis (NBK430960)
- Peritonsillar abscess (quinsy): pus between tonsillar capsule and superior pharyngeal constrictor → trismus, muffled "hot potato" voice, uvular deviation.
- Retropharyngeal abscess: deep neck space infection → potential airway compromise and mediastinal spread; often needs CT and surgical drainage.
- Preseptal (periorbital) vs orbital cellulitis: preseptal = eyelid only; orbital = orbital tissues → pain with eye movement, ophthalmoplegia, proptosis, decreased vision; orbital cellulitis risks abscess, cavernous sinus thrombosis.
Reference: NBK507901
- Conjunctivitis: viral = watery, preauricular adenopathy; bacterial = purulent discharge; contact-lens associated = high risk of Pseudomonas keratitis.
- Corneal abrasion / foreign body: corneal epithelial defect, photophobia. Corneal ulcer (infectious keratitis) = epithelial defect + stromal infiltrate — sight-threatening in contact lens users.
Reference: NBK532960; NBK539689
- Retinal detachment: rhegmatogenous most common — flashes, floaters, curtain/sectoral visual loss; ocular ultrasound may show mobile, hyperechoic membrane.
- Acute angle-closure glaucoma: sudden IOP rise → severe pain, mid-dilated fixed or minimally reactive pupil, blurred vision, nausea/vomiting. Precipitated by anticholinergic/antihistamine drugs (e.g., diphenhydramine), SSRIs, TCAs, etc.
Reference: NBK430857
- Epistaxis: anterior (Kiesselbach's plexus ~90%) vs posterior (Woodruff's/sphenopalatine artery) — posterior bleeds more severe and harder to control.

Arabic summary (concise):
- التهابات الأذن الخارجية، متوسطة، وانتشارها (الماستويديتس)، التهاب الحنجرة العلوي

(الإبقلوتايتس)، وخمج العين/القرنية قد يهددان المجرى البولي البصري أو الهواء؛ يجب تقييم عوامل الخطر (سباحة، عدسات لاصقة، مناعية ضعيفة) بعناية.

2) Recognition and differential (key discriminators and red flags)
Red flags — immediate airway/vision/brain threats:
- Airway: tripod position, drooling, stridor, muffled voice (epiglottitis, retropharyngeal abscess, Ludwig's angina) — avoid provocative exam maneuvers (tongue blade).
- Ey

EXAM TRICK / تريكة امتحانية
Head, Ear, Eye, Nose & Throat — condensed emergency revision notes (High-yield synthesis of ENT/ophthalmology/aural/vestibular problems; integrated red flags, discriminators, algorithms, key treatments, disposition, pitfalls and exam tips)

Core concepts and pathophysiology
Anatomy/compartments matter: middle ear → otitis media → (rarely) mastoid - air cells → mastoiditis; external canal → otitis externa; orbit vs preseptal vs ;orbital cellulitis; submandibular/sublingual spaces → Ludwig's angina paranasal sinuses → sinusitis; retina/vitreous/optic nerve → different vision-loss .mechanisms
:Infectious patterns -
Acute otitis media (children): most common pathogen Streptococcus - .pneumoniae; H. influenzae and Moraxella catarrhalis common alternatives
Otitis externa: often Pseudomonas and Staph aureus; in contact-lens wearers - .think Pseudomonas for corneal ulcers
Mastoiditis: usually complication of AOM — organisms similar to AOM; can - lead to post-auricular erythema/induration, loss of postauricular crease, pinna .displacement
Neutropenic/immunocompromised sinusitis: Pseudomonas and fungi - .more likely (mucor/aspergillus)
/Rhinocerebral mucormycosis: angioinvasive fungal infection in diabetics - /DA-ketoacidosis/steroid use → tissue necrosis, black eschars, rapid orbital .brain invasion
:Eye pathology physiology -

- Corneal ulcer = epithelial defect with stromal involvement → pain, photophobia; contact-lens users at risk.
- Retinal detachment = separation of neurosensory retina from RPE; “floaters, flashes, curtain or peripheral field loss”; macula-on (urgent repair) vs macula-off (vision often poorer; urgent but not immediate in same way).
- Central retinal artery occlusion (CRAO) = ischemic retinal pallor with cherry-red spot → sudden, painless, monocular vision loss.
- Vitreous hemorrhage = “spiderwebs”/floaters, often diabetic retinopathy complication.
- Optic neuritis = painful eye movement, afferent pupillary defect, often associated with MS.
- CMV retinitis (AIDS) = fluffy white, perivascular retinal lesions +/- hemorrhage → progressive painless vision loss.
- Vestibular physiology:
- Peripheral vestibulopathy (vestibular neuritis, BPPV, Ménière): positional patterns, unidirectional horizontal nystagmus, positive head impulse in peripheral causes.
- Central causes (CNS stroke, acoustic neuroma): persistent symptoms, direction-changing nystagmus, neurologic signs.
Arabic summary: أساسيات مرضى الرأس/أذن/عين/أنف/حنجرة تعتمد على تشريح المناطق، المؤشرات الجرثومية المميزة، والتمييز بين أسباب محيطية ومركزية للأعراض.

Recognition and differential (red flags and discriminators)
- Ear pain:
- AOM: fever, otalgia, bulging/erythematous TM. Pathogen: S. pneumoniae most common.
- Otitis externa: canal edema, discharge, pain with pinna manipulation; if TM perforated → avoid aminoglycoside drops.
- Mastoiditis: post-auricular swelling/erythema, loss of postauricular crease, pinna displacement → post-AOM progression; red flag for surgical/ENT evaluation.
- Eye pain / red eye / vision loss:
- Painful vision loss: corneal ulcer, acute angle-closure glaucoma (severe pain, nausea, halos), optic neuritis (pain on eye move

Head, Ear, Eye, Nose & Throat — condensed high-yield revision notes (HIPPO style)

Core concepts and pathophysiology
- Conjunctivitis

- Viral: adenovirus most common; watery discharge, itching, preauricular LAD; often unilateral then bilateral; highly contagious (daycare, pools). (NCBI NBK470271)
- Bacterial: mucopurulent discharge, crusting; Staph, Strep, H. influenzae.
- Allergic: bilateral itching, rhinorrhea, sneezing.
- Lid lesions
- Hordeolum (stye): acute painful, erythematous abscess of eyelid margin (infectious, often Staph aureus).
- Chalazion: chronic, painless firm nodule of meibomian gland (granulomatous), conservative Rx.
- Cornea and globe
- Corneal abrasion: epithelial defect that stains with fluorescein; contact-lens wearers → high risk Pseudomonas.
- Keratitis/corneal ulcer: stromal infiltrate/opacity; vision threatened → urgent ophthalmology.
- Herpetic keratitis: dendritic ulcer on fluorescein.
- Endophthalmitis: intraocular infection (hypopyon), often post-trauma or post-op; vision threatening.
- Hyphema: blood in anterior chamber after trauma; risk for rebleed and raised IOP.
- Acute angle-closure glaucoma: pupillary block → sudden IOP rise; severe ocular pain, halos, steamy cornea, mid-dilated non-reactive pupil, nausea/vomiting. (NCBI NBK430857)
- Retinal detachment: flashes, floaters, “curtain”/peripheral shadow → surgical emergency.
- Ear
- Acute otitis media (AOM): tympanic membrane bulging, erythema; in AOM + purulent conjunctivitis suspect H. influenzae (use amox-clav). (NBK470332)
- Otitis externa: painful canal, tragal tenderness; malignant otitis externa = skull-base osteomyelitis (Pseudomonas) in diabetics/immunocompromised → IV antipseudomonal therapy. (NBK560735)
- Tympanic membrane (TM) perforation: trauma or infection; usually conservative care (dry ear), topical antibiotics if contaminated.
- Barotrauma/perforation in diving: treat with topical antibiotics if contaminated.
- Nose & sinuses
- Acute rhinosinusitis: usually viral (<10 days). Supportive Rx (saline, intranasal

steroids); antibiotics only if prolonged (>10 days), worsening or severe. (NBK470383)
- Epistaxis: anterior (Kiesselbach) common; posterior (sphenopalatine) severe, often in anticoagulated/elderly → posterior packing, ENT, admit.
- Salivary glands
- Sialadenitis (often bacterial, Staph aureus): painful swelling of gland, worse with meals, possible purulent duct drainage → antibiotics, hydration, sialogogues, massage. (NBK562211)
- Vestibular
- BPPV: brief positional vertigo (<1 minute), triggered by head turns; Dix-Hallpike positive (rotatory nystagmus) → Epley maneuver.
- Meniere's: episodic vertigo ± tinnitus ± aural fullness, progressive low-frequency sensorineural hearing loss; treat salt restriction, diuretics (e.g., HCTZ). (PMC4348125)

> Arabic (موجز): المفاهيم الأساسية: التهاب الملتحمة غالباً فيروسي؛ خدش القرنية عند مرتدي العدسات يعرض لـPseudomonas؛ الزائدة الحِسِية (chalazion) غير مؤلمة؛ الزاوية المغلقة حالة طارئة داخل العين؛ نزيف أنفي خلفي غالباً من الشريان الشفوي الأنفي (sphenopalatine).

Recognition and differential (key discriminators, red flags)
- Painful red eye — immediate triage i

Core concepts and pathophysiology
- Hordeolum (stye): acute pyogenic infection of eyelid glands/eyelash follicles (usually Staphylococcus). External = Zeis/Moll follicle; internal = meibomian gland. Typically painful, erythematous pustule that may spontaneously drain with symptom relief. (Ref: NCBI Bookshelf — NBK441985)
- Acute bacterial rhinosinusitis: usually follows viral URI → mucosal edema → impaired drainage → secondary bacterial infection (common pediatric pathogens: S. pneumoniae, nontypeable H. influenzae, Moraxella). Clinical diagnosis; imaging not routinely required.
- Mastoiditis: spread of middle-ear infection into mastoid air cells → inflammation, possible coalescent mastoiditis with bone erosion, abscess, intracranial spread. Signs include post-auricular erythema/swelling, auricle protrusion, systemic toxicity.
- Cholesteatoma: keratinizing squamous epithelial collection in middle ear/ mastoid due to eustachian dysfunction and TM retraction pocket; can erode bone,

cause chronic infection, hearing loss, and intracranial complications. (Ref: NBK448108)
- Herpes zoster ophthalmicus (HZO): VZV reactivation in trigeminal (V1) distribution. Nasociliary branch (Hutchinson sign — tip of nose lesions) predicts ocular involvement (cornea, conjunctiva).
- Penetrating ocular trauma / intraocular foreign body → open-globe risk and posttraumatic endophthalmitis (notably Bacillus cereus with metallic material); management differs from blunt/no-penetration eye injuries. (Ref: NBK545184)

الشرح بالعربية (مختصر):
- دمل الجفن عدوى بكتيرية حادة لمسام الجفن.
- التهاب الجيوب الأنفية البكتيري يحدث غالباً بعد إنفلونزا وأنف فيروسي.
- التهاب الخشاء امتداد لالتهاب الأذن الوسطى ويحتاج تصوير ومضاد حيوي وجراحة عند التعقيد.
- الكوليستياتوما تجمع خلايا جلدية خلف غشاء الطبلة يسبب تآكل العظم وفقد سمع.
- هربس زوستر العيني يدمر عصب ثلاثي التوائم الفرع العيني، وجود آفة على طرف الأنف ينبه لمشاركة القرنية.
- إصابة العين النافذة/جسم غريب معدني خطر عدوى عنيفة ويتطلب تدخل عيني عاجل.

Recognition and differential
- Eye—eyelid lump:
- Painful, erythematous pustule at lid margin → Hordeolum (vs chalazion: painless, chronic, noninfectious lipogranuloma).
- Diffuse eyelid erythema/edema with systemic signs → preseptal vs orbital cellulitis (orbital = painful eye movements, proptosis, impaired EOM, decreased vision).
- Red eye + vesicular rash in V1 distribution → HZO; check for Hutchinson sign (nasociliary involvement).
- Acute ear pain + postauricular swelling/auricular protrusion → mastoiditis (vs uncomplicated otitis media which lacks postauricular findings).
- Chronic ear drainage, retracted TM, white debris, conductive hearing loss → cholesteatoma.
- Sinusitis:
- Viral URTI: symptoms <10 days and improving.
- Bacterial sinusitis suspicion: persistent symptoms >10 days without improvement, severe onset (high fever with purulence ≥3–4 days), or worsening after initial improvement (“double-sickening”).

الشرح بالعربية (مختصر):
- ألم جفن وقمة صديدية → دمل (هوردوليوم).
- الهربس على توزيع V1 مع آفة على طرف الأنف → احتمال إصابة العين كبير.
- ألم أذن مع انتفاخ خلف الأذن وبروز الأذن → التهاب الخشاء.
- إفرازات مزمنة من الأذن وتراجع طبلة الأذن → كوليستياتو

7. Arabic rapid-review summary

تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Head, Ear, Eye, Nose, Throat Disorders.

CHAPTER 07: TOXICOLOGIC DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Toxicokinetics basics: absorption (rapid for alcohols), distribution (lipid vs water soluble), metabolism to active/toxic metabolites (eg, ethylene glycol → glycolate/oxalate; methanol → formate), elimination (renal, hepatic, dialyzable). Antidotes either (a) bind toxin (idarucizumab → dabigatran), (b) block formation of toxic metabolites (fomepizole → alcohol dehydrogenase inhibitor), (c) replace deficient/coagulation factors (prothrombin concentrate), or (d) reverse receptor effects (naloxone → opioids).
- Dialyzability: many small, low-protein drugs/poisons are dialyzable. Commonly dialyzable agents (important ED list): salicylates, toxic alcohols (methanol, ethylene glycol), lithium, metformin, isoniazid (INH), theophylline, some beta-blockers (atenolol, metoprolol), topiramate, acyclovir, phenobarbital, dabigatran (partial). Hemodialysis removes parent compound and some metabolites; used when toxic metabolites or severe clinical deterioration occur.
- Acid–base patterns help identify toxins:
- Mixed respiratory alkalosis + metabolic acidosis → salicylates.
- High osmolar gap + anion gap metabolic acidosis → ethylene glycol or methanol (depending on clinical clues).
- Elevated osmolar gap without anion gap or acidosis → isopropyl alcohol (ketonemia/ketonuria present).
- Toxidromes: cluster signs to narrow differential (opioid, anticholinergic, cholinergic, sympathomimetic, serotonin syndrome, NMS).

Key references: Pollack et al., NEJM 2015 (idarucizumab); 2020 ACC guidance on OAC bleeding; Tintinalli (Rosen’s) chapters; NCBI toxicology reviews.

> خلاصة مبسطة (عربي): المفاهيم الأساسية تركز على كيفية دخول السمّ للجسم، تحوله إلى نواتج سامة، وإزالته (بما في ذلك دور الغسيل الكلوي)؛ الأنماط الحمضية والسمّية والتوكسيدرومات تساعد على التمييز السريع وتوجّه المضادّات.

Recognition and differential
- Rapid diagnostic “red flags” in ED:
- Altered mental status + respiratory depression → consider opioids (naloxone), diphenoxylate/atropine in children (also naloxone).
- Mixed respiratory alkalosis + metabolic acidosis ± tinnitus, fever → salicylates.
- Osmolar gap ↑ with normal anion gap and ketonemia/ketonuria → isopropanol.
- Osmolar gap ↑ with anion gap metabolic acidosis + calcium oxalate crystals in urine + renal failure → ethylene glycol.
- Visual disturbances and metabolic acidosis → methanol.
- Wide QRS, terminal R in aVR → tricyclic antidepressant (TCA) with sodium-channel blockade.
- Hyperreflexia, clonus, myoclonus, diaphoresis, hyperthermia → serotonin syndrome (SSRIs, MAOIs, combinations).
- “Lead-pipe” rigidity, bradyreflexia, slower onset over days → NMS (antipsychotics).
- Hypoglycemia delayed after ingestion, especially in children → sulfonylurea (glyburide) — beware recurrent/late hypoglycemia.
- Nausea/vomiting 6–24 hours after mushroom ingestion → amatoxin (risk of fulminant hepatic failure).
- Body-packers: asymptomatic but radiopaque packets on x-ray → whole-bowel irrigation; risk of sudden toxicity if packet ruptures.

> التمييز السريع (عربي): احزر التوكسيدروم من العلامات المميزة (مثل الشخير للمواد الأفيونية، الارتعاش والسجلات للمتلازمة السيروتونينية، وغيرها) وفكّر في

Core concepts and pathophysiology
- Toxidrome-first approach: identify pattern (anticholinergic, cholinergic, sympathomimetic, opioid, serotonergic, sedative-hypnotic, dopaminergic, etc.) to guide immediate therapy while diagnostics are pending.
- Mechanisms emphasized in these scenarios:
- Enzyme inhibition: organophosphates/carbamates inhibit acetylcholinesterase → excess ACh at muscarinic + nicotinic synapses (SLUDGE, bronchorrhea, seizures). Pralidoxime can reactivate AChE if given before “aging.”
- Ion-channel / receptor blockade: TCAs block fast Na+ channels → wide QRS, arrhythmia; local anesthetics cause Na+ channel blockade and can progress to cardiac collapse (LAST).
- Mitochondrial inhibition: cyanide blocks cytochrome oxidase → anaerobic metabolism, profound lactic acidosis, cherry-red skin.

- Oxidation of hemoglobin: oxidized Fe3+ (metHb) cannot carry O2 → pulse ox ~85% that does not correct with supplemental O2; co-oximetry required.
- Enzyme cofactor depletion: isoniazid → functional pyridoxine (B6) deficiency → refractory seizures.
- Receptor antagonism vs overstimulation: cocaine/meth/amphetamines cause sympathetic hyperactivity; SSRIs/MAOI interactions → serotonin syndrome (hyperreflexia, clonus) vs neuroleptic malignant syndrome (lead-pipe rigidity, hyporeflexia).
- Disruption of glucose/insulin: non-dihydropyridine and dihydropyridine CCBs impair insulin release → hyperglycemia in overdose (useful discriminator vs β-blocker which can cause hypoglycemia in children).
- Heavy metals/toxic ingestion: lead → GI, neurologic effects and basophilic stippling; chelation thresholds differ by age.
- General priorities: protect airway, support breathing & circulation, prevent further exposure (decontamination), give targeted antidote when indicated.

> العربية — Arabic (core concepts):
> ابدأ بتحديد نمط "السمّيّة" لتوجّه العلاج الفوري. ترافق التسمّمات آليات مختلفة: تثبيط الإنزيمات (مبيدات الفوسفات العضوية)، حجب قنوات الأيونات (TCA، موضعيّات التخدير)، تثبيط التنفّس الخلوّي (السيانيد)، أكسدة الهيموغلوبين (الميتهيموغلوبين)، ونقص عوامل الإنزيم (إيزونيازيد → نقص فيتامين B6). الأولوية: ABC، إزالة المصدر، وإعطاء الترياق الخاص عند اللزوم.

Recognition and differential
- Key toxidrome clues / discriminators:
- Cholinergic (organophosphates): salivation, lacrimation, urination, defecation, emesis, bronchorrhea/bronchospasm, bradycardia; seizures and miosis common.
- Sympathomimetic (cocaine, amphetamines): agitation, diaphoresis, mydriasis, hyperthermia, severe hypertension, tachycardia; treat with benzodiazepines (avoid β-blockers — risk unopposed α).
- Serotonin syndrome: agitation, hyperreflexia, inducible/clonus, diaphoresis, hyperthermia developing quickly after serotonergic agent + MAOI/linezolid; treat with benzos, supportive; cyproheptadine if refractory.
- TCA: anticholinergic features + wide QRS (Na+ block), hypotension, seizures — QRS >100 ms or instability → sodium bicarbonate.
- Methemoglobinemia: cyanosis, chocolate-brown blood, SpO2 ≈85% unresponsive to O2, recent exposure to oxidizing agents (benzocaine, nitrates) →

methylene blue.
- Cyanide: exposure from smoke/fire + severe lactic acidosis (very hig

Core concepts and pathophysiology
- General emergency approach to poisoning: ABCs (airway protection ± early intubation), decontamination (remove clothes, wash skin), toxidrome recognition, targeted antidote/adjunctive therapy, enhanced elimination or extracorporeal removal when indicated, disposition and specialty consult (toxicology, transplant, dialysis, psychiatry).
- Toxidromes and key mechanisms covered:
- Sympathomimetic (cocaine, amphetamines): catecholamine excess → agitation, hypertension, tachycardia, dilated pupils. Initial therapy: benzodiazepines to reduce sympathetic outflow. Avoid pure β-blockers (risk of unopposed α effect). (Ref: NBK430757)
- Cholinergic excess / organophosphates: AChE inhibition → DUMBELS (diarrhea, urination, miosis, bronchorrhea/bronchospasm, emesis, lacrimation, salivation), bradycardia, fasciculations, seizures. Antidotes: atropine + pralidoxime + benzodiazepines. (Ref: NBK470430)
- Beta-blocker toxicity: bradycardia, hypotension, bronchospasm, hypoglycaemia. Mechanism: decreased cAMP from β-receptor antagonism. Antidotes: glucagon (first-line), then high-dose insulin euglycemia therapy (HIET), calcium, vasopressors; atropine for bradycardia. (Ref: Rosen’s, NBK448097)
- Calcium-channel blocker (CCB) toxicity: hypotension and bradycardia; treat with IV calcium, HIET, vasopressors.
- Tricyclic antidepressant (TCA) cardiotoxicity: sodium channel blockade → QRS widening, ventricular arrhythmias, hypotension, anticholinergic features. First-line: sodium bicarbonate for QRS widening/acidosis; if refractory, Class IB antiarrhythmics (lidocaine) may be used. Avoid other Class I agents (IA, IC). (Refs: NBK430931; Clin Toxicol 2010)
- Acetaminophen: metabolism produces NAPQI; toxicity when glutathione depleted → hepatocellular necrosis. Antidote: N-acetylcysteine (replenishes glutathione). Single acute vs chronic/therapeutic-error management differs (Rumack–Matthew nomogram applies only to single acute ingestions with levels ≥4 h post-ingestion). (Ref: NBK537183; NBK441917)
- Amanita (amatoxin) mushroom poisoning: amatoxin inhibits RNA polymerase II → halts protein synthesis → delayed (6–24+ h) GI symptoms followed by fulminant hepatotoxicity. Consider early charcoal (reduces absorption, enterohepatic

circulation) and anticipate transplant consultation. (Ref: NBK431052)
- Salicylate (aspirin): mixed early respiratory alkalosis (stimulates respiration) and metabolic acidosis later; severe toxicity → CNS effects, seizures, pulmonary edema. Treatments: sodium bicarbonate (urinary alkalinization), activated charcoal early (including multi-dose for extended-release or pharmacobezoar), and hemodialysis for severe cases. (Ref: Tintinalli; Levitan)
- Ethylene glycol (antifreeze): early elevated osmolar gap → later anion gap metabolic acidosis as toxic metabolites accumulate; calcium oxalate crystals and renal injury. Treat with fomepizole (or ethanol) ± hemodialysis. (Ref: Tintinalli)
- Digoxin: cardiac glycoside → Na-K ATPase inhibition, arrhythmias classically include bidirectional ventricular tachycardia and high-degree AV block. Specific therapy: digoxin-immune Fab. (Ref: Rosen's)
- Warfarin: in

Core concepts and pathophysiology
- Toxidromes are patterns of signs/symptoms produced by classes of poisons (anticholinergic, cholinergic, sympathomimetic, opioid, sedative–hypnotic, serotonergic, neuroleptic, etc.). Recognize pattern first, confirm agent second.
- Key mechanisms to recall:
- Enzyme blockade (cyanide, hydrogen sulfide → cytochrome oxidase inhibition → cellular hypoxia).
- Receptor antagonism/agonism (opioids → μ-receptor; anticholinergics → muscarinic blockade; neuroleptics → dopamine D2 blockade).
- Ion channel blockade (TCA, local anesthetics, some antiarrhythmics → Na+ channel block → QRS widening, arrhythmia).
- Metabolic conversion to toxic metabolites (ethylene glycol → glyoxylic/oxalic acids; methanol → formic acid; salicylates → uncoupling oxidative phosphorylation → mixed acid–base disturbance).
- Protein binding and volume of distribution dictate efficacy of dialysis and lipophilic toxin sequestration (lipid emulsion).
- Principles of antidotal therapy: (1) antagonize target receptor or enzyme, (2) chelate/remove toxin, (3) support physiology (airway/circulation/temperature, treat complications), (4) prevent further absorption/decontaminate.
- Sources: Rosen's Emergency Medicine; Tintinalli's Emergency Medicine; review chapters cited in QBank explanations.

> Arabic: المبادئ الأساسية: نمط السمّية → آلية → علاج داعم/مضاد/إزالة السمّ. استعمال مصادر مثل Rosen’s وTintinalli.

Recognition and differential (clinical discriminators)
- Salicylates (aspirin)
- Mixed primary respiratory alkalosis + anion-gap metabolic acidosis (resp alkalosis early → hyperventilation + metabolic acidosis later).
- Odor (methyl salicylate = “mint” topicals), tinnitus, hyperthermia, agitation.
- Acetaminophen
- Early nonspecific GI symptoms; delayed transaminitis/hepatic failure by 24–72 h after significant ingestion.
- Use Rumack–Matthew nomogram for single acute ingestion (4–24 h). For repeated supratherapeutic ingestion, treat more liberally.
- Local anesthetic systemic toxicity (LAST; e.g., lidocaine, bupivacaine)
- Early CNS: perioral numbness, tinnitus, agitation → seizures → cardiovascular collapse, arrhythmias.
- ECG: ventricular dysrhythmias; treat with lipid emulsion + ACLS modifications (avoid vasopressin, CCBs, β-blockers).
- Tricyclic antidepressant (TCA)
- Anticholinergic and Na-channel blockade → wide QRS, hypotension, seizures. Sodium bicarbonate for cardiotoxicity.
- Anticholinergic (e.g., diphenhydramine, antihistamines, some antidepressants)
- “Mad, blind, hot, dry, red, full” (delirium, mydriasis, hyperthermia, anhidrosis, flushed, urinary retention). Consider physostigmine selectively (avoid in TCA cardiotoxicity).
- Serotonin syndrome vs Neuroleptic malignant syndrome (NMS)
- Serotonin: hyperreflexia, clonus, tremor, hyperthermia; precipitated by SSRIs + serotonergic drugs (e.g., tramadol, cocaine).
- NMS: “lead-pipe” rigidity, high fever, altered mental status, autonomic instability; often after dopamine antagonists (e.g., haloperidol). Dantrolene can be helpful in severe NMS.
- Beta-blocker and calcium-channel blocker toxicity
- Bradycardia, hypotension, shock. Glucagon (bypasses β-receptor

Core concepts and pathophysiology
- Toxidromes result from predictable physiologic effects of classes of poisons (anticholinergic, serotonergic, opioid, sympathomimetic, cholinergic, neuroleptic).

Recognize by constellation of mental status, autonomic signs, pupils, skin, neuromuscular findings.
- Caustic ingestions:
- Alkalis (e.g., bleach, lye): liquefactive necrosis → deep tissue penetration, saponification of fats, progressive injury, risk of delayed perforation and stricture formation.
- Acids: coagulative necrosis → surface coagulum that may limit deeper penetration but still can cause severe injury.
(Ref: NCBI Bookshelf: caustic injury; Rosen's Emergency Medicine)
- Toxic alcohols:
- Methanol → metabolized to formic acid → optic nerve toxicity (visual symptoms), high osmolar gap, high anion-gap metabolic acidosis.
- Ethylene glycol → glycolate/oxalate → renal injury, calcium oxalate crystals.
- Both: treat with fomepizole to block alcohol dehydrogenase; folinic/folate for methanol; pyridoxine/thiamine for ethylene glycol; consider dialysis for severe cases.
(Rosen's; methanol treatment principles)
- Cyanide (common from smoke in fires): inhibits cytochrome oxidase → stops oxidative phosphorylation → profound lactic acidosis and rapid cellular hypoxia.
- Marine toxins:
- Ciguatera (reef fish: barracuda, grouper, snapper, moray eel): GI symptoms early; neurologic features include hot–cold temperature reversal, paresthesias, "loose teeth," bradycardia; onset within hours; supportive care.
- Scombroid: preformed histamine in spoiled fish → flushing, urticaria, bronchospasm; mimics allergic reaction; responds to antihistamines.
(Ref: NCBI NBK482511)
- Drugs requiring specific decontamination/antidotes:
- Theophylline: stimulant effects (tachycardia, tremor) — multi-dose activated charcoal reduces enterohepatic recirculation; dialysis for life-threatening toxicity (arrhythmia, seizure) and very high levels.
(Ref: NCBI NBK532962)
- Important pathologic contrasts:
- Serotonin syndrome: hyperreflexia, clonus, myoclonus; rapid onset after serotonergic agent(s).
- Neuroleptic malignant syndrome (NMS): "lead-pipe" rigidity, hyporeflexia, slower onset over days after antipsychotics, high CK.

- Anticholinergic toxidrome: delirium/hallucinations, dry flushed skin, mydriasis, urinary retention, hyperthermia.

> Arabic summary (concise):
> - السموم تُنتج مظاهر نمطية (توكسدورم). القلويات تسبب نخر تسييلي، الأحماض نخر تخثري. الميثانول → حمض فورميك → ضرر بصري؛ الإيثيلين جلايكول → حصى/فشل كلوي. السيانيد يوقف الفسفرة التأكسدية → حُماض لبني شديد. سيجوتوكسين → انقلاب الإحساس الساخن/البارد. التيُوفيلين يستجيب للفحم المنشط المتكرر.

Recognition and differential (key discriminators and red flags)
- Hot–cold temperature reversal + early GI symptoms after reef fish → ciguatera.
- Bitter almond odor (unreliable) or history of enclosed-space fire + high lactate/metabolic acidosis → suspect cyanide.
- Vision loss or “snow-field” vision + high osmolar gap ± high anion gap → suspect methanol.
- Ataxia, optic disc hyperemia, abdominal pain with high osmolar gap/metabolic acidosis → methanol.
- Flushing, urticaria, bronchospasm minutes after fish

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Toxicologic Disorders.

CHAPTER 08: EM PROCEDURES AND SKILLS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Procedure risk vs physiologic reserve: choose techniques/drugs that minimize hemodynamic compromise in unstable/septic patients (etomidate, ketamine preferred for hemodynamic stability; avoid agents that cause vasodilation/ bradycardia in shock) (NBK535364).
- Local anesthetics: two chemical classes — esters (procaine, benzocaine, chloroprocaine, tetracaine) vs amides (lidocaine, bupivacaine, ropivacaine, mepivacaine). Ester allergy → avoid other esters; amide names usually contain two "i"s (memory aid, not absolute) (Becker & Reed 2012).
- POCUS/FAST principles: FAST detects free intraperitoneal or pericardial fluid (dependent spaces); less sensitive for retroperitoneal injuries. E-FAST adds lung windows to detect pneumothorax/pleural fluid (Roberts & Hedges; Cochrane review).
- Pleural/abdominal drainage: paracentesis and thoracentesis are diagnostic + symptom-relieving; complications include pneumothorax (most common after thoracentesis; ↓ with US guidance) and re-expansion pulmonary edema if large/ rapid removal.
- Ventilator lung-protective physiology: low tidal volume ventilation (4–8 mL/kg, commonly 6 mL/kg) and permissive hypercapnia for ARDS; plateau pressure >30 cmH2O associated with barotrauma risk.
- Electrolyte effects on ECG: hypokalemia → U waves; hyperkalemia → peaked T waves → PR prolongation, P-wave loss, QRS widening → sine wave and arrest (cardiac ECG changes are not perfectly sensitive—treat based on level/clinical context).

> Arabic summary (core):
> - اختر إجراءات وأدوية تقلل التأثير الهيموديناميكي عند المريض غير المستقر. التمييز الكيميائي للمخدرات الموضعية مهم (إسترات مقابل أَميدات). فحص الموجات فوق الصوتية بالترتيب يختلف بحسب نوع الرضة؛ حماية الرئتين أثناء التهوية الآلية ضرورية. تغيرات ECG ليست حساسة دائماً، عالج بحسب المستوى والسياق.

Recognition and differential (red flags & discriminators)
- Sepsis/unstable patient needing procedures: red flags = hypotension, tachycardia, lactic acidosis; choose hemodynamically neutral induction agents (etomidate, ketamine).
- Pneumothorax vs contusion vs pleural effusion on lung US:
- Absence of lung sliding and presence of lung point → pneumothorax.
- B-lines (vertical comet tails) → interstitial syndrome / contusion; their presence effectively excludes pneumothorax at that site.
- Curtain sign is normal at costophrenic recess.
- FAST/E-FAST window priorities:
- Penetrating torso trauma: cardiac window first (subxiphoid/parasternal) to rule out tamponade.
- Blunt trauma: RUQ (Morison's pouch) first; most sensitive dependent view = suprapubic for free fluid.
- Epistaxis:
- Anterior (Kiesselbach's plexus): visible anterior bleeding, responds to topical vasoconstrictor, direct pressure, cautery, anterior packing.
- Posterior: bleeding into posterior pharynx, bilateral/large-volume bleed, failure of anterior measures → posterior packing/balloon, ENT/IR consult.
- Genital emergencies:
- Paraphimosis: painful swollen glans with constricting foreskin band → manual reduction urgently to restore circulation; if refractory, hyaluronidase/needle decompression or dorsal slit.
- Compartment syndrome: severe pain out of proportion, pain with pass

Core concepts and pathophysiology
- Point-of-care ultrasound (POCUS) is a rapid, radiation-free adjunct for many ED procedures: differentiates peritonsillar cellulitis vs abscess; identifies pericardial vs pleural fluid on FAST; confirms pleural effusions, gallstones/cholecystitis, free intraperitoneal fluid (FAST), bladder volume before suprapubic catheter, and abdominal aortic aneurysm extent.
- FAST cardiac window (parasternal long or subxiphoid): identify pericardial effusion vs pleural effusion by relation to the descending aorta — pericardial fluid lies anterior/above the aorta; pleural fluid lies posterior/below it.
- Hemorrhagic pericardial effusion → hemopericardium → tamponade: tamponade physiology = impaired diastolic filling → decreased preload and cardiac output; echo signs = RA/RV diastolic collapse, swinging heart, plethoric IVC.

- Tension pneumothorax = one-way air entry into pleural space → mediastinal shift, decreased venous return → hypotension and hypoxia.
- Asthma failure physiology: severe bronchospasm, air trapping, dynamic hyperinflation → ventilatory failure; warning signs include “silent chest,” rising PaCO2, altered mental status, exhaustion.
- Fluid physiology in acute pericardial tamponade: patients are preload-dependent; small fluid bolus may transiently improve perfusion pre-op.
- Pediatric airway sizing and approaches: uncuffed ETT size (years/4)+4 is commonly used (note age applicability and cuffed vs uncuffed practice variations).
- Central line considerations in hypothermia: avoid high (IJ/subclavian) lines if possible because guidewire stimulation can trigger arrhythmia; femoral access preferred.

> Arabic (concise): مفاهيم أساسية: التَّصوير بالموجات فوق الصوتية في الطوارئ يسرّع التشخيص ويُدلّ على الإجراء. الانصباب التأموري يسبب انهيار غرف القلب اليمنى؛ الضغط الصدري المتوتر يحتاج تفريغ فوري. قواعد اختيار أنبوب الطفل وحُقَن السوائل مهمة.

Recognition and differential (red flags, discriminators)
- Peritonsillar abscess vs cellulitis: red flags = trismus, “hot potato” or muffled voice, unilateral tonsillar bulge and uvular deviation. POCUS (intraoral or transcutaneous) can show discrete fluctuance/collection and adjacent vessels.
- Pericardial tamponade: hypotension, muffled heart sounds, JVD (Beck’s triad), pulsus paradoxus, tachycardia; POCUS: RA/RV collapse, large pericardial effusion, plethoric IVC.
- Tension pneumothorax: acute deterioration on ventilator, unilateral decreased/absent breath sounds, tracheal deviation (late), JVD, hypotension — treat immediately.
- Massive hemothorax / penetrating chest trauma with pericardial fluid on FAST: consider pericardial tamponade from cardiac injury → OR definitive.
- Impending respiratory failure in asthma: altered mental status, exhaustion, hypoxia, rising or normalizing/↑ PaCO2 (loss of expected hypocapnia), silent chest; minimal air movement and inability to speak.
- Retrobulbar hemorrhage (orbital compartment syndrome): proptosis, decreased vision, severe pain, tense orbit; IOP >35 mmHg is an indication for emergent lateral canthotomy.
- Umbilical cord prolapse: palpated cord in vagina or visualized — immediate fetal compromise; presenting fetal part must be el

EXAM TRICK / تريكة امتحانية

Core concepts and pathophysiology

- Confirmatory testing hierarchy: for tubes and invasive devices, radiographic or direct visualization is the gold standard; bedside tricks (air insufflation auscultation, aspiration of contents, external measurements) are unreliable and exam traps. (Roberts & Hedges)
- Pressure/volume problems in airway and ventilated patients are often mechanical (equipment, tube position, breath-stacking) rather than primary cardiac — rapidly exclude ventilator-related causes during arrest. (DOPES mnemonic)
- Loss of CSF after lumbar puncture → low intracranial pressure headache that is postural; tends to improve with conservative care but may require epidural blood patch if persistent >24 hours.
- Asthmatic physiology: obstructive disease with prolonged expiratory time and risk of dynamic hyperinflation → ventilator strategies emphasize long expiratory time, low RR, low/minimal PEEP, smaller TV per kg (or 6–8 mL/kg PBW), permissive hypercapnia to avoid barotrauma.
- Focal thoracic bleeding (massive hemoptysis) requires rapid lung isolation: intubate the “good lung” (usually right mainstem when bleeding is left-sided) to protect ventilated lung.
- Acute surgical problems (testicular torsion, peritonsillar abscess, thrombosed external hemorrhoid, large Bartholin abscess) need timely bedside procedures when operative teams are delayed — technique and anatomical landmarks are critical to avoid catastrophic complications.

Arabic takeaway (concise):

- الأفضل تأكيد محاذاة الأنابيب بالأشعة. مشاكل التهوية الحادة عند المريض المفترض أنبوبه غالباً سببه الجهاز. الصداع بعد البزل عادة يُعالج تحفظياً ثم رقعة دموية إذا استمر >24 ساعة. تنفس المرضى الربويين يحتاج وقت زفير طويل و permissive hypercapnia. تسريب كبير للدم من صدر يتطلب عزل الرئة السليمة.

Recognition and differential (red flags, discriminators)

- Misplaced nasogastric/feeding tube: sudden respiratory distress after placement, coughing, oxygen desaturation — consider tracheobronchial placement. But absence of symptoms does not exclude misplacement — radiograph is definitive.
- Post-lumbar-puncture headache: postural (worse sitting/standing, better lying

flat), onset within 48 hours (can be early or delayed), no fever/meningeal signs unless infection.
- Epistaxis in anticoagulated patient: assess hemodynamics and ongoing bleeding. Therapeutic anticoagulation alone does not mandate reversal if bleeding controllable with local measures; reverse only for uncontrollable/hemodynamic bleeding.
- Intubated patient who arrests (PEA/asystole): think DOPES — displaced tube, obstruction (kink/mucus), pneumothorax, equipment failure, stacking breaths (auto-PEEP).
- Testicular torsion: sudden severe unilateral testicular pain, high-riding testis, absent cremasteric reflex → surgical emergency. Doppler ultrasound supportive but do not delay reduction/operation.
- Peritonsillar abscess: muffled “hot potato” voice, trismus, uvular deviation away from affected side — differentiate from tonsillitis or parapharyngeal abscess.
- Massive hemoptysis: >100–200 mL/24 hr (clinical context matters) or any volume causing airway compromise; progressive hemoptysis and respiratory distress are red flags — nee

Core concepts and pathophysiology
- Airway/ventilation in bleeding/respiratory failure
- Massive hemoptysis: blood floods and aspirates into non-bleeding lung → hypoxia, airway obstruction, and asphyxia. Immediate priorities = protect airway, isolate bleeding lung, maintain oxygenation, limit contamination of contralateral lung (place bleeding side down). (NBK538278)
- ARDS (acute lung injury from pancreatitis, sepsis, trauma): noncardiogenic pulmonary edema, shunt physiology → low compliance, V/Q mismatch. Lung-protective ventilation decreases ventilator-induced lung injury (low Vt, higher PEEP, permissive hypercapnia).
- Trauma ultrasound (FAST/eFAST)
- FAST looks for free fluid as a proxy for hemoperitoneum/hemothorax: RUQ (Morison’s pouch), LUQ (splenorenal), subxiphoid cardiac, suprapubic. eFAST adds bilateral anterior pleura for pneumothorax/hemothorax.
- Sensitivity depends on operator, patient hemodynamics and fluid volume; sensitivity approaches 100% in hypotensive trauma patients but is lower in stable patients. Small volumes (<100–400 mL) may be missed. (Roberts & Hedges)
- Pericardial vs pleural effusion on echo
- Relationship to descending thoracic aorta: pericardial effusion lies anterior

(above) the descending aorta in the parasternal long view; pleural effusion lies posterior (below). (NBK470584)
- Pericardial tamponade and ED thoracotomy/pericardiotomy
- Tamponade: intrapericardial pressure impairs diastolic filling → hypotension, JVD, muffled heart sounds (Beck triad; may be subtle in trauma). Pericardiotomy during resuscitative thoracotomy decompresses tamponade; ED thoracotomy can also permit open cardiac massage and aortic cross-clamp for ongoing exsanguination.
- Obstetric resuscitation
- Perimortem cesarean (resuscitative hysterotomy) aims primarily to improve maternal venous return by relieving aorto-caval compression and to potentially save viable fetus. If gestation >~24 wk (fundus well above umbilicus), begin hysterotomy by 4 minutes after maternal arrest to optimize outcomes. (ACLS/ perimortem C-section guidelines)

> Arabic summary (core concepts)
> - نزف قصبي كبير: ضع الجانب النازف لأسفل واعزل الرئة السليمة. التهوية الوقائية في ARDS: حجم تيار منخفض وPEEP عالي. فحص FAST: RUQ (موريسون)، LUQ، قلب، فوق العانة؛ eFAST يضيف الجنبين للبلورا. الانصباب التاموری يظهر أمام الشريان الأورطی النازل على الإيكو.

Recognition and differential
- Massive hemoptysis: definition clinically variable — large volume or life-threatening bleeding with airway compromise, hypoxia, or hemodynamic instability. Differential: bronchiectasis, lung cancer, TB, bronchial artery bleeding, pulmonary AVM, PE, trauma.
- Red flags: active gurgling/aspiration, oxygen desaturation, inability to protect airway, hypotension → immediate airway control and lung isolation.
- Tension pneumothorax
- Rapid deterioration, unilateral hyperresonance, absent breath sounds, JVD, hypotension/cardiac arrest.
- Hemoperitoneum in trauma
- Positive FAST in unstable patient → presumed significant intra-abdominal bleed until proven otherwise.
- Pericardial tamponade vs pleural effusion
- Echo discriminator: fluid anterior to descending aorta = pericardial; p

Core concepts and pathophysiology
- Pleural effusion: accumulation of fluid in pleural space → dyspnea from lung

compression, decreased ventilation and V/Q mismatch. On ultrasound appears anechoic (or complex), obliterates normal lung artifacts; RUQ (subxiphoid/diaphragmatic) views may show free fluid above the diaphragm and the “spine sign” (visualization of vertebral bodies above the diaphragm) indicating pleural fluid. Distinguish from pericardial effusion by location relative to cardiac structures. (Ref: NBK448189)
- Pericardial effusion/tamponade: fluid within pericardial sac causing impaired diastolic filling → hypotension, JVD, muffled heart sounds; requires pericardiocentesis if tamponade physiology present.
- Pulmonary hypertension (chronic): increased pulmonary vascular resistance → RV pressure overload and hypertrophy. ECG may show right-axis deviation, RV hypertrophy/strain patterns including tall R in V1, RBBB, P pulmonale. (Ref: NBK482463)
- Acute urinary retention and suprapubic catheter (SPC): SPC is bladder decompression when urethral catheterization impossible/contraindicated. Avoid SPC if bladder malignancy is suspected because of risk of tumor tract seeding. (Ref: NBK482179)
- Pediatric acute upper airway obstruction (eg, epiglottitis): rapidly progressive obstruction from supraglottic inflammation → tripod position, drooling, stridor, respiratory distress. Airway management differs from adults: maintain calm, keep child upright, avoid provoking examination; definitive airway often requires experienced operator and surgical backup. (Ref: Roberts & Hedges)

> Arabic (مختصر): الفَائِضُ الجنبيّ يَضغطُ على الرئة مَعَ علامة العمود الفقري فوق الحجاب الحاجز. فَشلُ قَلبٍ تَحْتَاجُ إلى نَزع سائل التامبون. القسطرة فوق العانة تَمنَعُ إذا كانَ شَكّ بورم المثانة. في الأطفال المصابين بإِپيگلوثيتِيس، لا تُزعج الطفل واحتِفظ بخِبرةٍ للتنبيب والجهة الجِراحية.

Recognition and differential
- Respiratory distress with pleural vs pericardial effusion:
- Pleural: unilateral/bilateral reduced breath sounds, dullness to percussion, ultrasound: anechoic collection above diaphragm, spine sign, lung/fluid interface posterior to descending aorta.
- Pericardial: muffled heart sounds, pulsus paradoxus, hypotension; echo: circumferential anechoic collection around heart, right atrial/ventricular diastolic collapse.
- ECG for pulmonary hypertension: look for tall R in V1, R axis deviation, RBBB, peaked P waves (P pulmonale), signs of RV strain.

- Urinary retention causes and indications for SPC:
- Indications for SPC: failed/contraindicated urethral catheter (urethral trauma/stricture, severe BPH, penile trauma, morbid obesity).
- Contraindication: suspected bladder malignancy (avoid tract seeding).
- Pediatric airway obstruction:
- Epiglottitis: high fever, toxic child, tripod posture, drooling, muffled voice, inspiratory/expiratory stridor worsened supine.
- Differentiate croup (barking cough, viral, responds to nebulized epinephrine) vs epiglottitis (toxic, drooling, rapid progression).

> Arabic (مختصر): تَمييز السوائل الجنبيّة عن التأمور بواسطة موضع السّائل على اليوزر. موجات القلب في فرط ضغط الدم الرئوي تُظهر إشارات للجنب الأيمن. قَولُوا لا

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to EM Procedures and Skills.

CHAPTER 09: ENDOCRINE, METABOLIC, AND NUTRITIONAL DISORDERS

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1. Clinical overview and pathophysiology

Endocrine, metabolic & nutritional disorders — high-yield emergency revision notes

Core concepts and pathophysiology
- Diabetes mellitus
- Type 1: autoimmune β-cell destruction → absolute insulin deficiency → prone to ketogenesis → DKA.
- Type 2: insulin resistance ± relative insulin deficiency → hyperglycemia, hyperosmolar states.
- Diagnostic thresholds commonly used in ED/clinic: random glucose ≥200 mg/dL with symptoms; fasting glucose ≥126 mg/dL; HbA1c ≥6.5%; 2-hr OGTT ≥200 mg/dL. (Use clinical context.)
- Diabetic emergencies
- DKA: insulin deficiency → lipolysis → free fatty acids → ketogenesis → high anion-gap metabolic acidosis, ketonuria/ketonemia; often younger/type-1; dehydration variable.
- HHS: severe hyperglycemia and hyperosmolarity without significant ketosis; often older/type-2; profound dehydration (typical deficits many liters) and neurologic dysfunction.
- Hypoglycemia
- Neurogenic (sympathetic symptoms) then neuroglycopenic signs. Sulfonylurea-induced hypoglycemia may recur; octreotide blocks insulin secretion for sulfonylurea ingestion.
- Electrolyte and acid-base
- Hyperkalemia: decreased renal excretion/icellular shifts → ECG changes (peaked T → PR prolongation → loss of P → wide QRS → sine wave) → risk VT/VF.
- Hypocalcemia: total calcium may be normal but ionized Ca falls with alkalosis (respiratory alkalosis increases protein binding → ↓ionized Ca → tetany, Chvostek/Trousseau signs).

- High-altitude physiology: chronic hypoxia → sustained hyperventilation → primary respiratory alkalosis; renal compensation lowers HCO_3^- (rule of thumb used clinically: chronic respiratory alkalosis → HCO_3 falls ~5 mmol/L per 10 mmHg $PaCO_2$ drop).
- Endocrine emergencies
- Thyroid storm: life-threatening thyrotoxicosis with fever, delirium, tachyarrhythmia; needs rapid multimodal therapy.
- Pheochromocytoma crises: episodic catecholamine surges → paroxysmal hypertension, headaches, diaphoresis, tachycardia; diagnosed by plasma/urine metanephrines.
- Adrenal insufficiency (Addison): primary adrenal cortex failure — most commonly autoimmune destruction in developed world → glucocorticoid/mineralocorticoid deficiency.
- Hypothalamic-pituitary lesions
- Prolactinoma: amenorrhea, galactorrhea, infertility; large lesions → bitemporal hemianopsia. Prolactin >200 ng/mL strongly suggests macroprolactinoma; MRI pituitary with contrast is imaging of choice.
- Acromegaly: GH-secreting pituitary adenoma; gradual coarse facial/hand changes and visual field defects; MRI pituitary.
- Nutritional & toxic metabolic states
- Wernicke encephalopathy: thiamine (B1) deficiency → triad of encephalopathy, ophthalmoplegia (nystagmus/VI palsy), ataxia; treat empirically with IV thiamine before any glucose.
- Urea cycle disorders/newborn hyperammonemia: present in first days after feeding with vomiting, lethargy, hyperventilation and elevated ammonia; urgent metabolic and supportive care.
- Diabetes insipidus
- Central DI: ADH deficiency → polyuria (very dilute urine osmolality <300 mOsm/kg) and hypernatremia; treatment = desmopressin (DDAVP).
- Nephrogenic DI: renal resistance to ADH → treat by addressing cause, low solute d

Endocrine, Metabolic & Nutritional Disorders — High-Yield Revision Notes
Structured for emergency medicine exam (FRCEM/HIPPO) and clinical practice. Dense, algorithmic, exam-smart. References where used: Rosen’s Emergency Medicine (Maloney/Glauser; Thiessen; Pfennig/Slovis), ADA DKA guidance, selected NCBI reviews cited in QBank items.

1) Core concepts and pathophysiology
- Glucose: hypoglycaemia in children — treat promptly; dextrose concentration choice driven by vein size and osmolarity. Pediatric dextrose bolus target = 0.5 g/kg IV. "Rule of 50s": concentration (%) × volume (mL/kg) ≈ 50. Practical regimens: neonates/infants D10 5 mL/kg (0.5 g/kg); toddlers/children D25 2 mL/kg; adolescents D50 1 mL/kg. (Rosen)
- Diabetic ketoacidosis (DKA): volume depletion + insulin deficiency → hyperglycaemia, ketosis, metabolic acidosis. First treat volume; correct potassium before insulin if low. Insulin infusion standard: 0.1 U/kg/hr (no routine bolus); hold insulin if K+ < 3.3 mmol/L. Corrected Na for hyperglycaemia: add 1.6 mEq/L for every 100 mg/dL glucose >100 mg/dL. Cerebral oedema is a rare, life-threatening complication in children (more common in young/newly diagnosed). (ADA, Rosen)
- Potassium disorders: hyperkalaemia causes progressive ECG changes — peaked T waves → P-wave flattening/PR prolongation → wide QRS → sine wave/VT/VF. Immediate membrane stabilisation (IV calcium) if ECG changes or QRS widening. Temporizing intracellular shift therapies: insulin + dextrose, nebulized beta2 agonist (albuterol). Removal: diuretics, cation exchangers, dialysis. Hypokalaemia causes U waves, flattened T waves, arrhythmia risk; symptoms often when K+ < ~3.0 mmol/L. (Rosen, Pfennig/Slovis)
- Calcium disorders: hypercalcaemia (malignancy, primary hyperparathyroidism, vitamin D excess) → polyuria, neuropsychiatric symptoms, constipation, nephrolithiasis. Initial ED therapy = isotonic saline hydration; follow with calcitonin and bisphosphonates; dialysis in severe renal/heart failure. Hypocalcaemia post-neck surgery (parathyroid/thyroidectomy) causes perioral/finger paresthesias, carpopedal spasm, tetany; treat symptomatic hypocalcaemia with IV calcium (calcium gluconate). (Rosen)
- Adrenal insufficiency: primary (Addison) — low cortisol and aldosterone → hypotension, hyperpigmentation, hyponatraemia, hyperkalaemia, hypoglycaemia. Secondary (exogenous steroid withdrawal or pituitary disease) — low ACTH → low cortisol but usually preserved aldosterone → less hyperkalaemia; hyponatraemia possible; hyperpigmentation absent. Random cortisol low suggests insufficiency; cosyntropin (ACTH) testing used later. (NCBI reviews, Rosen)
- Thyroid storm/ severe thyrotoxicosis: life-threatening hypermetabolic state. Sequence of ED therapy: beta-blockade (propranolol or esmolol if unstable) → antithyroid drug (PTU preferred acutely because it reduces peripheral T4→T3 conversion and blocks synthesis; methimazole used for long term; PTU higher

hepatotoxicity) → iodine (only after antithyroid drug given) → glucocorticoids (reduce T4→T3 conversion, treat possible adrenal insufficiency). Avoid NSAIDs for fever (they displace T4 from binding globulin); acetaminophen preferred.

Endocrine, Metabolic & Nutritional Disorders — High-Yield Emergency Revision (Integrated, de-duplicated from HIPPO QBank items)

Core concepts and pathophysiology
- Wernicke encephalopathy: thiamine (B1) deficiency in malnourished/heavy drinkers → classic triad ophthalmoplegia/nystagmus, ataxia, confusion. Hypomagnesemia impairs response to thiamine. (Rosen's EM)
- Thyroid disorders:
- Subacute (De Quervain) thyroiditis: post-viral, painful tender thyroid, transient thyrotoxicosis → elevated ESR, negative thyroid autoantibodies; self-limited.
- Thyrotoxicosis / thyroid storm: excess circulating T3/T4; high-output cardiac state that can progress to systolic (low-output) heart failure if untreated.
- Pregnancy: propylthiouracil (PTU) preferred in first trimester (and in thyroid storm) because it inhibits peripheral T4→T3 conversion; iodine given only after antithyroid drug started.
- Adrenal disorders:
- Adrenal crisis (primary or secondary after steroid withdrawal) → hypotension, hyponatraemia, hyperkalaemia, hypoglycaemia; urgent IV glucocorticoids required.
- Etomidate can cause transient adrenocortical suppression (important when considering repeated/continuous use).
- Pituitary apoplexy → acute loss of ACTH → secondary adrenal insufficiency; presents with severe headache, visual/CN deficits, nausea/vomiting, hypotension.
- Catecholamine-secreting tumours (pheochromocytoma): episodic headaches, diaphoresis, tachycardia, hypertension; biochemical diagnosis with plasma/urine metanephrines; pre-op alpha blockade required.
- Calcium–parathyroid axis:
- Symptomatic hypercalcaemia: causes include primary hyperparathyroidism (↑PTH) and malignancy (↓/low PTH). Symptoms: "stones, bones, groans, thrones, psychic overtones." ECG can show shortened QT.
- Hypoparathyroidism most commonly follows thyroidectomy (causes hypocalcaemia, positive Chvostek sign).
- Electrolyte disturbances:

- Hypomagnesaemia (common with alcoholism) → arrhythmias (prolonged PR, flattened T waves, widened QRS, predisposition to torsades). Mg required for some thiamine responses.
- Hypokalaemic periodic paralysis (familial or thyrotoxic): intracellular K shift → profound weakness; treat carefully with K repletion (oral if no respiratory/ECG compromise).
- Diabetes emergencies:
- DKA: insulin deficiency → hyperglycaemia, ketosis, metabolic anion gap acidosis. First priority = isotonic fluid resuscitation; check serum potassium BEFORE insulin (insulin drives K intracellularly). Bicarbonate rarely indicated.
- Metformin-associated lactic acidosis (MALA): severe lactic acidosis, elevated lactate; no specific antidote — supportive care; consider dialysis.
- Lithium complications: nephrogenic diabetes insipidus (polyuria/polydipsia) — diagnosis by water deprivation + vasopressin challenge (no urine osmolality rise in nephrogenic DI); lithium can also increase serum calcium/PTH.
References: Rosen’s Emergency Medicine; relevant NCBI/Endocrine guidance (Rosen’s chapters; Endocrine Emergency Guidance).

> Arabic (موجز): المفاهيم الأساسية: نقص الثيامين يسبب متلازمة فيرنيك (علاج بالثيامين+مغنيسيوم)، التهابات الغدة الدرقية تحت الحادة ذات ألم ومؤقتة، أزمة قشرية فوق كلوية

Endocrine, Metabolic & Nutritional Disorders — High-Yield Emergency Revision

Core concepts & pathophysiology
- Electrolyte shifts are often life-threatening because of brain/cardiac excitability (Na, K, Ca, Mg) and require immediate recognition of symptom severity (seizures, arrhythmia, coma).
- Hyponatremia: cerebral edema risk when acute/rapid fall → seizures/coma. Chronic hyponatremia: brain adapts; risk of osmotic demyelination (central pontine myelinolysis, ODS) if corrected too rapidly.
- Potassium: total-body deficit vs transcellular shift. Insulin, β-agonists, and alkalosis drive K+ into cells (can precipitate hypokalemia). Hypomagnesemia prevents effective K+ repletion.
- Calcium/PTH: hypoparathyroidism → ↓PTH → ↓Ca^{2+}, ↑PO_4 → neuromuscular irritability (positive Chvostek/Trousseau, carpopedal spasm).
- Acute endocrine emergencies: thyroid storm (life-threatening thyrotoxicosis),

myxedema coma (severe hypothyroidism with hypothermia, AMS), adrenal crisis (primary or secondary → hypotension, hyperpigmentation in primary Addison).
- Metabolic hyperglycemic crises: DKA = insulin deficiency + ketogenesis + anion gap metabolic acidosis; HHS = extreme hyperglycemia + hyperosmolarity with minimal ketones; both need aggressive fluids ± insulin and electrolyte management.
- Nutritional deficiencies: B1 (thiamine) → Wernicke/Korsakoff; B3 (niacin) → pellagra (dermatitis, diarrhea, dementia); B12 → megaloblastic anemia + posterior column dysfunction.

> Arabic summary — المفاهيم الأساسية
> اختلالات الألكتروليت والاضطرابات الهرمونية قد تؤدي إلى أعراض مهددة للحياة (نوبات، غيبوبة، اختلال نبضات القلب). التصحيح السريع قد يسبب أضرار (مثل اعتلال الميالين الجسري عند تصحيح الصوديوم بسرعة). في الطوارئ، اعمل على استقرار المريض مع تصحيح سريع للأسباب الحادة ثم علاج السبب الأساسي.

Recognition & differential (red flags, discriminators)
- Hyponatremia: severity by symptoms, not just number.
- Red flags: seizures, obtundation, coma → treat emergently.
- Differentiate by osmolality and volume status:
- Hypotonic euvolemic + urine osmolality inappropriately high (>100 mOsm/kg) and urine Na >30 → SIADH (common causes: small cell lung cancer, drugs, CNS disease).
- Psychogenic polydipsia → very dilute urine (low urine osmolality).
- GI losses (vomiting → metabolic alkalosis; diarrhea → non-AG metabolic acidosis).
- SIADH: hyponatremia + low plasma osmolality + inappropriately concentrated urine.
- Seizures in a patient with severe hyponatremia → treat with hypertonic saline regardless of etiology.
- Hypocalcemia (post-neck surgery): perioral numbness, carpopedal spasm, positive Chvostek/Trousseau; labs show ↓Ca2+, ↑phosphate, low PTH.
- Thyroid emergencies:
- Myxedema coma: hypothermia, AMS, bradycardia, hypoglycemia, hyponatremia.
- Thyroid storm: fever, tachycardia/AF, delirium, GI/liver dysfunction.
- Adrenal insufficiency:
- Primary (Addison): hyperpigmentation, hypotension, hyponatremia, hyperkalemia.

- Secondary (e.g., Sheehan): hypopituitarism features (failure to lactate) with hypotension — give corticosteroids emergently.
- Pheochromocytoma: paroxysmal headache, sweating, palpitations, severe labile HTN → screen with plasma

Endocrine, Metabolic & Nutritional Disorders — high-yield ED revision (condensed; QBank synthesis)

Core concepts and pathophysiology
- Hyponatremia: brain adapts to chronic low Na by losing osmolytes → risk of osmotic demyelination (central pontine myelinolysis) if corrected too rapidly. Safe correction commonly cited: limit 8–12 mEq/L per 24 h (use lower limit, e.g., ≤8 mEq/L/day, for high-risk or chronic cases). Acute symptomatic hyponatremia (seizure, coma) requires prompt controlled rise using hypertonic saline. (Rosen's EM)
- Diabetes insipidus (DI): failure to concentrate urine due to ADH axis problems.
- Central DI = deficient ADH secretion (posterior pituitary).
- Nephrogenic DI = renal resistance to ADH (lithium, hypercalcemia, genetic).
- Both → polyuria, polydipsia, hypotonic urine (low urine osmolality), possible hypernatremia if free water cannot be replaced.
- Congenital adrenal hyperplasia (CAH), classically 21-hydroxylase deficiency: impaired cortisol (± aldosterone) synthesis → if salt-wasting form: hyponatremia, hyperkalemia, hypotension/dehydration, metabolic acidosis; females show virilization (ambiguous genitalia), males may present later with salt-wasting shock. Hyperkalemia causes peaked T waves/arrhythmia risk. (NCBI review)

Recognition and differential (ED red flags & discriminators)
- Hyponatremia severity:
- Red flags: seizures, obtundation, respiratory depression, acute severe headache, persistent vomiting — treat as severe/symptomatic.
- Chronic/insidious: nausea, headache, gait disturbance, mild confusion.
- Distinguish causes using volume status, urine Na/osmolality:
- Hypovolemic hyponatremia: low ECF (vomiting, diarrhoea, diuretics) → urine Na low if extrarenal losses, high if renal losses.
- Euvolemic hyponatremia: SIADH, hypothyroid, glucocorticoid deficiency; urine osmolality >100 mOsm/kg, urine Na >30–40.
- Hypervolemic hyponatremia: CHF, cirrhosis, nephrosis.
- Polyuria/polydipsia differential:

- Diabetes insipidus (urine osmolality low, serum osmolality high/normal).
- Primary polydipsia (psychogenic): very dilute urine but concentrates with water deprivation.
- Osmotic diuresis (hyperglycaemia): high serum glucose, glycosuria.
- Neonatal shock with electrolyte abnormalities:
- CAH (21-hydroxylase) — salt wasting, hyperkalemia, hyponatremia; infants (especially males) may present with lethargy, vomiting, hypotension, arrhythmia.
- Sepsis, inborn errors of metabolism, congenital cardiac lesions less likely to show hyperkalemia pattern but must be in differential.

Diagnostics (what to order and how to interpret)
- Hyponatremia workup:
- Immediate: serum Na, plasma osmolality, serum glucose, serum uric acid, TSH, cortisol if indicated.
- Urine: urine osmolality and urine Na (after initial fluid resuscitation if hypovolemic).
- Acute symptomatic → do not delay hypertonic therapy for extensive testing.
- DI workup:
- Serum Na and plasma osmolality; urine osmolality and specific gravity.
- Water deprivation test (specialist setting) → failure to concentrate → desmopressin (DDAVP) challenge: rise in urine osmolality indicates central DI; no change → nephrogenic DI.
- Con

7. Arabic rapid-review summary

تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Endocrine, Metabolic, and Nutritional Disorders.

CHAPTER 10: THORACIC-RESPIRATORY DISORDERS

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Upper airway: larynx level obstruction produces inspiratory stridor; trachea/bronchi produce expiratory stridor/wheeze. Important pediatric causes: croup (viral laryngotracheobronchitis, subglottic narrowing), epiglottitis (supraglottic infection → rapid obstruction), bacterial tracheitis (post-viral bacterial infection of the subglottic/tracheal mucosa). (Rosen's; NCBI Bookshelf)
- Lower airway: bronchiolitis (infants, usually RSV → small-airway inflammation, mucus, air-trapping), acute asthma (bronchospasm, reversible airflow obstruction, risk of status asthmaticus "silent chest"), COPD/emphysema (alveolar destruction, hyperinflation, blebs → risk of spontaneous pneumothorax).
- Pleuropulmonary vascular: pulmonary embolism (thromboembolus → V/Q mismatch, RV strain; severity: massive = hemodynamic instability → reperfusion therapy; submassive = RV strain without shock), ARDS (noncardiogenic pulmonary edema, diffuse infiltrates, ↓compliance; hypoxemia with reduced PaO2/FiO2).
- Airway devices/iatrogenic: recent tracheostomy can erode into innominate artery → sentinel bleed → tracheo-innominate fistula (life-threatening).
References: NCBI Bookshelf chapters on croup, bronchiolitis, PE, ARDS; Rosen's Emergency Medicine.

> الملخص العربي (الأساسيات والفيزيولوجيا)
> - تضيق المسالك الهوائية العليا يسبب أزيز استنشاقي (stridor)؛ أسباب الأطفال الرئيسية: croup، التهاب الحنجرة الحلقي (epiglottitis)، و التهاب الرغامي البكتيري.
> - برونشيوليتيس = رضح في الشعب الهوائية الصغيرة (غالباً RSV). الانصباب الرئوي الحاد (ARDS) = وذمة لا قلبية مع نقص تأكسج واضح.
> - نزيف حول التراكيستومي خلال الأسابيع الأولى = إنذار خطر لتكون ناسور شرياني ربوبي (traheo-innominate).

Recognition and differential
- Stridor in children:
- Inspiratory stridor, barking cough, low-grade fever → croup (steeple sign on AP neck x-ray possible).

- High fever, drooling, tripod position, muffled "hot potato" voice, rapid progression → epiglottitis (thumbprint sign on lateral neck x-ray). Do NOT perform throat exam in uncontrolled setting.
- Recent viral illness, high fever, toxic child, poor response to racemic epi → bacterial tracheitis.
- Wheeze / decreased air entry:
- Focal/unilateral wheeze or sudden onset while eating → foreign body aspiration (may have normal AP/lateral CXR; decubitus/expiratory films show air-trapping).
- Diffuse expiratory wheeze, hyperinflation → asthma or bronchiolitis (age differentiates: bronchiolitis <2 yrs, RSV; asthma older child/adult).
- Silent chest (minimal airflow, poor air movement) = severe obstruction/status asthmaticus → impending respiratory failure.
- Chest pain / dyspnea with risk factors (recent surgery, immobility, malignancy) → pulmonary embolism. Look for tachycardia, hypoxia, signs of DVT, ECG changes (sinus tachy, possible S1Q3T3 or anterior T-wave inversions), CXR may show Hampton's hump (wedge infarct).
- Hypoxemic respiratory failure with bilateral infiltrates after sepsis/aspiration/trauma → consider ARDS (noncardiogenic pulmonary edema).
- Hospital-acquired pneumonia (HAP) risk factors (recent hospitalization, ICU stay) → cover Pseudomonas and MRSA empirically.

الملخص العربي (التعرف والتمييز)
- صف

Thoracic — Respiratory Disorders — High-Yield Emergency Revision
(Integrated synthesis of common HIPPO QBank concepts: PE/hemodynamic PE, pulmonary HTN, DAH/vasculitis, ARDS, infections — pertussis/aspiration/abscess/empyema/pneumonia, hemoptysis, COPD/bronchiectasis/asbestosis, croup)

> Core concepts and pathophysiology
> → Pulmonary embolism (PE): thromboembolus from venous system -/pulmonary arterial obstruction → increased RV afterload → acute RV dilation strain, hypoxemia, V/Q mismatch. EKG may show sinus tachycardia, nonspecific ST/T changes, S1Q3T3 pattern (insensitive/specific), new RBBB, or anterior T-wave inversions. CXR often normal. (Ref: NCBI Bookshelf — PE)
> Pulmonary hypertension (PH): sustained elevation of pulmonary arterial - pressure (mPAP >20 mmHg on RHC). Pathways include pulmonary vascular

remodeling, hypoxic vasoconstriction (COPD), left heart disease, chronic thromboembolic disease, connective-tissue disease. RV failure signs develop as disease advances. (Ref: NCBI)

- Diffuse alveolar hemorrhage (DAH): capillary/alveolar bleeding from vasculitis - immune disease; presents with hemoptysis, hypoxia, diffuse ,(e.g., ANCA/GPA) ;infiltrates, anemia. High-dose IV steroids are the immediate treatment plasmapheresis considered in select causes. (Ref: NCBI)
- ARDS: acute diffuse inflammatory lung injury → increased permeability = edema, bilateral infiltrates, non-cardiogenic pulmonary edema; hallmark .reduced PaO2/FiO2 (P/F) ratio (<300 mmHg defines mild or worse) Management is supportive ventilation, treat underlying cause. (Ref: NCBI)
- Aspiration/aspiration pneumonia and lung abscess: oropharyngeal aspiration - ,mixed aerobic + anaerobic infection; risk factors include alcohol, seizures → dysphagia/CREST (esophageal dysmotility). Lung abscess cavities often after aspiration. (Ref: NCBI)
- Empyema: infected pleural fluid; loculated collections → require drainage - .antibiotics + (thoracentesis/chest tube)
- ,Pertussis: Bordetella pertussis → paroxysmal cough, post-tussive vomiting possible subconjunctival hemorrhage; confirm with PCR or culture within ~3 /weeks; treat with macrolides (adult outpatient alternatives exist). (Ref: CDC (AAFP
- Chronic airway diseases: chronic bronchitis (clinical: productive cough ≥3 - ,months x2 years), emphysema, bronchiectasis (permanent airway dilation recurrent purulent sputum, “tram-tracks”). Alpha-1 antitrypsin deficiency: early panacinar emphysema and cirrhosis association. (Ref: NCBI)

Arabic take-home (core concepts):

الانسداد الرئوي: جلطة وريدية تصل للشريان الرئوي تؤدي لإجهاد البطين الأيمن. فرط ضغط رئوي: ضغط متوسط للشريان الرئوي >20 مم زئبق. نزيف الحويصلات: حاجة لستيرويد وريدي عاجل إذا ناجم عن فرفرية مناعية. متلازمة الضائقة التنفسية الحادة تحكمها نسبة PaO2/FiO2 <300.

Recognition and differential (key presentations and red flags)

- Suspect PE when acute dyspnea ± pleuritic chest pain ± hemoptysis plus risk factors: recent surgery/immobilization, long travel, thrombophilia, cancer, estrogen use. Red flags for massive (high-risk) PE: sustained hypotension, shock, severe hypoxia, syncope, cardiac arrest.

- RV strain signs: jugular venous distension, peripheral edema (later), loud P2, right axis deviati

Core concepts and pathophysiology
- Diffuse alveolar hemorrhage (DAH): capillary/alveolar bleeding due to vasculitis/autoimmune inflammation → rapid respiratory failure, bilateral alveolar consolidation; immunosuppression (high-dose steroids) is first-line adjunct to supportive care (intubation/oxygen, control bleeding) (Park JA, Int J Mol Sci 2021).
- Acute lung injury / ARDS: diffuse alveolar-capillary membrane injury → noncardiogenic pulmonary edema, refractory hypoxemia, stiff lungs; treat with lung-protective ventilation (low Vt, Pplat <30 cm H2O), PEEP, prone ventilation for severe cases (Berlin definition, NIH/NCBI reviews).
- Pulmonary embolism (PE): thrombotic occlusion → V/Q mismatch, RV strain; massive PE = hemodynamic instability (hypotension, shock) and may need systemic thrombolysis; submassive = RV dysfunction without hypotension. Empiric anticoagulation is indicated when pretest probability is moderate–high and imaging is delayed (StatPearls; Vyas & Goyal).
- Airway disease: asthma (bronchospasm, reversible airflow obstruction), status asthmaticus may require intubation — high risk of breath-stacking/auto-PEEP; croup (laryngotracheobronchitis) = viral upper airway inflammation causing inspiratory stridor and barky cough; bronchiolitis (infants <2) = viral small-airways disease with wheeze/retractions.
- Opportunistic lung infections: Pneumocystis jirovecii (PCP) in CD4 <200 → progressive dry cough, hypoxia, diffuse interstitial infiltrates; treat TMP-SMX ± adjunctive steroids if hypoxic.
- Occupational pneumoconioses: asbestosis → pleural plaques/pleural thickening and interstitial fibrosis (lower lobes), risk mesothelioma; silicosis → upper-lobe nodules and progressive fibrosis.
- Cystic fibrosis (CF): defective CFTR → thick secretions, recurrent pneumonia (Pseudomonas), pancreatitis; diagnostic sweat chloride abnormality; manage pulmonary exacerbations with double anti-pseudomonal coverage.
- Infectious disease highlights: influenza → treat with neuraminidase inhibitors (oseltamivir commonly) for severe/high-risk disease; pertussis → macrolide (azithromycin).

> Arabic summary — المفاهيم الأساسية:
> - نزف الحويصلات الرئوية: التهاب/اعتلال أوعية → علاج ستيرويدات.

- الفشل التنفسي الحاد/ARDS: وذمة غير قلبية، تهوية وقائية للرئتين.
- الانصمام الرئوي الضخم: صدمة → علاج مذيب للجلطات.
- أمراض مجرى الهواء: الربو، التهاب القصيبات (الأطفال)، خناق (croup).
- الالتهابات الانتهازية: PCP عند مرضى نقص المناعة.
- أمراض مهنية وراثية: الأسبست (تلّيات غشاء الجنب)، السيليكوس (عقيدات فوقية)، التليف الكيسي: اختبار العرق وإدارة ضد Pseudomonas.

Recognition and differential (red flags, discriminators)
- Red flags requiring immediate action:
- Tripoding, accessory muscle use, one-word sentences, marked hypoxia → impending respiratory failure/intubate.
- Hemodynamic instability with RV strain on echo → massive PE → consider thrombolysis.
- Bilateral “white-out” with hemoptysis → think DAH; early steroids + supportive care.
- Refractory hypoxemia on high-flow oxygen in sepsis → ARDS → intubate + lung-protective strategy.
- Diagnostic discriminators (common exam cues):
- Currant-jelly sputum, alcohol use → Klebsiella

Thoracic — Respiratory Disorders — Consolidated high-yield revision notes (De-duplicated from multiple HIPPO QBank items; references: Berlin ARDS definition NBK436002; Empyema/pleural disease NBK544279, NBK459237; RSV/bronchiolitis NBK441959; PCP NBK560530; PERC papers Kline et al.; CDC pertussis guidance)

1) Core concepts and pathophysiology
- Acute Respiratory Distress Syndrome (ARDS)
- Non-cardiogenic diffuse alveolar injury → increased permeability pulmonary edema, inflammation and V/Q mismatch.
- Berlin definition: acute onset ≤1 week of known clinical insult (sepsis, aspiration, trauma, transfusion, inhalational injury), bilateral radiographic opacities not fully explained by effusions/atelectasis/heart failure, and impaired oxygenation (PaO_2/FiO_2 <300 on PEEP ≥5 cm H_2O). (NBK436002)
- Cardiogenic (hydrostatic) pulmonary edema (CHF)
- Elevated left-heart pressures → cardiomegaly, vascular congestion, cardiogenic interstitial/alveolar edema; BNP/evidence of volume overload aid differentiation.
- Pneumothorax

- Air in pleural space → lung collapse; spontaneous (primary/secondary), traumatic, iatrogenic, or tension physiology (one-way valve → hemodynamic compromise).
- Pleural infection / Empyema
- Parapneumonic effusion that becomes infected; loculated fluid, thick pus, may track into chest wall (empyema necessitans, commonly TB).
- Decubitus films show layering; indicators for drainage include frank pus, positive Gram stain/culture, pleural fluid pH low, low glucose.
- Bronchitis / Bronchiolitis / Viral pneumonias
- Acute bronchitis usually viral → supportive care. Bronchiolitis (infants, RSV) → small airways obstruction, supportive care only.
- Pertussis (Bordetella pertussis)
- Paroxysmal cough with inspiratory “whoop”; macrolide is first-line; TMP-SMX if macrolide allergy.
- Pneumocystis jirovecii (PCP)
- Immunocompromised (HIV) diffuse interstitial pneumonia; elevated LDH common; hypoxemia may require adjunctive steroids when severe (see diagnostics/management).
- Status asthmaticus and ventilator barotrauma
- Severe bronchospasm → air-trapping, dynamic hyperinflation; mechanical ventilation can generate barotrauma → tension pneumothorax.

> Arabic (concise): المفاهيم الأساسية: ARDS = وذمة رئوية غير قلبية التهابية بعد إصابة؛ CHF = وذمة هيدروستاتية مع قلب متضخم؛ استرواح صدر = هواء بالغشاء الجنبي؛ وجود صديد بالجنبة = استمياج/تصريف عاجل.

2) Recognition and differential (key discriminators / red flags)
- ARDS vs cardiogenic pulmonary edema
- ARDS: recent risk factor (sepsis, aspiration, trauma), PaO2/FiO2 <300, bilateral diffuse infiltrates, normal/nearly normal cardiac size, no clinical evidence of fluid overload or left-sided failure.
- CHF: cardiomegaly, pleural effusions, orthopnea, elevated BNP, response to diuretics.
- Tension pneumothorax
- Sudden hypotension, distended neck veins, tracheal deviation (late), unilateral absent breath sounds; after positive pressure ventilation suspect immediately.
- Small spontaneous pneumothorax (primary)

- Tall thin young patients, pleuritic chest pain, dyspnea; if rim <2 cm and clinically stable → observation.
- Empyema vs simple parapneumonic effusion
- Empyema: fe

Core concepts and pathophysiology
- Pneumothorax: air in pleural space → lung collapse; tension physiology causes mediastinal shift and cardiovascular compromise. Primary spontaneous in tall thin smokers/young; secondary on background lung disease (eg COPD). (See NCBI Bookshelf: pleural disease.)
- Pulmonary embolism (PE): thromboembolus → acute RV afterload increase → RV strain, hypoxemia, possible hemodynamic collapse. Spectrum: small-subsegmental → submassive (RV strain without hypotension) → massive (hemodynamic instability/shock).
- Diffuse alveolar hemorrhage (DAH): capillary/alveolar bleeding from vasculitis, autoimmune disease, drugs → acute respiratory failure and bilateral alveolar infiltrates; hemoptysis variable.
- Fat embolism syndrome: long-bone fracture/orthopedic surgery → fat droplets → pulmonary capillary obstruction + inflammatory response; classic triad = respiratory distress, neurologic signs, petechial rash.
- Bronchiolitis (RSV common): viral small-airway inflammation in infants → wheeze, increased work of breathing; supportive care mainstay.
- Asthma / exercise-induced bronchoconstriction (EIB): reversible bronchospasm; pre-exercise short-acting β2 agonist prevents EIB.
- Aspiration pneumonitis vs aspiration pneumonia vs lung abscess: chemical pneumonitis occurs immediately after gastric content aspiration (often early, may not require antibiotics); bacterial aspiration pneumonia arises later (>48 h) with anaerobic/upper-airway flora and can progress to lung abscess (cavitation, air-fluid level).
- Empyema: infected pleural fluid (pus) often complicates pneumonia; requires diagnostic thoracentesis and usually drainage (tube thoracostomy).
- Septic pulmonary emboli: right-sided endocarditis/IVDU → multiple cavitary bilateral lesions; systemic sepsis and murmurs often present.
- Occupational pneumoconioses (eg silicosis, coal worker’s): chronic inhalational exposure → progressive interstitial fibrosis; CXR shows innumerable small rounded opacities; clubbing and progressive dyspnea.
- Upper airway pediatric emergencies: epiglottitis = acute supraglottic edema with

drooling, tripod position — airway protection first. Croup = subglottic inflammation; AP neck x-ray may show “steeple sign.”

> Arabic (مُلخّص): نبضات جوهرية: الانهيار الرئوي عبر الهواء في الجنبة (الاسترواح الصدري)، الانسداد الوعائي في الانصمام الرئوي، النزف الحويصلي المنتشر، متلازمة الانصمام الدهني، التهاب الشعب الهوائية عند الرضع، الالتهاب الشفطي والقيحي، والخطر الاختناقي في التهاب الحنجرة عند الأطفال.

Recognition and differential (red flags & discriminators)
- Sudden pleuritic chest pain ± dyspnea, tachycardia, hypoxemia → consider PE, pneumothorax, spontaneous pneumothorax more likely with asymmetric breath sounds and hyperlucent hemithorax on CXR.
- ECG in PE: most common abnormality = sinus tachycardia. Classic S1Q3T3 and RBBB indicate RV strain but are neither sensitive nor specific.
- Massive PE: hemodynamic instability (shock/hypotension) or severe hypoxemia = immediate reperfusion consideration.
- Hemoptysis: common causes = bronchitis, bronchiectasis, malignancy, TB, DAH. Massive hemoptysis (threat to airway) → airway control, localize bleed, urgent br

Thoracic–Respiratory Disorders — high-yield revision notes (deduplicated)

Core concepts and pathophysiology
- Typical vs atypical pneumonia
- Typical (eg Streptococcus pneumoniae): abrupt onset, high fever, productive cough, lobar consolidation on CXR; common cause of post-influenza secondary bacterial pneumonia.
- Atypical (eg Mycoplasma, Chlamydia, Legionella): more indolent or systemic features, diffuse/patchy infiltrates.
- Post-influenza secondary bacterial pneumonia
- Classic biphasic course: viral illness → improvement → recurrent fever, productive cough within about 7 days.
- Common pathogens: S. pneumoniae (most common), Staphylococcus aureus, H. influenzae, Moraxella catarrhalis.
- Legionnaires’ disease (Legionella pneumophila)
- Water/aerosol sources (cooling towers, hot tubs, centralized AC). Presents with high fever, cough, myalgias, headache, and gastrointestinal symptoms (diarrhea); may have hyponatraemia.
- Diagnostic: urinary antigen for L. pneumophila (serogroup 1), sputum culture on

special media.
- Tuberculosis
- Primary TB: may show hilar/mediastinal lymphadenopathy; reactivation (post-primary) often causes upper-lobe cavitary lesions.
- Pancoast (superior sulcus) tumour
- Apical lung malignancy that invades brachial plexus/cervical sympathetic chain → shoulder/arm neuropathic pain, ulnar neuropathy, muscle wasting, and Horner's syndrome.
- Foreign body aspiration (children)
- Most aspirated objects are radiolucent → normal CXR possible. Classically unilateral expiratory wheeze, sudden onset.
- Massive pulmonary embolism (PE)
- Hemodynamic instability, syncope, hypoxaemia; bedside echo with RV dilation and septal bowing indicates RV strain → consider immediate reperfusion therapy.
- Hyperoxia test (neonatal cyanosis workup)
- Give 100% FiO2 for 10 min: PaO2 >150 mmHg argues for pulmonary cause; PaO2 ≤150 mmHg (or minimal rise) suggests cyanotic congenital heart disease.

> Arabic summary (مفاهيم أساسية)
> - التمييز بين الالتهاب الرئوي النمطي وغير النمطي مهم سريرياً. عدوى بكتيرية ثانوية بعد الإنفلونزا غالباً بسبب S. pneumoniae. Legionella تسبب إسهال وتنتقل عبر أنظمة تكييف/مياه. ورم بانكوست يظهر بألم واعتلال أعصاب اليد و/أو متلازمة هورنر. الفشل التنفّسي الحاد مع اتساع البطين الأيمن يتطلب إعادة تروية سريعة.

Recognition and differential
- Post-influenza biphasic illness → secondary bacterial pneumonia (most likely S. pneumoniae).
- Legionella clues → high fever, diarrhoea, recent exposure to contaminated water/AC, multiple cases in a building.
- TB risk factors → incarceration, crowded settings, weight loss, night sweats; primary TB → hilar/mediastinal lymphadenopathy; reactivation → upper lobe cavitation.
- Pancoast tumour → smoker or occupational exposure, apical mass, ipsilateral upper limb neuropathy (ulnar distribution), Horner's.
- Child with sudden onset cough/wheeze, unilateral signs, normal CXR → high suspicion for aspirated foreign body.
- Suspected massive PE → syncope, hypotension, severe hypoxaemia, tachycardia,

RV strain on echo/ECG (new R axis deviation, S1Q3T3 possible).
- Hemoptysis differential (US): acute bronchitis most common cause; worldwide TB common.

Arabic summary (التعرّف والتفري

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Thoracic-Respiratory Disorders.

CHAPTER 11: MUSCULOSKELETAL DISORDERS (NONTRAUMATIC)

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1. Clinical overview and pathophysiology

Core concepts and pathophysiology
- Degenerative (mechanical)
- Osteoarthritis (OA): cartilage wear → joint-space narrowing, osteophytes, crepitus; pain worse with activity, morning stiffness <30–60 min. First-line analgesic: acetaminophen (NSAIDs risk if GI bleed). (Rosen's EM Ch.102)
- Rotator cuff tear vs tendinopathy: tears → true weakness of abduction or external rotation not explained by pain/atrophy.
- Adhesive capsulitis ("frozen shoulder"): prolonged immobilization → capsular adhesions → loss of both active and passive ROM.
- Inflammatory / autoimmune
- Rheumatoid arthritis / Juvenile idiopathic arthritis (JIA): symmetric small-joint synovitis; JIA common oligoarticular form in young children; ANA often positive, RF often negative in oligoarticular JIA.
- Spondyloarthropathies (ankylosing spondylitis): inflammatory back pain improving with exercise, sacroiliitis, "bamboo spine"; common extra-articular associations include inflammatory bowel disease.
- Crystal arthropathies
- Gout: monosodium urate (MSU) crystals = needle-shaped, negatively birefringent under polarized light; classic first MTP (podagra); precipitated by diuretics (e.g., HCTZ), alcohol.
- Pseudogout (CPPD): calcium pyrophosphate = rhomboid-shaped, positively birefringent; often knee or wrist; x-ray may show chondrocalcinosis.
- Infectious bone & joint disease
- Septic arthritis: hematogenous or direct spread → purulent joint; urgent diagnosis to prevent joint destruction.
- Osteomyelitis (hematogenous or contiguous): diabetics, IVDU, prosthetic devices; MRI most sensitive for early disease.
- Necrotizing soft-tissue infection (including Fournier gangrene): rapidly progressive, severe pain out of proportion, crepitus, skin necrosis; often

polymicrobial (type I) in diabetics; strep/staph/clostridial or polymicrobial mixes.
- Disseminated gonococcal infection (DGI): triad of tenosynovitis, dermatitis (pustular lesions), polyarthralgias — treat with ceftriaxone.
- Neurologic compression syndromes
- Lumbar spinal stenosis: neurogenic claudication — bilateral leg pain worse with walking/extension, improves with sitting/flexion; MRI diagnostic.
- Cauda equina syndrome: compression of sacral nerve roots → bilateral sciatica, saddle anesthesia, bowel/bladder dysfunction, decreased sphincter tone — surgical emergency; most commonly due to large herniated lumbar disc.

> Arabic (concise)
> - ملخَّص فسيولوجي: تُمثّل الاضطرابات العضلية الهيكلية غير الرضحية أمراضاً تنكسية، التهابية، بلورية، عِدَوِيَّة أو ضغطية عصبيّة. يجب التمييز سريرياً وباستخدام تصوير ومحلول المفصل.

Recognition and differential
- Acute monoarthritis (hot swollen joint): differential = septic arthritis, crystal arthritis (gout/pseudogout), reactive arthritis, traumatic hemarthrosis. Red flags for septic arthritis: fever, systemic toxicity, joint very tender with pain on passive ROM, synovial fluid WBC high (see Diagnostics).
- Key discriminators
- Synovial fluid crystals: MSU (gout) = needle, negative birefringence; CPPD (pseudogout) = rhomboid, positive birefringence. Presence of crystals does NOT exclude septic arthritis.
- Synovial WBC: inflammatory >25,000; >50,000 suggests bacterial;

Musculoskeletal — non-traumatic: condensed, high-yield emergency revision notes
(Integrated from HIPPO QBank items; key references: Rosen’s Emergency Medicine; NCBI reviews cited in QBank)

1. Core concepts and pathophysiology

- Categories covered: inflammatory myopathies (polymyositis, dermatomyositis), crystal arthropathies (gout, CPPD/pseudogout), degenerative disease (osteoarthritis, adhesive capsulitis), pediatric hip disorders (SCFE, Legg–Calvé–Perthes, developmental dysplasia), infective bone/soft-tissue processes (vertebral osteomyelitis, spinal epidural abscess, septic bursitis/joint), mechanical low back pain and radiculopathy, rhabdomyolysis, spondyloarthropathies (ankylosing

spondylitis), common bursitides (olecranon, anserine), carpal tunnel syndrome.
- Key pathophysiology highlights:
- SCFE: displacement through hypertrophic physis in obese adolescent; shear across proximal femoral growth plate.
- Legg–Calvé–Perthes: idiopathic avascular necrosis of femoral capital epiphysis (peak diagnosis ~4–8 y; median ~7 y).
- Vertebral osteomyelitis/epidural abscess: hematogenous seeding (IV drug use common), contiguous spread, or direct inoculation (spine procedures); Staphylococcus aureus most common.
- Rhabdomyolysis: muscle necrosis → myoglobinuria (dipstick blood + no RBC), CK markedly elevated; electrolyte shifts (hyperK, hyperPO4, hypoCa early), risk AKI and compartment syndrome.
- CPPD (pseudogout): calcium pyrophosphate deposition → chondrocalcinosis on x-ray; synovial fluid shows weakly/positively birefringent, rhomboid crystals.
- Dermatomyositis: immune-mediated myopathy with characteristic skin findings (Gottron papules); diagnosis requires muscle biopsy.
- Carpal tunnel: median nerve compression at wrist; night pain, thenar numbness; positive Phalen/reverse-Phalen/Tinel.

> Arabic summary (concise):
> - المفاهيم الأساسية: انقسام بين اضطرابات التهابية، بلورية، تنكسية، معدية وميكانيكية. مثلاً SCFE عند المراهق السمين، Perthes عند الأطفال الصغار، التهاب نخاعي فقري/خراج فوق الجافية مرتبطان بحقن وريدية وStaph aureus، والرابدوميوlysis يؤدي لبيلة مي oglobin وAKI.

2. Recognition and differential (red flags / discriminators)

- Red flags in low back pain — urgent imaging/consult:
- Neurologic deficit (progressive motor weakness, saddle anesthesia, new bowel/bladder dysfunction → cauda equina)
- Severe/progressive neurological deficits, fever, IV drug use, recent spine procedure, spinal tenderness with constitutional symptoms, known malignancy (new back pain)
- Suspicion of spinal epidural abscess/vertebral osteomyelitis → MRI
- Distinguish inflammatory vs mechanical back pain:
- Inflammatory (ankylosing spondylitis): onset <40 y, >3 months, morning stiffness, improves with exercise, possible uveitis.
- Mechanical/acute muscular pain: sudden, localized, improves with rest/NSAIDs;

absence of red flags → conservative care.
- Hip pain in children/adolescents:
- Leg pain referred from hip (e.g., SCFE) — adolescent, obesity, antalgic gait, limited internal rotation.
- Legg–Calvé–Perthes — younger child (median ~7 y), limp, hip ROM restriction; x-ray shows fragmentation/flattening of femoral head.
- Developmental dyspl

Musculoskeletal disorders — nontraumatic (high-yield ED revision)
Concise, exam-focused synthesis of repeated QBank concepts (sources: Rosen's Emergency Medicine; NCBI Bookshelf reviews cited in stems; Edlow Ann Emerg Med back-pain review).

Core concepts and pathophysiology

- Inflammatory vs mechanical vs infectious vs metabolic causes present to the ED as pain/swelling/limp/back pain/monoarthritis.
- Cytokine storm complication: systemic juvenile idiopathic arthritis (Still disease) → risk of macrophage activation syndrome (MAS) — life-threatening hyperinflammatory state with fevers, cytopenias/↑LFTs, hyperferritinemia, multiorgan dysfunction.
- Crystal arthropathies: gout = monosodium urate (needle-shaped, strongly negatively birefringent); pseudogout = calcium pyrophosphate (rhomboid, weakly/positively birefringent).
- Spine infections: vertebral osteomyelitis, discitis, spinal epidural abscess (SEA). Risk factors: IVDU, immunosuppression, recent spine procedure (post-op S. epidermidis risk), recent bacteremia, TB → Pott disease (spinal TB) →
- Rhabdomyolysis: muscle necrosis → myoglobinuria → heme-positive dipstick with few/absent RBCs → AKI risk.
- Necrotizing soft-tissue infection/myonecrosis: gas-forming organisms (eg, Clostridium) → crepitus/subcutaneous gas, fulminant sepsis.
- Overuse/insertional disorders: plantar fasciitis (medial anterior calcaneus insertion; positive windlass test), trochanteric bursitis (lateral hip tenderness, worse with direct pressure), Osgood-Schlatter (adolescent tibial tubercle traction apophysitis).
- Tendon rupture associations: fluoroquinolones increase risk of tendon rupture (eg, Achilles, patellar tendon).
- Arabic summary (core): ملخص: أمراض الجهاز العضلي غير الرضّية تنقسم إلى التهابية،

> ميكانيكية، معدية، واستقلابية. حالات طارئة بارزة: متلازمة تنشيط البلاعم (MAS) عند مرضى التهاب المفاصل الطفولي النظامي، والعدوى الفقرية، واعتلال العضلات التحللي (rhabdomyolysis)، وعدوى الأنسجة الناخرة.

Recognition and differential (red flags & discriminators)
- Acute monoarthritis: always consider septic arthritis. Red flags: fever, inability to bear weight, WBC synovial >50,000/mm3, purulent fluid, positive gram stain/ culture.
- Crystal vs septic: crystals present do NOT exclude septic arthritis — high synovial WBC, systemic features, or gram stain positive → treat as septic.
- Back pain red flags → urgent imaging: fever, IVDU, immunosuppression, recent spine surgery, progressive neuro deficits, saddle anesthesia, bowel/bladder dysfunction (suspect SEA, vertebral osteomyelitis, cauda equina, metastasis).
- Cauda equina suspicion: bilateral leg weakness/heaviness, saddle anesthesia, urinary retention/overflow, decreased anal tone → immediate MRI and neurosurgical consult.
- Spine infection discriminators: constitutional symptoms (fever, weight loss), focal midline tenderness, elevated ESR/CRP (may be normal early), plain films insensitive; MRI is test of choice.
- Rhabdomyolysis clues: recent immobilization/overexertion/crush, sympathomimetic toxicity, urine dipstick positive for blood with no RBCs → check CK, electrolytes, renal function.
- Tendon rupture: inability to plantarflex on calf squeeze (Thompson test) for

Core concepts and pathophysiology
- Broad approach: non-traumatic musculoskeletal (MSK) emergencies range from superficial soft-tissue infections and overuse/tendinopathies to spine emergencies (fracture, epidural hematoma, cauda equina), necrotizing soft-tissue infections, and pediatric hip disorders. Early recognition of “red flags” (neurologic deficit, systemic toxicity, rapid progression, inability to weight-bear, signs of sepsis, bowel/ bladder dysfunction) directs urgent imaging, surgery, or admission. (Rosen’s Emergency Medicine; NCBI Bookshelf)
- Infections:
- Paronychia: bacterial infection of nail fold; erythema, swelling, possible purulent drainage. Can become abscess or extend to felon if pulp involved.
- Felon: closed-space abscess of the finger pad under the volar fat pad → intense throbbing pain and risk of ischemia.

- Herpetic whitlow: HSV infection → grouped clear/turbid vesicles; usually NO frank purulence. Incision and drainage contraindicated.
- Necrotizing fasciitis/Fournier gangrene: rapidly progressive subcutaneous/fascial infection with severe pain out of proportion, bullae, crepitus, systemic toxicity; often polymicrobial; diabetes a common comorbidity. CK and AST can rise with muscle/fascia involvement. LRINEC score exists but has limitations. (NCBI)
- Tendinopathy/entrapment:
- De Quervain tenosynovitis: stenosing tenosynovitis of abductor pollicis longus + extensor pollicis brevis at 1st dorsal compartment → radial styloid pain; positive Finkelstein test.
- Carpal tunnel: median nerve compression at wrist → paresthesia of thumb–index–middle fingers, nocturnal symptoms, positive Tinel/Phalen.
- Spine:
- Compression fracture risk increased with age and osteoporosis; ground-level falls in elderly warrant CT for fracture evaluation.
- Cauda equina syndrome / epidural hematoma: saddle anesthesia, bilateral weakness, bowel/bladder dysfunction; epidural hematoma risk increased after spinal procedures and with anticoagulation.
- Sciatica (radiculopathy): unilateral leg pain with positive straight-leg raise, normal neuro exam and ambulation → outpatient care often appropriate.
- Pediatric hip disorders:
- SCFE (slipped capital femoral epiphysis): adolescent obese child, gradual limp, hip/knee pain; requires surgical fixation—avoid weight bearing.
- Legg-Calvé-Perthes (idiopathic avascular necrosis of femoral head): younger children, insidious limp, non-weight bearing and ortho referral; plain films and MRI as indicated.

> Arabic (مختصر مَرْكَزي عن الفسيولوجيا المرضية)
> - يجب التعرف مبكراً على علامات الخطر: ضعف عصبي، حمى شديدة، ألم متزايد بسرعة، عدم القدرة على حمل الوزن أو احتباس البول.
> - التهابات الأصابع: الالتهاب حول الظفر paronychia = purulent؛ بثور شفافة = herpetic whitlow؛ felon = خُرّاج وسطي في وسادة الإصبع.
> - حالات العمود الفقري: كسر انضغاطي عند كبار السن بعد سقوط بسيط → تصوير مقطعي؛ متلازمة ذيل الحصان/ورم دموي فوق الجافية → طوارئ جراحية.

Recognition and differential (clinical discriminators and red flags)
- Hand/finger infections:

- Paronychia: tenderness/swelling localized to nail fold; possible purulent drainage. I&D indicated if fluctuance/abscess. Risk factors: wet work, diabetes, immuno

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Musculoskeletal Disorders (Nontraumatic).

CHAPTER 12: HEMATOLOGIC DISORDERS

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1. Clinical overview and pathophysiology

Hematologic disorders — consolidated high-yield emergency revision notes

Core concepts and pathophysiology
- Sickle cell disease complications
- Splenic sequestration: acute pooling of large fraction of blood in an enlarged spleen → hypovolemia, anemia, high reticulocyte count (pre-autosplenectomy). Can be life-threatening.
- Aplastic (pure red-cell) crisis: marrow arrest of erythropoiesis caused most commonly by parvovirus B19 → severe anemia with low reticulocyte count.
- Acute chest syndrome: new pulmonary consolidation plus respiratory symptoms in sickle cell patients; pathogenesis = pulmonary vaso-occlusion from bone-marrow/fat emboli, infection or in-situ sickling → rapid respiratory decompensation risk.
- Tumor lysis syndrome (TLS)
- Rapid tumor cell lysis → hyperuricemia, hyperphosphatemia, hyperkalemia, secondary hypocalcemia, AKI from uric acid and calcium-phosphate precipitation.
- Transfusion reactions
- Febrile non-hemolytic (most common), allergic/anaphylactic, acute hemolytic (ABO incompatibility; life-threatening), TRALI (non-cardiogenic pulmonary edema within 6 hrs), TACO (volume overload).
- Coagulopathies / microangiopathies
- DIC: consumptive coagulopathy in sepsis/trauma/malignancy → bleeding + thrombosis; labs: ↑PT/PTT, ↑D-dimer, ↓fibrinogen, thrombocytopenia, schistocytes.
- TTP: ADAMTS13 deficiency → large vWF multimers, platelet microthrombi → thrombocytopenia, microangiopathic hemolytic anemia, neurologic/renal signs, fever sometimes.
- ITP: immune platelet destruction → isolated thrombocytopenia, mucocutaneous bleeding; otherwise normal CBC and smear.
- Hematologic malignancies & paraproteinemias
- ALL: most common childhood leukemia; marrow blasts >20% → pancytopenia,

systemic symptoms.
- AML: can present with gingival hyperplasia, cytopenias, blasts; Auer rods may be seen.
- Multiple myeloma: monoclonal M-protein, lytic bone lesions, anemia, renal failure, hypercalcemia; hyperviscosity syndromes possible.
- Thrombophilia
- Factor V Leiden: common inherited cause of recurrent VTE; management of acute VTE is standard anticoagulation; consider long-term anticoagulation if unprovoked/recurrent.

> Arabic (core concepts): ملخصات أساسية: مضاعفات فقرات المنجلية، متلازمة تحليل الورم، تفاعلات نقل الدم، اضطرابات التخثر والثلاسيميا واللوكيميا والورم النقوي المتعدد — كل حالة لها فيزيولوجيا واضطرابات مختبرية مميزة.

Recognition and differential (key clinical red flags and discriminators)
- Splenic sequestration vs aplastic (parvovirus) crisis:
- Sequestration: tender LUQ, splenomegaly, hypotension, pallor, high retic count.
- Aplastic crisis: recent viral prodrome (erythema infectiosum), severe anemia with low retic count, no splenomegaly.
- Acute chest syndrome vs pneumonia vs PE:
- ACS = new pulmonary consolidation in a patient with SCD + respiratory symptoms; can mimic CAP but treat as ACS (incentive spirometry, antibiotics, transfusion as indicated).
- TLS: timing soon after cytotoxic therapy (hours–days), rapid uric acid ↑, K+, PO4↑, Ca2+ ↓ with AKI and oliguria.
- TRALI vs TACO:
- TRALI: acute hypoxia, fever, hypotension, noncardiogenic bilateral infiltrates within 6 hrs of transfusion; no

SOURCE-DERIVED FALLBACK NOTES:
ptomatic or they may have weight loss, fever, and night sweats. Laboratory studies show lymphocytosis with a white blood cell count >20,000 per microL. The presence of smudge cells on a peripheral blood smear confirms a diagnosis of CLL.
D. Chronic myelogenous leukemia
Chronic myelogenous leukemia (CML) results from an excess production of mature myeloid cells. It occurs in three phases: chronic, accelerated, and acute. Fever,

anorexia, low grade fever, and weight loss are common. Symptoms typically develop gradually. Splenomegaly is also common. Philadelphia chromosome is pathognomonic for CML, and is p
Q813 | STEM: Which clinical scenario is a patient most at risk for developing Transfusion Related Acute Lung Injury (TRALI)? | ANSWER: Septic Shock | EXPLANATION: Explanation
A. During general anesthesia for surgery
TRALI is more likely to develop in sepsis, mechanical ventilation, or massive transfusion patients.
B. First blood transfusion
A first-time blood transfusion is not a major risk factor for TRALI. TRALI is more likely to develop in patients with sepsis, mechanical ventilation, or massive transfusion.
C. Prior history of blood transfusions
A prior history of blood transfusion is not a major risk factor for TRALI. TRALI is more likely to develop in patients with sepsis, mechanical ventilation, or massive transfusion.
D. Septic Shock
Transfusion-related acute lung injury (TRALI) is a syndrome in which there is noncardiogenic pulmonary edema associated with hypoxia that occurs during or after a transfusion. TRALI is more likely to develop in patients with sepsis, mechanical ventilation, or massive transfusion. Management of TRALI is to stop the transfusion, notify the blood bank to screen the donor blood for antileukocyte antibodies, anti-HLA or anti-neutrophil-specific antibodies, and supportive oxygenation.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK507846/
Q814 | STEM: A 21-year-old man with a history of sickle cell disease comes to the emergency department with generalized body pain, chest pain, coughing, and shortness of breath for two days. Vital signs are: BP 109/73 mmHg, P 132/min, T 103.3F (39.6C), RR 23/min, O_2 Sat 88% room air. Your physical examination notes diffuse tenderness to palpation across the chest and extremities but no rashes. You note crackles to the right mid-lung zone. The chest x-ray is shown. Intravenous fluid resuscitation, pain medication, and oxygen are administered, with oxygen saturation improving to 97% on 10L facemask. Bedside hemoglobin is 8.0 g/dL. He is able to tolerate oral medications and food. Hematology has been consulted. Which of the following is the next best step in the management of this

patient? | ANSWER: Administer ceftriaxone and azithromycin IV and admit | EXPLANATION: Explanation
A. Administer azithromycin PO and discharge with next-day follow-up
This patient is presenting with clinical concern for acute chest syndrome, not simply community-acquired pneumonia. Patients with acute chest syndrome should be admitted with consideration of blood vs exchange transfusion, in addition to broad-spectrum antibiotics (third-generation cephalosporin +

Core concepts and pathophysiology
- Hemostasis and blood-cell disorders grouped by problem type:
- Red-cell (anemia, hemoglobinopathies, polycythemia): decreased O2-carrying capacity, marrow production problems, hemolysis, or blood loss. Secondary polycythemia from chronic hypoxia (COPD) via ↑EPO. (NCBI Bookshelf)
- Platelet disorders: decreased number (thrombocytopenia) vs. dysfunction (bleeding despite normal count). Immune-mediated (ITP), consumptive (TTP, DIC), drug-induced (HIT).
- Coagulopathies/factor deficiencies: congenital (hemophilia A/B) or acquired (warfarin, liver disease). Replacement or reversal strategies differ.
- White-cell / marrow failure / malignancy: neutropenia → high risk bacterial infections; leukemias/lymphomas with marrow infiltration, cytopenias, splenomegaly.
- Oncologic metabolic complications: tumor lysis syndrome (TLS) → hyperK, hyperphosphatemia, hyperuricemia, hypocalcemia → AKI, arrhythmia. (NCBI Bookshelf)

> Arabic: المفاهيم الأساسية: اضطرابات خلايا الدم تنقسم إلى مشاكل كريات حمراء وصفائح وعوامل تَخْثُّر وكريات بيضاء؛ كل فئة لها فيزيولوجيا وعلاج طارئ مختلف.

Recognition and differential (red flags and discriminators)
- Thrombocytopenia: petechiae/mucosal bleeding → consider ITP (isolated thrombocytopenia, post-viral in children), TTP (MAHA + thrombocytopenia + neuro/renal/fever), DIC (bleeding + prolonged PT/PTT, underlying sepsis/obstetric/trauma), HIT (heparin exposure + ↓platelets + new thrombosis). Key discriminator: presence of schistocytes/hemolysis → microangiopathic (TTP, HUS, DIC). (NCBI)
- Vaso-occlusive sickle cell crisis: severe bone pain (chest, long bones, back); triggers: infection, dehydration. Red flags: hypoxia, chest pain, focal neuro deficits, priapism → consider exchange transfusion. Parvovirus B19 exposure → aplastic

crisis (abrupt severe anemia). Dactylitis in infants 6–18 months = bone infarct. (NCBI)
- Febrile neutropenia: fever + ANC <500 (or expected to drop <500) → life-threatening; draw cultures then start broad-spectrum anti-pseudomonal antibiotics within 60 min. (In oncology guidelines and Rosen's/NCBI)
- TTP vs HUS vs DIC:
- TTP: ADAMTS13 deficiency → platelet-rich thrombi → neuro signs common; schistocytes, ↑LDH, low haptoglobin. Urgent plasmapheresis.
- HUS: more renal-predominant; often post-diarrheal in kids.
- DIC: consumptive coagulopathy with prolonged PT/PTT, low fibrinogen; triggered by sepsis, placental abruption, trauma, malignancy. (NCBI)
- HIT: look for platelet fall ~5–10 days after heparin (or sooner with prior exposure) + new thrombosis. Thrombosis is common despite low platelets. Use 4T pretest probability, send PF4/heparin antibody, functional assay. (Arepally, NCBI)
- TLS: prodrome of nausea, muscle cramps, tetany, AKI after cytotoxic therapy or spontaneously with high tumor burden → check uric acid, K, PO4, Ca and creatinine. (NCBI)

> Arabic: كيفية التمييز: صفائح منخفضة مع انحلال دموى → TTP/HUS/DIC؛ حمى مع عدل عدلات منخفضة → طوارئ؛ ألم عظمي لدى مريض بالمنجلية → أزمات وعائية.

Diagnostics (what to order fast and why)
- Immediate bedside/lab priorities in bleeding/thrombocytopenia:
- CBC with platelet count, peripheral smear (schis

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Hematologic Disorders.

CHAPTER 13: IMMUNE SYSTEM DISORDERS

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
oncepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
D. Systemic steroids
Systemic steroids are useful for allergic angioedema. While steroids are commonly given to patients with hereditary angioedema, the efficacy is unclear.
Q863 | STEM: A 33-year-old woman presents to the emergency department with cough and malaise that has been persistent for months. She also reports swollen lumps in her armpit and neck that have been present for the same period of time. She has no previous medical history and denies tobacco use. Vital signs are normal. Chest X-ray is obtained (see figure). Which of the following is true regarding this condition? | ANSWER: Anterior uveitis can be the presenting symptom | EXPLANATION: Explanation
A. Anterior uveitis can be the presenting symptom
This radiograph shows bilateral hilar adenopathy, and in a young woman with respiratory symptoms and malaise, sarcoidosis is the most likely diagnosis. Sarcoidosis is a non-caseating granulomatous disease that many involve multiple organs. Hypercalcemia and hypercalciuria are commonly seen. Complete heart block can occur when there is granulomatous involvement of the conducting system. Anterior uveitis is the presenting symptom in 5% of patients. Steroid therapy is the treatment of choice.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK430687/
B. Complete heart block is not a known complication
This patient's presentation is consistent with sarcoidosis. Complete heart block can occur when there is granulomatous involvement of the conducting system.
C. Hypocalcemia is common
This patient's presentation is consistent with sarcoidosis. Hypercalcemia and hypercalciuria are commonly seen.
D. Ulcerative colitis is associated with this disease
This patient's presentation is consistent with sarcoidosis. Anterior uveitis (not ulcerative collitis) is the presenting symptom in 5% of patients.

Q864 | STEM: A 48-year-old woman comes to the emergency department because she has facial, lip, and tongue swelling as well as voice changes that developed over the past hour. Current medications include lisinopril. She has no history of similar reactions in the past. Which of the following is the most likely cause of this patient's symptoms? | ANSWER: Bradykinin excess | EXPLANATION: Explanation
A. Bradykinin excess
This patient has signs of severe angioedema. There are two main pathophysiological causes of angioedema: histamine and bradykinin-mediated. Histamine-mediated angioedema, the most common form, is precipitated by IgE and is typically triggered by allergens such as foods and medications, but it can also be triggered by exercise, insect stings, or even extreme temperatures. Similar to allergy-induced anaphylaxis, medical therapy includes epinephrine, antihistamines, and corticosteroids.
Bradykinin-induced angioedema is typically drug-induced (classically by angiotensin-converting enzyme [ACE] inhibitors) or hereditary. Bradykinin acts as a vasoactive substance that causes swelling in the endothelial cells of the airway structures. Excess production, increased release, or reduced breakdown can all lead to bradykinin accumulation. In ad

SOURCE-DERIVED FALLBACK NOTES:
evere cases of HUS may. Neither condition results in abnormal coagulation studies.
Further information:https://www.ncbi.nlm.nih.gov/books/NBK556038/Shappell E, Miloslavsky EM. Chapter 105. Systemic Lupus Erythematosus and the Vasculitides. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen
Q888 | STEM: A 29-year-old female with a history of lupus presents to the emergency department with one week of progressive shortness of breath. She also reports several episodes of coughing up bloody sputum, low-grade fever, and joint swelling. Vital signs are: HR 105/min, RR 28/min, BP 145/90 mmHg, O_2 sat 91% on room air, and T 98.6ºF (37ºC). After the initial assessment, the patient acutely decompensates and begins coughing up large volumes of bright red blood. She is intubated for airway protection. The initial chest X-ray is shown below. What is the most critical next step in the management of this patient? | ANSWER: High-dose steroids | EXPLANATION: Explanation
A. Angiography
This patient likely has diffuse alveolar hemorrhage which is most commonly an autoimmune condition. Angiography would be helpful in traumatic injury but in

this case, the process is wide-spread and angiography is unlikely to be helpful.
B. Bronchoscopy
While bronchoscopy may be helpful in some cases of alveolar hemorrhage due to an isolated lesion such as a malignancy, this patient is likely suffering from auto-immune mediated diffuse alveolar hemorrhage. Steroids are the more appropriate treatment.
C. High-dose steroids
This patient likely has immune-mediated diffuse alveolar hemorrhage (DAH) and will require high-dose steroids. Diffuse alveolar hemorrhage is characterized by disruption of the alveolar-capillary membrane. Cough, hemoptysis, fever, and shortness of breath are the most common presenting signs and symptoms. Radiographic findings are characteristically diffuse patchy infiltrates. CT will show ground glass or consolidative opacities. Bronchoalveolar lavage is used for diagnosis. Angiography and bronchoscopy are unlikely to have a meaningful impact on this disease acutely.Source: Park MS. Diffuse Alveolar Hemorrhage. Tuberc Respir Dis. 2013;74(4)151-162. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3651925/.https://www.ncbi.nlm.nih.gov/books/NBK557515/
D. Left lateral decubitus position
The disease process is bilateral so placing the patie
Q889 | STEM: A 13-year-old boy presents to the emergency department for evaluation of rash and polyarthralgias for the past 3 days. The mother states that he is otherwise healthy except for a URI 2 weeks previous which resolved on its own. The patient endorses pain in the right elbow and left knee worse with movement and not relieved by rest. On physical exam the patient has a diffuse rosette-shaped pattern of macular-papular-hemorrhagic lesions concentrated on the ankles bilaterally, the dorsal aspect of the hands, and the buttocks (see image below). His right elbow and left knee are minimally swollen without overlying erythema, tenderness to palpation, or warmth to touch. His vitals are stable, he is afebrile and otherwise well appearing. You perform a CBC and BMP, which are within normal lim

SOURCE-DERIVED FALLBACK NOTES:
ith mucoid or watery urethral discharge and dysuria.
Reactive arthritis is defined by the classic triad of arthritis, urethritis, and conjunctivitis ("can't see, can't pee, can't climb a tree"). Typical presentation includes acute oligoarthritis of the lower extremities, sacroiliac joint, and lumbar spine. Reactive arthritis is very common in patients with HIV. Therefore, patients

with newly diagnosed reactive a
Q900 | STEM: A 68-year-old female presents to the emergency department with progressive left-sided headache over the past two weeks. The pain was gradual in onset but has progressed significantly and is now severe. It is particularly painful when the patient chews her food. Physical exam reveals an uncomfortable-appearing female, however, her physical exam, including a complete neurologic exam, is unremarkable other than tenderness to palpation over the lateral aspect of her forehead. Laboratory analysis, including a complete blood count and metabolic panel, is unremarkable other than an elevated erythrocyte sedimentation rate. Which of the following comorbidities is most likely to be found in this patient? | ANSWER: Polymyalgia rheumatica | EXPLANATION: Explanation
A. Lupus erythematosus
This patient’s presentation is concerning for temporal arteritis/giant cell arteritis. PMR occurs in about 50 percent of patients with GCA. Lupus is not associated with temporal arteritis.
B. Polymyalgia rheumatica
This patient's presentation is concerning for temporal arteritis/giant cell arteritis. PMR occurs in about 50 percent of patients with GCA. The cause is unknown, however, both PMR and GCA are associated with HLA-DR4. PMR is characterized by stiffness, joint pain, and muscle tenderness, as well as systemic findings such as weakness, depression, and low-grade fever. The initial treatment of choice for both diseases is systemic glucocorticoids.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK459376/Shappell E, Miloslavsky EM. Chapter 105. Systemic Lupus Erythematosus and the Vasculitides. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen’s Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
C. Polymyositis
This patient’s presentation is concerning for temporal arteritis/giant cell arteritis. PMR occurs in about 50 percent of patients with GCA. Polymyositis is not associated with temporal arteritis.
D. Raynaud's phenomenon
This patient’s presentation is concerning for temporal arteritis/giant cell arteritis. PMR occurs in about 50 percent of patients with GCA. While there are ca
Q901 | STEM: A 75-year-old female presents to the emergency department with severe right-sided headache over the past week. The headache was gradual in onset but has continued to worsen over the past several days. The patient reports

that her headache is particularly painful when chewing and she is unable to eat due to the pain. She denies fever, nausea, or vomiting. The physical exam is negative for any focal neurologic deficits but reveals tenderness to palpation on the lateral aspect of the forehead. Failure to treat this disease appropriately may lead to which of the following

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Immune System Disorders.

CHAPTER 14: NERVOUS SYSTEM DISORDERS

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
essionals From the American Heart Association/American Stroke Association [
Q1010 | STEM: A 28-year-old woman presents with frequent headaches for the past four months, occurring on more days of the month than not. She says the headaches feel band-like around her head and get worse as the day progresses. She does not wake up with a headache, nor do headaches wake her from sleep. Nonsteroidal anti-inflammatory drugs have relieved the pain, but she is worried about taking them so frequently. Vital signs and physical examination shows no abnormalities. Which of the following is the most appropriate prophylactic medication to prescribe for this patient? | ANSWER: Amitriptyline | EXPLANATION: Explanation
A. Amitriptyline
This patient has chronic tension headaches which are defined as greater than three months of tension headaches for ≥15 days/month. Amitriptyline is the most well-studied and proven efficacious medication for the prevention of tension headaches.Reference: https://www.ncbi.nlm.nih.gov/books/NBK562274/
B. Phenytoin
Phenytoin is used for seizure prophylaxis, not prophylaxis of tension headaches.
C. Propranolol
Propranolol is one of the most effective and most used medications for the prophylaxis of migraine headaches.
D. Sumatriptan
Sumatriptan is a medication used to treat migraine and cluster headaches. It is not used for prophylaxis of tension headaches.
Q1011 | STEM: A 10-year-old boy is brought to the emergency department by his parents because he has had worsening weakness in his extremities over the past 24 hours. He had a mild upper respiratory illness last week but seemed to recover. Vital signs are within normal limits. The patient appears well. Focused neurologic examination shows 3/5 strength in elbow and wrist flexion and extension, knee extension, and plantar and dorsiflexion. Decreased patellar reflexes and

numbness extending into the mid abdomen are noted. Which of the following test results is most likely to be expected in this patient? | ANSWER: Polymerase chain reaction test of a nasopharyngeal swab that is positive for enterovirus D68 | EXPLANATION: Explanation
A. Blood cultures that are positive for Staphylococcus aureus
Blood cultures would be obtained as part of this patient's evaluation but are unlikely to offer a diagnostic clue in the case of acute flaccid myelitis (AFM). Staphylococcus aureus would not be an expected cause of AFM.
B. Polymerase chain reaction test of a nasopharyngeal swab that is positive for enterovirus D68
This patient has symptoms and physical examination findings concerning for acute flaccid myelitis (AFM). AFM is a rare but serious condition that affects the nervous system, specifically the spinal cord, and can cause sudden muscle weakness, loss of reflexes, and sometimes paralysis. It is most commonly seen in children and often follows a respiratory or gastrointestinal illness.
In the 1960s, poliovirus was implicated in the epidemic outbreaks of this condition. However, poliovirus has been essentially eradicated. Outbreaks in the 2000s were tied to enterovirus D68. This virus is difficult to isolate but can be identified via nasopharyngeal swabs.
Imaging

SOURCE-DERIVED FALLBACK NOTES:
on studies and EMG, which play a large role in confirming more common variants of Guillain Barre, play a limited role in MFS. Elevated CSF fluid protein concentration helps confirm the diagnosis of MFS. The prognosis is generally excellent, as MFS is commonly self-limited.
The differential diagnosis for ophthalmoplegia/bulbar abnormalities most commonly include multiple sclerosis, botulism, myasthenia gravis, MFS, and intracranial pathology (hemorrhage/infarct/tumor/aneurysm).
Further information: https://www.ncbi.nlm.nih.gov/books/NBK507717/
B. Nerve conduction study
This patient is presenting with Miller-Fisher Syndrome (MFS), a variant of Guillain-Barre syndrome. Nerve conduction studies and EMG (electromyography), which play a large role in confirming more common variants of Guillain Barre, play a limited role in MFS.
C. Serum assay for botulinum toxin
This patient is presenting with Miller-Fisher Syndrome (MFS), a v

Q1036 | STEM: A 35-year-old male from Connecticut presents with facial weakness. He recently spent three months hiking the Appalachian Trail. He reports one week of gradually worsening symptoms. He is most concerned because he cannot close his eyes completely, and food falls out of his mouth when chewing. Prior to the onset of these symptoms, he had subjective fevers, myalgias, and a strange rash on his leg. Physical examination reveals decreased elevation of both eyebrows, incomplete closure of both eyes, flattening of the bilateral nasolabial folds, and inability to smile. The remainder of his neurologic examination is normal. What is the next best step in management? | ANSWER: Doxycycline | EXPLANATION: Explanation

A. Acyclovir

This patient is presenting with bilateral Bell's palsy, and the likely cause is Lyme disease. As such, acyclovir is unlikely to be helpful.

B. CT head

This patient is presenting with Bell's palsy. CT head is unnecessary in this case. It would be indicated if the patient had lower face involvement only and there was a concern for a stroke.

C. Doxycycline

This patient presents with bilateral Bell's palsy. A common acquired cause of this condition is Lyme disease. While Bell's palsy is most commonly unilateral and caused by a herpes virus infection, Lyme disease is unique in that it may cause a bilateral seventh nerve palsy, as in this patient.

Bell's palsy symptoms include abrupt onset facial paralysis with the forehead included, with possible loss of taste to the anterior two-thirds of the tongue, and hyperacusis. Bell's palsy secondary to Lyme disease is diagnosed via serologic titers and is treated with PO doxycycline.

Further information: Yang A, Dalal V. Bilateral Facial Palsy: A Clinical Approach. Cureus. 2021 Apr 25;13(4):e14671. PMID: 34079670.

D. IV ciprofloxacin

This patient is presenting with Bell's Palsy likely due to Lyme disease. Ciprofloxacin is unlikely to be helpful in this case.

Q1037 | STEM: A 47-year-old female with a history of diabetes presents to the emergency department with 1 week of a progressive severe headache. She describes the headache as a throbbing pain involving her entire forehead associated with nausea. She denies any fever, recent in

SOURCE-DERIVED FALLBACK NOTES:
gement | EXPLANATION: Explanation
A. Atrophy of the brain
This patient's presentation of gait impairment, cognitive deficits, and urinary incontinence is suggestive of normal pressure hydrocephalus (NPH). NPH is not characterized by brain atrophy.
B. Brain mass
This patient's presentation of gait impairment, cognitive deficits, and urinary incontinence is suggestive of normal pressure hydrocephalus (NPH). A brain mass is more likely to cause focal deficits.
C. Normal head CT
This patient's presentation of gait impairment, cognitive deficits, and urinary incontinence is suggestive of normal pressure hydrocephalus (NPH) which will not result in a normal head CT
D. Ventricular enlargement
Normal pressure hydrocephalus (NPH) is characterized by the following characteristics: wet, wacky, and wobbly. More specifically, it includes a triad of progressive dementia, ataxia, and urinary frequency or incontinence. It is a reversible cause of dementia and results from defective uptake of CSF. Head CT will reveal ventricular enlargement.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK542247/
Q1061 | STEM: An 11-year-old boy is brought to the emergency department because he has had a fever, headache, sensitivity to light, arthralgia in the small joints of the hands, and altered mentation for the past three days. His parents say their son was at Boy Scout camp last week in Ohio and came home with lots of noticeable bug bites. Temperature is 39.0°C (102.2°F). Encephalitis is suspected. Which of the following viruses is most likely causing the symptoms in this patient? | ANSWER: Arbovirus | EXPLANATION: Explanation
A. Arbovirus
This patient has suspected encephalitis. Arboviruses are a wide-ranging group of viruses that are spread by mosquito vectors, as well as ticks and sandflies. Examples in North America include eastern and western equine encephalitis, St. Louis encephalitis, and La Crosse and West Nile encephalitis. La Crosse encephalitis is common and can be severe in children. It usually presents with mild to severe meningoencephalitis symptoms (i.e., headache, fever, nuchal rigidity, photophobia, seizures, altered mentation), but subsequent neurologic

deterioration may occur.Reference: https://www.ncbi.nlm.nih.gov/books/NBK560866/
B. Coxsackievirus
Coxsackievirus virus is spread primarily by direct contact (e.g., respiratory droplets, hands) and not through an arthropod vector.
C. Enterovirus
Enterovirus is spread primarily by direct contact (e.g., respiratory droplets, hands) and not through an arthropod vector.
D. Paramyxovirus
Paramyxovirus (mumps virus) is spread primarily by direct contact (e.g., respiratory droplets, hands) and not through an arthropod vector.
Q1062 | STEM: A 68-year-old male with a history of diabetes and hypertension presents with right facial numbness and droop as well as right arm numbness/weakness for the past 3.5 hours. A CT head is performed on arrival and is negative for any acute pathology. Vital signs are: BP 195/105 mmHg, HR 90/min, RR 14/min, and O_2 sat 98%. Glucose is 410 mg/dL (22.6 mmol/L). Which of the following is a contraindication to tPA in this patient? | ANSWER: S

SOURCE-DERIVED FALLBACK NOTES:
al abscess, as he has evidence of intravenous drug abuse on physical exam, midline back pain, and elevated ESR. MRI of the whole spine with and without gadolinium is the gold standard for making the diagnosis (sensitivity and specificity >90%). Skip lesions, or noncontiguous abscesses, may occur in 15% of cases.
Further information: Long B, et al. High risk and low prevalence diseases: Spinal epidural abscess. Am J Emerg Med. 2022;53:168-172. PMID: 35063888
D. X-ray of the thoracic spine
The patient likely has a spinal epidural abscess, as he has evidence of intravenous drug abuse on physical exam, midline back pain, and elevated ESR. X-ray will not show an epidural abscess and will miss this patient's pathology.
Q1087 | STEM: A 65-year-old male is brought into the emergency department with weakness and difficulty speaking. He was last known well last night around 11 pm. His spouse states that when he did not get out of bed by 7 am, she checked on him, and he was unable to speak or move the right side of his body. Physical exam is significant for flaccid paralysis of his right upper and lower extremities. He is attempting to speak but is making incoherent sounds. The remainder of the motor and sensory exam is normal. Point of care blood glucose is 123 dL. CT head is

unremarkable with no evidence of hemorrhage. What is the most appropriate next step to be performed by the emergency medicine provider? | ANSWER: Computed tomography angiogram (CTA) head and neck | EXPLANATION: Explanation
A. Admit to medicine with neurology consult
This patient requires emergent additional workup with CT angiogram given symptoms concerning for large vessel occlusion. It is imperative that this be performed immediately as the patient has devastating impairments from his stroke and he is within the treatment window for thrombectomy.Admission to hospitalist with neurology consultation would be appropriate in the setting of transient ischemic attack (TIA) or an ischemic stroke outside the window for intravenous thrombolysis. In cases where the patient is appropriate for intravenous thrombolysis, neurology consultation should be obtained by the emergency medicine provider.
B. Computed tomography angiogram (CTA) head and neck
The patient's symptoms are concerning for a large vessel occlusion of the middle cerebral artery (MCA). CT head without contrast does not show evidence of hemorrhage, the next study in the emergency department should be a CT angiogram of the head and neck. CT angiogram of the head enables visualization of the occlusive thrombus in anticipation of a mechanical thrombectomy. In cases of large vessel occlusion, thrombectomy can be performed up to 24 hours after last known well. For those practicing in a hospital without endovascular capability sometimes intravenous thrombolysis is performed prior to transfer with anticipation of definitive thrombectomy a
Q1088 | STEM: A 55-year-old male with a history of diabetes and hypertension presents to the emergency department two hours after a sudden onset of right hand weakness and right-sided facial droop. He is able to provide a full history without difficulty. He is alert a

SOURCE-DERIVED FALLBACK NOTES:
reatment for this condition.
Tension-type headaches are the most common type of primary headaches. The average duration is 4 to 6 hours. The headaches are bilateral and described as dull, pressing, or band-like tightening. Routine physical activity, such as walking, does not exacerbate the pain. The pain is generally better in the morning and worsens later in the day.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK562274/
Q1102 | STEM: A 45-year-old female with a past medical history of migraine

headaches presents to the emergency department with a severe frontal headache. The patient reports that she was exercising this morning when she felt a sudden onset severe headache associated with nausea and vomiting. This occurred 8 hours ago. She notes that typically her migraines are occipital, gradual in onset, and fairly mild. Physical exam reveals an uncomfortable-appearing female with no focal neurologic deficits. CT scan of the brain is unremarkable. Which of the following studies is the next best step in establishing the diagnosis? | ANSWER: Lumbar puncture | EXPLANATION: Explanation
A. CT angiogram
This patient's history is concerning for subarachnoid hemorrhage. While a CT angiogram may be necessary to evaluate for aneurysms in the setting of a positive LP, the LP is necessary to evaluate for bleeding and to help distinguish between bleeding aneurysms and incidental findings of aneurysms.
B. Lumbar puncture
This patient's sudden onset headache brought on by activity is concerning for subarachnoid hemorrhage (SAH). ~99% of SAH cases are detected on head CT if performed within a 6-hour window. However, if the patient presents outside of the 6-hour window, a lumbar puncture is indicated.
Lumbar puncture allows for direct visualization of the cerebral spinal fluid (CSF) specifically looking for red blood cells in CSF. Red blood cell count that does not decline from tube 1 to tube 4 is diagnostic. If a traumatic lumbar puncture is suspected, then the presence of xanthochromia, which is the presence of bilirubin in the cerebral spinal fluid, can help differentiate traumatic tap versus subarachnoid blood. Xanthochromia will result in the CSF having a yellowish appearance.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK518975/
C. MR angiogram
This patient's history is concerning for subarachnoid hemorrhage. While a MR angiogram may be necessary to evaluate for aneurysms in the setting of a positive LP, the LP is necessary to evaluate for b
Q1103 | STEM: A 33-year-old man comes to the emergency department because of numbness and weakness of both legs for the past day and inability to walk. Two weeks ago, he was treated for atypical pneumonia with a course of antibiotics. Temperature is 37°C (98.6°F), pulse rate is 82/min, respirations are 16/min, and blood pressure is 138/82 mmHg. Physical examination shows bilateral lower extremity motor weakness and hyporeflexia. There is no saddle anesthesia, and

rectal tone is intact. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate treatment for this patient's condition? | ANSWER: Intravenous administr

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Nervous System Disorders.

CHAPTER 15: OBSTETRICS AND GYNECOLOGY

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
d be inappropriate.
B. Inform patient of complete abortion and schedule OB follow-up
While the patient may have had a complete abortion, the patient is below the discriminatory zone; therefore, this diagnosis can not be made at this point as the patient may have a viable pregnancy that is not visualized yet. A repeat HCG is needed within 48 hours.
C. Methotrexate
No ectopic pregnancy was visualized and patient is below the discriminatory zone. The diagnosis of ectopic pregnancy can not be made at this point so administering methotrexate would be inappropriate.
D. Recheck BHCG in 48 hours
The cause of bleeding and abdominal pain in this patient is indeterminate. She may have a normal intrauterine pregnancy, an unvisualized ectopic at this time, or may have had a spontaneous abortion. This patient is below the discriminatory zone, which is defined as a quantitative beta-hCG of 1500. Below this level, pregnancy may not be adequately visualized. This patient is clinically stable and does not need immediate intervention. Given these factors, the most appropriate next course of action is to recheck the beta-hCG. In normal pregnancie
Q1127 | STEM: A 46-year-old patient presents two weeks after being seen by her PCP for non-painful, heavy, and irregular menstrual periods for the past year. She comes today due to ongoing bleeding and hoping to get results from the prior visit. Records reveal a normal pelvic examination. Labs included a normal hCG, thyroid function, prolactin, liver, and coagulation studies. Complete blood cell count was normal except for mild anemia. The results of a Pap smear, endometrial sampling, and transvaginal ultrasound were all normal. Today, vital signs are normal. Pelvic exam reveals minimal dried blood in the vaginal vault. Hemoglobin is unchanged. Which of the following is the most appropriate next step in management? | ANSWER: Estrogen-progestin oral contraceptive pills | EXPLANATION: Explanation

A. Desmopressin
Desmopressin is used as a last resort to treat abnormal uterine bleeding in patients who have a coagulation disorder. It rapidly increases levels of von Willebrand factor and factor VIII for about 6 hours.
B. Dilation and curettage
Dilation and curettage is an appropriate diagnostic step to use in a patient who fails to respond to hormonal management for abnormal uterine bleeding to evaluate for structural/physical causes and for evaluation of the endometrial lining. However, a D&C is not typically used as a therapeutic measure for AUB because it has not been shown to be very effective for future menstrual cycles. Of the options listed, a hormonal contraceptive option are the first line of treatment for abnormal uterine bleeding.
C. Estrogen-progestin oral contraceptive pills
Hormonal contraception is the first-line treatment for abnormal uterine bleeding. Estrogen-progestin oral contraceptive pills work by suppressing endometrial development, re-establishing predictable bleeding patterns, decreasing menstrual blood flow, and lowering the risk of iron deficiency anemia.Further information: https://www.ncbi.nlm.nih.gov/books/NBK532913/
D. Tranexamic acid
Tranexamic acid prevents fibri

SOURCE-DERIVED FALLBACK NOTES:
ockberger RS, Gausche-Hill M, et al. Rosen’s Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
C. Trichomonas vaginalis
Trichomonas vaginalis is the caustive organism of trichomonas. Trichomoniasis can be diagnosed by identifying the mobile protozoan parasite on wet mount rather than clue cells.
D. Candida albicans
Candida albicans is a common cause of vaginal yeast infections. Budding yeast and hyphae may be seen on a wet mount.
Q1151 | STEM: A 23-year-old woman presents with vaginal pruritis and white cottage cheese-like discharge for four days. Which of the following conditions is the most likely, given this patient's clinical manifestations? | ANSWER: Candidiasis | EXPLANATION: Explanation
A. Candidiasis
This is the most common condition to present with white cottage discharge.

Trichomoniasis usually presents with green and frothy vaginal discharge. Gardnerella's presentation varies.
References: https://www.ncbi.nlm.nih.gov/books/NBK560624/McKinzie J. Chapter 84. Sexually Transmitted Infections. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen's Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
B. Gardnerella vaginosis
This bacteria causes BV and presents with grayish vaginal discharge that is sometimes frothy and often has a fishy odor.
C. Gonorrhea
Gonorrhea may be asymptomatic in women or may present with purulent discharge, vaginitis, and cervicitis.
D. Trichomoniasis
This presents with green, frothy discharge.
Q1152 | STEM: A 28-year-old gravida 1 para 1 woman, four weeks postpartum, presents to the emergency department with breast pain. She first noticed some mild swelling two days ago but came in today because she also developed a fever, body aches, fatigue, and severe tenderness to palpation of her left breast. The patient has been exclusively breastfeeding her newborn son and has noticed decreased milk production from the affected breast since her symptoms began. Vital signs are BP, 128/87 mmHg; P, 80/min; R, 16/min; and T, 38.3°C (101°F). Oxygen saturation is 98% on room air. On examination, the patient is non-toxic. The left breast is tender, warm, erythematous, and has a small area of fluctuance lateral to the nipple. An ultrasound is performed of the left breast with images shown below. In addition to antibiotics, which of the following is the next best step in management? | ANSWER: Incision and drainage | EXPLANATION: Explanation
A. Admission
While patients with large breast abscesses or signs of sepsis may require admission to the hospital, this patient is stable. Provided that incision and drainage can take place in the emergency department, admission is not required.
B. Discontinue breast-feeding
In order to promote the movement of blockages through the breast ducts, patients are encouraged to continue breastfeeding throughout the mastitis spectrum. To prevent transmission of infection to the child, women are commonly recommended to use a breast pump when fever, cellulitis, or abscess are involved.

C. Incision and drainage
This patient's presentation suggests lactational mastitis, and the ultrasound i

SOURCE-DERIVED FALLBACK NOTES:
which is within the discriminatory zone needed to identify an intrauterine pregnancy on transvaginal ultrasound but not transabdominal ultrasound. Ectopic pregnancy should still be strongly suspected, and a transvaginal ultrasound should be ordered next.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK539860/Henn M, Lall M. Chapter 173. Complications of Pregnancy. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen’s Emergency Medicine Concepts and Clinical Pr
Q1177 | STEM: A 24-year-old pregnant female at 22 weeks gestation with a history of migraine presents to the emergency department with 48 hours of headache. The headache came on gradually but has continued to progress, prompting her to seek evaluation. She denies fever, infectious symptoms, or trauma. Temperature is 37.1°C (98.8°F), pulse rate is 85/min, respirations are 14/min, and blood pressure is 145/80 mmHg. Oxygen saturation is 98% on room air. Physical exam, including a complete neurologic exam, is unremarkable. The fetal heart rate is 140/min. The patient is given acetaminophen in triage with the resolution of her headache and is completely asymptomatic at this time. Which of the following is the most appropriate next step in management? | ANSWER: Urinalysis | EXPLANATION: Explanation
A. CT head
This patient's elevated blood pressure is most concerning for pre-eclampsia and should prompt further laboratory studies.
B. Discharge home with supportive care
This patient's elevated blood pressure is concerning for pre-eclampsia and should prompt further studies.
C. MRI brain
This patient's elevated blood pressure is most concerning for pre-eclampsia and should prompt further studies.
D. Urinalysis
Preeclampsia should be a consideration in all pregnant patients presenting with headaches. The diagnostic workup should include a urinalysis to evaluate the presence of proteinuria, a complete blood count to evaluate for thrombocytopenia, and a complete metabolic panel to assess for impaired liver function and renal function. Elevated blood pressure with proteinuria is typically

thought to be diagnostic of preeclampsia, but it may not always be present. For those with new-onset hypertension but no proteinuria, other new-onset symptoms such as thrombocytopenia, renal insufficiency, pulmonary edema, impaired liver function, or new-onset headache may be used for diagnosis. Further information: https://www.ncbi.nlm.nih.gov/books/NBK570611/
Q1178 | STEM: A 15-year-old female comes to the emergency department with abdominal pain, nausea, and vomiting for the past 8 hours. She has no significant past medical history but recently discovered she is pregnant. Temperature is 37.1°C (98.8°F), pulse rate is 105/min, respirations are 22/min, and blood pressure is 95/50 mmHg. Physical exam demonstrates moderate suprapubic abdominal pain without rebound or guarding. The cervical OS is closed, and there is no vaginal bleeding. Serum beta-HCG is 5,000 mIU/mL. Your bedside ultrasound (shown below) does not identify a gestational sac. Which of the following is the most appropriate next step in management? | ANSWER: Insert two large bore IV's, type and screen, a

SOURCE-DERIVED FALLBACK NOTES:
brown purulent vaginal discharge and moderate left adnexal tenderness. Urinalysis and urine pregnancy tests are both negative. Pelvic ultrasound is negative and shows intact vascular flow to both ovaries. Which of the following is the most appropriate management of this patient? | ANSWER: Admit and administer cefoxitin 2g IV q6 hours and doxycycline 100mg PO/IV q12 hours | EXPLANATION: Explanation
A. Admit and administer cefoxitin 2g IV q6 hours and doxycycline 100mg PO/IV q12 hours
This patient has the presumptive diagnosis of pelvic inflammatory disease (PID). As recommended by the Centers for Disease Control and Prevention, empiric treatment of PID should be initiated when a sexually active young woman presents with lower abdominal/pelvic pain and no other identifiable cause of illness other than PID + one or more of the following: 1) cervical motion tenderness; 2) uterine tenderness; 3) adnexal tenderness. A temperature greater than 101F (38.3C) and the presence of purulent vaginal discharge increase the likelihood of PID. Inpatient versus outpatient treatment depends on pregnancy status, clinical presentation, and the reliability of follow-up. Patients should be admitted for IV antibiotics if pregnant, fever, suspected pelvic abscess/ tubo-ovarian abscess (TOA), persistent nausea and vomiting, or poor follow-up/ medication compliance. The patient in this case is homeless, with questionable

follow-up, and unable to tolerate fluids, thus it is best to admit for IV antibiotics. Other CDC recommended empiric antibiotic regimens include:ceftriaxone 1g q24 hours + doxycycline 100mg q12 hours + metronidazole 500mg q12 hoursORcefotetan 2g q12 hours & doxycycline 100mg q12 hours
Further information: https://www.ncbi.nlm.nih.gov/books/NBK499959/McKinzie J. Chapter
Q1205 | STEM: A 25-year-old woman comes to the emergency department because she has had worsening right lower quadrant and pelvic pain as well as nausea and fever over the past three days. Temperature is 39.0°C (102.2°F), pulse rate is 106/min, respirations are 18/min, and blood pressure is 108/68 mmHg. The patient appears ill. Physical examination shows bilateral tenderness to palpation in the lower quadrant, with the right quadrant being more tender than the left. Pelvic examination shows mucopurulent discharge and severe unilateral right adnexal tenderness with fullness on bimanual examination. CT scan of the abdomen and pelvis shows an appendiceal diameter of 9 mm, complex right adnexal mass with internal echoes, left fallopian tube diameter measuring 8 mm, right fallopian tube diameter measuring 12 mm, and moderate free pelvic fluid. Which of the following is the most likely diagnosis? | ANSWER: Tubo-ovarian abscess | EXPLANATION: Explanation
A. Appendicitis
Both pelvic inflammatory disease (PID) and acute appendicitis can have similar and overlapping findings on CT scan. This patient does not have left fallopian tube enlargement suggestive of PID and has classic findings of tubo-ovarian abscess including a complex right adnexal mass with internal echoes and free pelvic fluid. The appendiceal inflammation is secondary and not due to acute appendicitis.

SOURCE-DERIVED FALLBACK NOTES:
oes require treatment of the partner as well as the patient.
D. Vaginal pH is typically >4.5
Trichomonas vaginalis is the causative agent of trichomoniasis. It is the most common non-viral sexually transmitted infection worldwide. Patients typically complain of yellow-green, frothy, and malodorous vaginal discharge, with protozoa seen on wet mount. "Strawberry cervix," or punctate hemorrhage around the cervix, is seen on exam. Vaginal pH becomes more alkalotic and is generally greater than 4.5. Treatment of trichomonas vaginitis is metronidazole, and partners should be treated as well.
Further information:McKinzie J. Chapter 84. Sexually Transmitted Infections. In:

Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen's Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelp
Q1212 | STEM: A 32-year-old woman, gravida 3, para 3, currently two months postpartum from a normal vaginal delivery, presents with right breast pain and erythema. She is breastfeeding her infant. Yesterday, she started having body aches and felt like she was coming down with the flu. Last night, she started to feel feverish, and she developed chills. After her shower this morning, she noticed a red, warm area on the upper outer aspect of her right breast that is tender to touch. The pain has worsened throughout the day. Temperature is 37.8°C (100.2°F), pulse rate is 72/min, respirations are 16/min, and blood pressure is 124/80 mmHg. Which of the following organisms is most commonly implicated in her suspected condition? | ANSWER: Staphylococcus aureus | EXPLANATION: Explanation
A. Escherichia coli
E. coli is a potential cause of mastitis but is not the most common. Staph aureus is the most common causative organism found in both mastitis and breast abscesses.
B. Staphylococcus aureus
S. aureus is the most common causative organism found in both mastitis and breast abscesses. S. aureus is a common bacteria that colonizes the skin.Reference: https://www.ncbi.nlm.nih.gov/books/NBK557782/
C. Streptococcus pneumoniae
Streptococcus pneumoniae is a potentional cause of mastitis, but is not the most common. Staph aureus is the most common causative organism found in both mastitis and breast abscesses.
D. Streptococcus pyogenes
Streptococcus pyogenes is a potential cause of mastitis but is not the most common. Staph aureus is the most common causative organism found in both mastitis and breast abscesses.
Q1213 | STEM: A 19-year-old female presents to the emergency department complaining of lower abdominal pain and fever for three weeks. She is a poor historian but admits to occasional unprotected sex and explains that she had similar but milder symptoms three weeks ago. At the time was given a "shot" in the arm and some pills at an outside clinic. She admits to taking the pills for only three days and stopped them because she was feeling better. Vital signs are: HR 112/min, BP 112/76 mmHg, RR 16/min, T 38.4ºC (101.1ºF), oxygen sat 100% RA. On

physical exam, she has exquisite symmetric lower abdominal tenderness and cervical motion tenderness. On speculum exam, you note white discharge from the cervix. She vomits twice during your inte

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Obstetrics and Gynecology.

CHAPTER 16: PSYCHOBEHAVIORAL DISORDERS

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
d re-emphasize the outpatient follow-up.
An actively suicidal patient would not be appropriate for outpatient follow-up.
C. Physically and chemically restrain the patient for self-harming behavior
The decision to restrain someone is often difficult and should not be taken lightly. The major driver in this decision is will restraints prevent harm to this patient or others. In this case, the patient has become agitated to the point of self-injurious behavior and should be restrained not only for his own safety but also for that of the hospital staff and other patients. The prior psychiatrist evaluation should not impact your decision. Psychiatric illness, like medical illness, can be dynamic and require differing levels of emergent intervention.
For more information: Guerrero P, Mycyk MB. Physical and Chemical Restraints (an Update). Emerg Med Clin North Am. 2020;38(2):437-451. PMID: 32336335
D. Place another emergent psychiatry consult
Placing another consult should not delay controlling this patients potentially harmful behavior.
Q1235 | STEM: A mother brings her 5-week-old daughter to the emergency department for "trouble sleeping." The infant's vitals and exam are unremarkable but the mother breaks down into tears and confesses she has not been able to sleep for over two weeks despite feeling incredibly tired. She does not even enjoy caring for her new baby or showing her to her family. She feels she is too tired to enjoy anything anymore. She is suffering from what disorder? | ANSWER: Postpartum depression | EXPLANATION: Explanation
A. Major depressive disorder
Although the patient fulfills the diagnostic criteria for major depressive disorder, because her onset of symptoms is within six weeks of delivery, the WHO classifies her condition as postpartum depression.
B. Postpartum blues
This patient's symptoms are too severe (with changes in sleep and energy level plus anhedonia) and have persisted for too long (over two weeks) to be considered

postpartum blues. Women experiencing baby blues tend to recover quickly, and PPD tends to last longer and severely affects the ability to return to normal function.
C. Postpartum psychosis
This patient is not exhibiting any symptoms of psychosis such as delusions, hallucinations, thought disorganization or disorganized behavior.
D. Postpartum depression
This patient's symptoms include depressed mood, anhedonia, sleep disturbances, and decreased energy level with a duration of greater than two weeks within six weeks of delivering her baby. She is suffering from postpartum depression and should be referred for a psychiatric evaluation. First-line treatment is psychotherapy and antidepressant medications.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK519070/
Q1236 | STEM: A 61-year-old man comes to the emergency department because he has had a headache and palpitations for the past 24 hours. Temperature is 37.0°C (98.6°F), pulse rate is 109/min and regular, and blood pressure is 168/72 mmHg. Oxygen saturation is 99% on room air. The patient appears disheveled. Physical examination shows slight diaphoresis and hand tremors. Pupils are equal, round, and reactive t

SOURCE-DERIVED FALLBACK NOTES:
mplaints. There can be multiple complaints and patients will have had multiple encounters with other providers without a clear diagnosis.
Q1260 | STEM: A 29-year-old man is brought to the emergency room by the police for a blood alcohol level check. The patient was arrested under suspicion of drunk driving after having run a red light and striking another vehicle. He refused a breathalyzer and is verbally abusive with staff. He appears to have no remorse regarding injuring the driver of the other vehicle. Upon further questioning, he states that alcohol is not really a problem for him and he can quit at any time. He denies feeling anxious or jittery whenever he has chosen to reduce his level of intake. He has had one other DUI in the past and feels like the police are targeting him. The patient endorses drinking tonight after being fired from his second job in six months. He believes he was really fired because his boss was a 'jerk.' He reports drinking only a few beers every other day during the week but will drink up to 20 drinks over the weekend and has done this for 8 years. The patient has also been arrested twice for assault and spent a portion of his youth in juvenile hall for violent behavior. What is the most likely diagnosis? | ANSWER: Antisocial

personality disorder | EXPLANATION: Explanation
A. Antisocial personality disorder
Patients with an antisocial personality disorder have a pattern of disregard for and violation of the rights of others. It is a deeply ingrained and rigid dysfunctional thought process that focuses on social irresponsibility with exploitive and criminal behavior and lack of remorse. Manifestations include failure to conform to the law, inability to sustain employment, manipulation for personal gain, and incapacity to form stable relationships.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK546673/
B. Bipolar disorder
Bipolar disorder is characterized by episodes of mania or hypomania as well as depressive episodes in some individuals. Bipolar I occurs when patients experience full manic episodes (7 days). Bipolar II occurs when individuals experience hypomania (4 days). This individual's history is not consistent with bipolar disorder.
C. Narcissistic personality disorder
Narcissistic personality disorder is a pattern of grandiosity, need for admiration, and lack of empathy.
D. Paranoid personality disorder
Paranoid personality disorder is a pattern of distrust and suspiciousness such that others' motives are interpreted as malevolent.
Q1261 | STEM: A male neonate who was born at full term at home 24 hours ago is brought to the emergency department via ambulance because he has poor feeding and loose stools. The patient's mother is disheveled and has multiple linear scars on both of her arms. Physical examination of the patient shows a high-pitched cry as well as sneezing and yawning. Which of the following is the most appropriate pharmacologic therapy for this patient? | ANSWER: Morphine | EXPLANATION: Explanation
A. Clonidine
Clonidine is an alpha-2 agonist that can be a useful adjunct in opioid withdrawal. It is used as a second medication in infants with severe ne

SOURCE-DERIVED FALLBACK NOTES:
ion. The remainder of the physical exam is normal. B-scan ocular ultrasound, CT scan, and labs are unremarkable. An emergent evaluation by the ophthalmologist also is unremarkable. What is the most likely cause of this patient’s vision loss? | ANSWER: Conversion disorder | EXPLANATION: Explanation

A. Conversion disorder
Conversion disorder is typically characterized by the sudden and often dramatic onset of a single nonpainful neurologic disorder. Any known organic etiology cannot explain the symptom. Conversion disorder will often be associated with recent severe stressors or conflict. The most common presentations include pseudoseizures, paralysis, numbness, aphonia, coordination disturbance, blindness, and tunnel vision. This disorder has a female predominance and typically appears in adolescence and early adulthood. Patients with conversion disorder have a nondeliberate symptom and will have a flatter affect than expected under the circumstances, termed la belle indifference. This is a diagnosis of exclusion, and all organic etiologies must be ruled out. The treatment involves identifying the stressor and addressing the underlying issue.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK551567/
B. Malingering
Malingering is typically found in connection with antisocial personality disorder. Patients will fabricate symptoms to attain 'secondary gain' and receive financial compensation, avoid military service, school, or work.
C. Retinal detachement
Patient will typically complain of progressively worsening vision loss (curtain-like) associated with a shower of floaters or flashes of light. B-san will show funnel-shaped abnormality.
D. Somatization
Q1279 | STEM: A 63-year-old homeless male with a history of polysubstance abuse presents to the emergency department with confusion and agitation. Temperature is 37.1°C (98.8°F), pulse rate is 133/min, respirations are 20/min, and blood pressure is 175/90 mmHg. Oxygen saturation is 98% on room air. On exam, the patient is agitated, tremulous with tongue fasciculations, and appears to be responding to internal stimuli. Over the next hour, the patient receives several doses of IV lorazepam with only minimal improvement in his symptoms. Which of the following medications would be the next appropriate agent in this patient's management? | ANSWER: Phenobarbital | EXPLANATION: Explanation
A. Haloperidol
Haloperidol often helps in the treatment of severely agitated patients or those with psychosis. Although this patient appears to be agitated and responding to internal stimuli, this is more likely a result of underlying alcohol withdrawal and should be treated as such.

B. Leviteracitam
Leviteracitam is an anti-epileptic. While this patient is at risk for developing seizure given his alcohol withdrawal, leviteracitam is not the appropriate next agent.
Phenobarbital is indicated in alcohol withdrawal for patients refractory to benzodiazepines.
C. Midazolam
This patient has already received high doses of benzodiazepines without improvement. Addition of a second class of medications is warranted at this time.
D. Phenobarbital
This patient is showing classic signs of d

7. Arabic rapid-review summary

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8. Source-question coverage map

This chapter consolidates all source questions assigned to Psychobehavioral Disorders.

CHAPTER 17: RENAL AND UROGENITAL DISORDERS

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
pulse rate is 100/min, and blood pressure is 92/60 mmHg. On examination, there is bilateral periorbital edema. No rashes are present. Examination of a urine sample shows it to be foamy. Serum albumin is 2.0 g/dL (normal 3.4 to 5.5 g/dL). Which of the following findings on dipstick urinalysis is most likely to be seen in this patient? | ANSWER: Proteinuria | EXPLANATION: Explanation
A. Bilirubinuria
Bilirubinuria is a sign of liver disease, and reflects elevated direct bilirubin in the serum.
B. Hematuria
Hematuria has several causes, including glomerular disease, renal cell carcinoma, malignant hypertension, nephrolithiasis, bladder cancer, and cystitis. Glomerular disease should be suspected when there is hematuria with red cell casts, 'cola colored' urine, and proteinuria.
C. Ketonuria
Ketonuria is an indication that the body is breaking down fatty acids for energy. This can be present in starvation, diabetic ketoacidosis, and hyperthyroidism.
D. Proteinuria
This patient has suspected nephrotic syndrome. Heavy proteinuria, hypoalbuminemia, edema, and lipiduria are the main characteristics of nephrotic syndrome. There are many causes of nephrotic syndrome, including minimal-change nephropathy (common cause in children), focal glomerulosclerosis, diabetes mellitus, and SLE. Treatment involves a low salt diet, ACE inhibitor or ARB, and loop diuretics.Further information: https://www.ncbi.nlm.nih.gov/books/NBK470444/
Q1303 | STEM: A 20-year-old male college football player presents with severe muscle pain in his thighs and lower back, weakness, and dark urine. Earlier in the afternoon, the patient and his teammate had an intense weightlifting workout that he states "almost made me pass out." He has no significant past medical history or allergies. He denied taking any performance-enhancing drugs or

supplements. He has a urine cup with brown urine at the bedside. The point-of-care urine dipstick is positive for blood but negative otherwise. What is the most appropriate initial step in the management of this patient? | ANSWER: Complete metabolic panel, urinalysis with microscopy, creatinine kinase, IV normal saline 2 liter bolus | EXPLANATION: Explanation
A. CBC, complete metabolic panel, urinalysis with microscopy, IV normal saline 2 liter bolus
It is important to get a creatinine kinase level on this patient with a classic presentation for rhabdomyolysis.
B. CBC, complete metabolic panel, urinalysis with microscopy, creatinine kinase, sodium bicarbonate in 1L D5W
This patient likely has rhabdomyolysis. No studies have shown alkaline diuresis to be more effective than saline diuresis, and as such bicarbonate is not indicated as an initial treatment in the emergency department.
C. CBC, complete metabolic panel, urinalysis with microscopy, tamsulosin, IV normal saline 2L bolus
This patient's history is most suggestive of rhabdomyolysis, not nephrolithiasis. Tamsulosin may be considered in cases of renal stone.
D. Complete metabolic panel, urinalysis with microscopy, creatinine kinase, IV normal saline 2 liter bolus
This patient's presentation is likely the result of rhabdomyolysis. The urine dipstick interprets myog

SOURCE-DERIVED FALLBACK NOTES:
testicular masses or pain on palpation is noted. Which of the following complications is most likely if this patient's condition is not surgically corrected? | ANSWER: Decreased fertility | EXPLANATION: Explanation
A. Decreased bone mass
While osteopenia is possible if a severe varicocele causes decreased Leydig cell function and low testosterone, this is not a common effect,
B. Decreased fertility
An unrepaired varicocele can impair sperm structure, function, and production and cause impaired fertility. It is thought that ~ 40% of infertile men have varicoceles.
References:https://www.ncbi.nlm.nih.gov/books/NBK448113/Germann CA. Chapter 85. Urologic Disorders. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen’s Emergency Medicine Concepts and Clinical Practice. 10th

ed. Philadelphia: Elsevier; 2023.
C. Testicular cancer
Varicocele is not a risk factor for testicular cancer; however, cryptorchidism, or an undescended testicle, is.
D. Urinary retention
Structural changes associated with varicoceles in the testicles do not cause abnormalities in urinary function.
Q1327 | STEM: A 55-year-old male with a past medical history significant for hypertension and depression presents with altered mental status. Per the patient's wife, the patient admitted to overdosing in an attempt to commit suicide; however, he would not admit to what substance. The patient exhibits slurred, incoherent speech and is unable to provide a history. Temperature is 37.1°C (98.8°F), pulse rate is 115/min, respirations are 31/min, and blood pressure is 179/93 mmHg. Oxygen saturation is 98% on room air. The physical exam is unremarkable. Basic metabolic panel reveals the following: Na 145 mEq/L, K 3.5 mEq/L, Cl 105 mEq/L, HCO_3 15 mEq/L, BUN 20 mg/dL, Cr 0.8 mg/dL. Urinalysis shows trace hematuria and oxalate crystals. Which of the following agents would be appropriate at this time? | ANSWER: Fomepizole | EXPLANATION: Explanation
A. Fomepizole
This patient presents with an unknown ingestion and an anion gap metabolic acidosis. The differential for anion gap metabolic acidosis can be remembered by the mnemonic MUDPILES: methanol, uremia, diabetic ketoacidosis, paraldehyde, isoniazid or iron, lactic acidosis, ethylene glycol, and salicylates. The presence of oxalate crystals on urine microscopy suggests ethylene glycol toxicity. Of note, this is a late finding and is only present in roughly 50% of cases. Symptoms of ethylene glycol poisoning include altered mental status, slurred speech, and ataxia, followed by cardiopulmonary findings such as hypertension, tachycardia, and ARDS. Pharmacotherapy is directed at preventing the further production of toxic metabolites. Fomepizole is the antidote for toxic alcohols, and it acts by inhibiting alcohol dehydrogenase to cease toxic alcohol metabolism. Ethylene glycol is dialyzable; however, with the use of fomepizole, dialysis is typically not indicated in the absence of renal dysfunction.
Further information:Cohen JP, Quan D. Chapter 185. Alcohols. In: Tintinalli JE, Ma OJ, Yealy DM, Meckler GD, Stapczynski JS, Cline DM, Thomas SH, eds. Tintinalli's Emergency Medicine: A Comprehensive Study Guide,

SOURCE-DERIVED FALLBACK NOTES:
ria, nephrotic syndrome, or azotemia.
D. Post-streptococcal glomerulonephritis
Post-streptococcal glomerulonephritis occurs 1-2 weeks following pharyngitis or 3-6 weeks following impetigo and is caused by a nephritogenic strain of group A Streptococcus. It comm
Q1352 | STEM: A 17-year-old uncircumcised male presents complaining of penile pain and nausea for the past day. He states that this has never happened before, but he is unable to put the foreskin back. Upon physical examination, you discover the foreskin trapped in a retracted position. Which of the following is the most likely based on this patient's symptoms and physical exam findings? | ANSWER: Paraphimosis | EXPLANATION: Explanation
A. Balanitis
Balanitis is inflammation of the glans penis and foreskin. It is characterized by pain, redness, swelling, and sometimes discharge or an unpleasant odor. Paraphimosis is the term for retracted foreskin in an uncircumcised male that can not be returned to its normal position.
B. Gonococcal infection
In males, gonococcal infection most commonly causes urethritis and epididymitis. Patients will have copious, purulent urethral discharge. More severe disease can cause penile edema and periurethral abscesses. Epididymitis presents as unilateral testicular pain and swelling.
C. Paraphimosis
Paraphimosis is retracted foreskin in an uncircumcised male that can not be returned to its normal position. Swelling of the head of the penis can occur, as venous flow is constricted. This is a true medical emergency if persistent and if signs of decreased blood flow are present, as it can result in penile necrosis and gangrene.Reference: https://www.ncbi.nlm.nih.gov/books/NBK459233/
D. Phimosis
Phimosis is the inability to retract the foreskin in an uncircumcised male. This can be normal in young children. Presence of phimosis or partial phimosis is the greatest risk factor for paraphimosis.
Q1353 | STEM: Which of the following organisms is the most common cause of uncomplicated lower urinary tract infections in adults? | ANSWER: Escherichia coli | EXPLANATION: Explanation
A. Enterobacter aerogenes

Enterobacter aerogenes is a Gram-negative rod, and is usually found in the gastrointestinal tract. In general, it does not cause disease in healthy individuals. However, it is an important nosocomial pathogen and can cause bacteremia, lower respiratory tract infections, skin and soft-tissue infections, urinary tract infections, endocarditis, intra-abdominal infections, septic arthritis, osteomyelitis, central nervous system, and ophthalmic infections.
B. Escherichia coli
Most urinary tract infections result from uropathogens that reside in the gut that contaminate the periurethral area and are able to colonize the urethra. The ascent of these bacteria results in the migration of these organisms into the bladder. A limited number of serotypes of Escherichia coli are responsible for the majority of urinary tract infections.
Reference:Germann CA. Chapter 85. Urologic Disorders. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen's Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
C. Mycopla

SOURCE-DERIVED FALLBACK NOTES:
oncepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
C. Postrenal acute renal injury
This patient has no obstructive signs.
D. Prerenal acute renal injury
While elevation of BUN and creatinine indicate general acute kidney injury, the appearance of casts and the ratio of BUN to creatinine of less than 20:1 indicates intrarenal renal failure. Potential etiologies include intrinsic AKI due to hypotension caused by dehydration or possibly rhabdomyolysis given the patient's extensive bruising.
Q1377 | STEM: A 31-year-old male with no known past medical history presents with penile pain and "stuff around the tip." He is uncircumcised. His vitals are within normal limits. Retraction of the foreskin reveals an erythematous and slightly swollen glans and foreskin with a thick white discharge within the foreskin that is not expressed from the urethra with manipulation. Aside from providing instructions on proper foreskin hygiene and a prescription for a topical anti-fungal, what is the most important step in management for this patient? | ANSWER: Check blood glucose and HIV status | EXPLANATION: Explanation
A. Check CD4 count
There is no mention of this patient being HIV+ or otherwise immunocompromised.

This is not a necessary test at this encounter. It would certainly be reasonable to check an HIV test, however.
B. Check blood glucose and HIV status
Balanitis is inflammation of the glans penis, while balanoposthitis involves both the glans and foreskin. In adults, the most common etiology of balanoposthitis is poor hygiene. However, it can be the presenting symptom of diabetes mellitus and HIV. Be sure to ask about polyuria and polydipsia, as well as sexual history, when appropriate.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK537143/
C. Check urinalysis
This patient has no complaint of dysuria and as such a urinalysis is unlikely to be helpful.
D. Culture discharge
It is reasonable to culture the discharge if the patient has recurrent balanoposthitis or symptoms refractory to topical azole treatment. It is not standard of care for a simple or first presentation.
Q1378 | STEM: A 61-year-old man presents with suprapubic fullness and discomfort, and an inability to urinate for nearly 24 hours. He has had increased urinary frequency, particularly at night, for the past few months. He has no history of renal stones, weight loss, or urinary tract infections. His blood pressure is 152/84 mmHg and the rest of his vital signs are within normal limits. He has a non-tender soft abdomen with a palpable bladder and 380 mL of urine in his bladder on bedside ultrasound after recently voiding. He has a normal neurological exam and large, firm prostate on the rectal exam. His creatinine is 0.88 mg/dL. What is the most appropriate management of this patient? | ANSWER: Place indwelling urinary catheter with attached leg bag, start tamsulosin and urology follow up outpatient | EXPLANATION: Explanation
A. Admit for catheter placement and biopsy
This patient's acute urinary retention and enlarged prostate are most likely due to benign prostatic hyperplasia. He may eventually require a biopsy or urologic inte

> **EXAM TRICK / تريكة امتحانية**
> SOURCE-DERIVED FALLBACK NOTES:
> Topic: Renal and Urogenital Disorders. Deduplicate these HIPPO QBank items into compact study notes. Use headings: Core concepts and pathophysiology; Recognition and differential; Diagnostics; Emergency management;

> Procedures/drugs/disposition; Exam tricks and pitfalls; Arabic take-home. Integrate useful source references present in the explanations. Do not repeat individual MCQs.

Q1384 | STEM: A 34-year-old man comes to the emergency department because he has had difficulty urinating as well as penile pain for the past day. He has no history of sexually transmitted infections or trauma to the penis. The patient is sexually active on a daily basis, and his most recent intercourse was approximately 18 hours ago. Which of the following is most likely to confirm the diagnosis? | ANSWER: Compression and repositioning of the foreskin | EXPLANATION: Explanation
A. Complete panel of sexually transmitted infections testing
Paraphimosis is a functional issue that needs to be corrected by manual decompression. STI testing is not indicated, as he has no symptoms of infection.
B. Compression and repositioning of the foreskin
This is both diagnostic and therapeutic for paraphimosis, which occurs when the foreskin is trapped behind the glans, compressing blood and urinary flow. For mild, uncomplicated paraphimosis, manual reduction of the foreskin over the glans of the penis can be achieved without analgesia or sedation. Alternatively, the glans can be compressed by wrapping it circumferentially with a 2-inch elastic bandage for five minutes.Reference: https://www.ncbi.nlm.nih.gov/books/NBK459233/
C. Needle aspiration of the glans penis
This patient has paraphimosis, a urologic emergency where the foreskin becomes trapped behind the glans penis. First-line management includes noninvasive procedures such as a manual reduction of the foreskin. When noninvasive procedures fail and a urologist is not available, advanced and invasive procedures are second line. Needle aspiration is an alternative invasive procedure to the penile dorsal slit procedure.
D. Penile dorsal slit procedure
This patient has paraphimosis, where the foreskin is retracted and causes strangulation of the glans penis. The initial management of this urologic emergen
Q1385 | STEM: Which of the following compositions of renal stones are usually the most radiolucent on a KUB plain film radiograph? | ANSWER: Uric acid | EXPLANATION: Explanation
A. Calcium oxalate

Calcium oxalate stones are relatively radiopaque. The order of decreasing radiopacity is calcium phosphate, calcium oxalate, struvite, cysteine, then uric acid.
B. Cysteine
The order of decreasing radiopacity is calcium phosphate, calcium oxalate, struvite, cysteine, then uric acid.
C. Struvite (ammonium magnesium phosphate)
The order of decreasing radiopacity is calcium phosphate, calcium oxalate, struvite, cysteine, then uric acid.
D. Uric acid
Uric acid stones are usually radiolucent and will not show on plain films. The order of decreasing radiopacity is calcium phosphate, calcium oxalate, struvite, cysteine, then uric acid.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK559227/Germann CA.

7. Arabic rapid-review summary

> تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Renal and Urogenital Disorders.

CHAPTER 18: SYSTEMIC INFECTIOUS DISORDERS

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1. Clinical overview and pathophysiology

SOURCE-DERIVED FALLBACK NOTES:
therapy. The antibiotic of choice is doxycycline.
Further information: The Clinical Assessment, Treatment, and Prevention of Lym
Q1406 | STEM: A 5-year-old female is brought in by her mother after multiple episodes of vomiting. She was diagnosed with influenza a week ago but seemed to be recovering well with over-the-counter antipyretics until today. Her vital signs are within normal limits, but she is confused. During the exam, she begins to seize. She has hepatomegaly, and her AST and ALT are markedly elevated. What other laboratory abnormalities should be expected in this patient? | ANSWER: Hypoglycemia, increased prothrombin time, metabolic acidosis | EXPLANATION: Explanation
A. Hyperglycemia, increased prothrombin time, metabolic acidosis
This patient's presentation is classic for Reye syndrome, which leads to hypoglycemia as opposed to hyperglycemia.
B. Hyperkalemia, hypoglycemia, metabolic acidosis
Hyperkalemia can accompany metabolic acidosis in certain scenarios, but it is not commonly associated with Reye syndrome.
C. Hypoglycemia, increased prothrombin time, metabolic acidosis
This patient's presentation is classic for Reye syndrome, which is brought on by aspirin use during or following a viral infection, most commonly influenza or varicella. Patients often have hepatomegaly, markedly elevated aminotransferase activity, hypoglycemia, increased PT, hyperammonemia, and metabolic acidosis.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK526101/Seethala R, Takhar S, Bullard-Berent J, Banks L. Chapter 119. Viruses. In: Walls RM, Hockberger RS, Gausche-Hill M, et al. Rosen’s Emergency Medicine Concepts and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
D. Metabolic acidosis, respiratory alkalosis, increased prothrombin time
A mixed metabolic acidosis and respiratory alkalosis is associated with acute salicylate toxicity. There is no known history in this case of a toxic level ingestion of aspirin or other salicylates, only a suggestion of possible aspirin use (OTC

symptom control) in the previous days.
Q1407 | STEM: A 35-year-old female presents to the emergency department with cough, fever, and foul-smelling sputum progressing over several days. The patient reports that approximately one week before the presentation she saw her primary doctor and was diagnosed with influenza. A chest x-ray is shown. What is the most common organism involved in this patient's presentation? | ANSWER: Streptococcus pneumoniae | EXPLANATION: Explanation
A. Bacteroides fragilis
B. fragilis is a common bacterial etiology of aspiration pneumonia, not for superinfection of viral pneumonia.
B. Klebsiella pneumoniae
The most common cause of post-influenza pneumonia is Streptococcus pneumoniae. Klebsiella pneumoniae can cause post-influenza pneumonia, but less commonly.
C. Staphylococcus aureus
While S. aureus, particularly community-acquired MRSA, should be considered in a patient with post-influenza pneumonia, S. pneumoniae is the most common pathogen in this case.
D. Streptococcus pneumoniae
The most common cause of post-influenza pneumonia is a bacterial infection, particularly Streptococcus pneumoniae.

SOURCE-DERIVED FALLBACK NOTES:
urther information: Acharya A, et al. Bacteriology and antibiotic susceptibility pattern of peritonsillar abscess. JNMA J Nepal Med Assoc. 2010;49(178):139-142.
C. Actinomyces
This patient presents with a peritonsillar abscess. The most common pathogen is Group A beta-hemolytic str
Q1431 | STEM: A 62-year-old woman presents with a rash on her upper abdomen that she noticed yesterday. Two days ago, she says she began experiencing shooting pain and tingling in the same area. Temperature is 37.0°C (98.6°F), pulse rate is 82/min, and blood pressure is 116/78 mmHg. Examination of the skin shows grouped vesicles starting to the left of the umbilicus and wrapping upwards around the side of the torso in a dermatomal distribution. There are no vesicles seen on the right abdomen or back. Which of the following is the most likely based on this patient's symptoms, history, and physical exam findings? | ANSWER: Herpes zoster | EXPLANATION: Explanation

A. Cutaneous herpes simplex virus
Herpes simplex virus (HSV) can cause a variety of skin eruptions, including oral lesions, genital lesions, herpetic gingivostomatitis, herpetic whitlow, erythema multiforme, and eczema herpeticum.
B. Dermatitis herpetiformis
Dermatitis herpetiformis is an autoimmune skin condition seen in patients with celiac disease that has no relation to viral infection. It appears as a vesicular rash that is intensely pruritic. Common locations include the extensor surfaces of the extremities and the buttocks.
C. Herpes gladiatorum
Herpes gladiatorum is caused by the herpes simplex virus and presents as a vesicular rash on the face, neck, extremities, or trunk. It most commonly contracted through contact sports, such as wrestling.
D. Herpes zoster
Herpes zoster, also known as shingles, is caused by the varicella-zoster virus and is a reactivation of the dormant virus following exposure earlier in life. The classic appearance is a vesicular rash that occurs along a dermatomal distribution, most often on the thorax. A prodrome of pain and paresthesia is
common. Complications include secondary bacterial infection, post-herpetic neuralgia, nerve palsy, and encephalitis. Antiviral therapy hastens the resolution of the rash, decreases pain and helps to prevent post-herpetic neuralgia.
References: https://www.ncbi.nlm.nih.gov/books/NBK441824/Seethala R
Q1432 | STEM: A 30-year-old man comes to the emergency department because he has development of a Jarisch Herxheimer reaction after he received intramuscular benzathine penicillin G for treatment of syphilis. Temperature is 38.5°C (101.3°F), pulse rate is 115/min, respirations are 18/min, and blood pressure is 94/62 mmHg. A photograph of the rash is shown. Which of the following is the best course of treatment? | ANSWER: Acetaminophen and reassurance | EXPLANATION: Explanation
A. Acetaminophen and reassurance
Jarisch Herxheimer reaction (JHR) is a self-limited systemic reaction. Management is supportive, including antipyretics and intravenous fluids if needed. Premedicating with antipyretics or corticosteroids has not been proven to prevent JHR. Treatment with corticosteroids has also not shown benefit. Severe cases

SOURCE-DERIVED FALLBACK NOTES:
rrhage/increased intracranial pressure.

D. Tenecteplase
Involvement of the upper and lower face as well as bilateral symptoms suggests peri
Q1456 | STEM: A 22-year-old patient comes to the emergency department because of fever, malaise, anorexia, and sore throat for the past three weeks. Temperature is 37.8 C (100 F). Physical examination shows bilateral tonsillar enlargement without exudate, scant palatal petechiae, and bilateral anterior and posterior cervical lymphadenopathy. The result of a rapid streptococcus antigen test is negative. Which of the following is the most likely diagnosis? | ANSWER: Infectious mononucleosis | EXPLANATION: Explanation
A. Diphtheria
Diphtheria is an infection caused by Corynebacterium diphtheriae, and its vaccine is very effective in disease prevention. Symptoms of diphtheria typically begin 2-7 days following infection, and include fever, sore throat, hoarseness, cough, headache, dysphagia, odynophagia, dyspnea, foul-smelling and blood-stained nasal discharge, and lymphadenopathy. The classic clinical finding is an adherent, dense gray pseudomembrane that covers the posterior aspect of the pharynx.
B. Infectious mononucleosis
Infectious mononucleosis has clinical manifestations that range from persistent exudative tonsillitis and marked cervical lymphadenopathy, to the more vague symptoms of fatigue and malaise. Fatigue and malaise are the most frequent complaints, followed by sore throat and fever. Physical examination frequently shows enlarged, erythematous, and exudative tonsils, but these findings are not always present later in the course of the illness. Posterior cervical lymphadenopathy can be suggestive, while anterior cervical lymphadenopathy can be quite marked, and is usually found more easily than the posterior chain enlargement. Additionally, a finding of splenomegaly or hepatomegaly can be suggestive of acute disease.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK470387/Seethala R, Takhar S, Bullard-Berent J, Banks L. Chapter 119. Viruses. In: Walls R
Q1457 | STEM: A 1-year-old boy is brought to the emergency department after he sustained a puncture wound to his forearm. He was trying to hold his pet rabbit, when the rabbit bit him approximately 1 hour ago. Medical history includes the administration of two tetanus vaccines. On physical examination, there is a 0.5 cm deep puncture wound without active bleeding and there is no evidence of erythema or streaking. In addition to irrigation, wound exploration, and

administration of a third tetanus vaccine, which of the following is the most appropriate next step? | ANSWER: Administer tetanus immune globulin | EXPLANATION: Explanation
A. Administer antibiotic prophylaxis
The most important next step for this patient is ensuring they receive TIG. In the absence of signs/symptoms of infection, antibiotic prophylaxis is not typically indicated. It can be considered for animal bites in the following scenarios: primary closure is required, wounds of the hands/face/genitals, close proximity to a bone/joint, underlying venous/lymphatic compromise, immunocompromised patients, deep wounds, crush injuries, deep cat

SOURCE-DERIVED FALLBACK NOTES:
CDC currently recommends 500 mg IM ceftriaxone (1 gram if >150 kg) to treat gonorrhea. Adding doxycycline is recommended if co-infection with chlamydia cannot be ruled out by a negative test result. In this case, gonorrhea has been identified as the lone pathogen.Reference: https://www.cdc.gov/std/treatment-guidelines/gonorrhea-adults.htm
D. Doxycycline
Doxycycline is the currently recommended treatment regimen for chlamydia.
Q1480 | STEM: An 18-month-old girl is brought in by her mother with a rash that appeared one day ago. The rash began on her abdomen but has now spread over her entire body. Four days ago, the patient had a high fever, but that resolved. Physical examination reveals a maculopapular, blanching rash that is widespread across the trunk, face, and extremities. She also has mild rhinorrhea. Her physical examination is otherwise negative. Which of the following is the pathogen which is causing this patient's illness? | ANSWER: Human herpes virus type 6 | EXPLANATION: Explanation
A. Coxsackie A16 virus
Coxsackie virus causes Hand-Foot-Mouth disease. It presents with painful lesions in the mouth and a skin rash on the hands and feet. Fever may also be present.
B. Human herpes virus type 6
Roseola (AKA: Sixth disease) is caused by human herpes virus type 6, and possibly others, and is classically characterized by 3-5 days of high fevers that resolves abruptly, followed by a characteristic erythematous macular to maculopapular rash that begins on the trunk and may spread to the arms and legs. Roseola is a disease of young children, most often between 7-13 months. Roseola is usually self-limited, and the most common complication is febrile seizure which may occur

in approximately 10% of patients. Reference: https://www.ncbi.nlm.nih.gov/books/NBK448190/
C. Parvovirus B19
Erythema infectiosum (AKA: Fifth disease) is caused by parvovirus B19 and is characterized by a prodrome of low-grade fever, malaise, sore throat, headache, and nausea, followed several days later by an erythematous "slapped cheek" facial rash. The facial rash typically fades after 2-4 days, but in the second stage of the disease process, pink patches and macules may develop in a lacy, reticular pattern, most often on the extremities. This rash also typically resolves within 1-6 weeks.
D. Rubella virus
Rubella (AKA: German measles) is a mild, infectious acute viral disease resul
Q1481 | STEM: A 60-year-old woman presents because she has a rash on her upper abdomen that she noticed yesterday. Two days ago, she says she began experiencing shooting pain and tingling in the same area. Vital signs are normal. Examination of the skin shows grouped vesicles starting to the left of the umbilicus and wrapping upwards around the side of the torso in a dermatomal distribution. Which of the following medications can be used to treat the most common complication of this patient's current infection? | ANSWER: Gabapentin | EXPLANATION: Explanation
A. Cryotherapy
This patient has herpes zoster which can be followed by postherpetic neuralgia. Very few studies have been done to look at the role of cryotherapy to the peripheral nerve, and its effects see

SOURCE-DERIVED FALLBACK NOTES:
and Clinical Practice. 10th ed. Philadelphia: Elsevier; 2023.
B. Infected pregnant women are at risk for congenital infection
While patients with acute viral illness generally should be instructed to practice good hygiene to minimize infecting surrounding people, Epstein-Barr Virus (EBV) has not been shown to cause congenital infections. Surprisingly, intrauterine infection with EBV is rare because <5% of pregnant women are susceptible to the virus. Strict isolation is not required.
C. Symptom burden and duration may be decreased with antiviral and prednisone taper
Q1507 | STEM: A 47-year-old female with a past medial history of HIV and medication noncompliance presents to the ED for headache of two weeks duration. Her last known CD4 count was 100 cells/mm3. Vitals are unremarkable.

CT head reveals a ring-enhancing lesion in the basal ganglia. Which of the following is the most appropriate treatment? | ANSWER: Pyrimethamine and sulfadiazine | EXPLANATION: Explanation
A. Acyclovir
Acyclovir would be useful in the treatment of herpes encephalitis, which has a predilection for the temporal lobe. It does not usually present as a ring-enhancing lesion
B. Nimodipine and blood pressure control
Nimodipine and blood pressure control would be appropriate treatments for subarachnoid hemorrhage.
C. Pyrimethamine and sulfadiazine
Toxoplasmosis is primarily a disease of the immunosuppressed and is a well-known opportunistic infection in HIV. It is the most common cause of encephalitis in AIDS and may produce headache, fever, and focal neurologic deficits. The Toxoplasma gondii parasite is often contracted while immunocompetent, with symptoms only arising with reactivation as CD4 counts drop below 200 cells/mm3. On CT scan, it classically produces a ring-enhancing lesion with a predilection for the basal ganglia. The first-line toxoplasmosis treatment is pyrimethamine, sulfadiazine, and folic acid.
Further information: Basavaraju A. Toxoplasmosis in HIV infection: An overview. Trop Parasitol. 2016;6(2):129-135. PMC5048699https://www.ncbi.nlm.nih.gov/books/NBK563286/
D. Vancomycin and ceftriaxone
Vancomycin and ceftriaxone would be appropriate initial regimens in treating meningitis.
Q1508 | STEM: A 35-year-old male presents to the emergency department with a fever and rash. He reports he had been out hiking with his dogs ten days ago and sustained a tick bit. Since then, he has developed nausea and vomiting, chills, myalgias, and a rash that began on his forearms and spread to his trunk. He denies significant past medical history. Vital signs are: BP 120/90 mmHg, HR 115/min, RR 16/min, T 102ºF (38.9ºC). On exam, he has nonpruritic flat erythematous macular lesions on his forearms, wrists, and spreading to his abdomen and trunk. Which of the following is the most like organism causing this disease process? | ANSWER: Rickettsia rickettsii | EXPLANATION: Explanation
A. Coxiella burnetii
Coxiella burnetti is a zoonotic disease, an obligate intracellular bacterium with

sheep, goats, and cattle as the primary reservoir. It causes Q fever, which is a flu-like illness with multiple complications that can develop if not treated, but

SOURCE-DERIVED FALLBACK NOTES:
m, she is ill-appearing, has a diffuse, blanching, macular erythroderma rash with desquamation, and a strawberry tongue. Labs are notable for creatinine of 2.2 mg/dL. What is the next best step in management? | ANSWER: Vaginal exam | EXPLANATION: Explanation
A. Give aspirin and start IVIG
Aspirin and IVIG are treatments for Kawasaki disease which is most commonly found in children younger than 5 years old. While Kawasaki should remain on the differential for this patient, toxic shock syndrome is more likely and a foreign body must be ruled out first.
B. Start broad-spectrum antibiotics
Until culture sensitivities are available, it is appropriate to begin broad-spectrum antibiotics such as clindamycin and vancomycin for both MRSA and MSSA. However, a complete exam, including a vaginal exam, would be the most appropriate next step, followed by starting empiric antibiotic therapy.
C. Transfer the patient to the burn unit
Depending on your institution, it may be appropriate to admit this patient to the burn unit for the management of diffuse skin damage and careful hemodynamic and electrolyte monitoring. Before doing so, a vaginal exam is still needed to determine whether a foreign body is present, causing this patient's condition.
D. Vaginal exam
In this post-menarcheal teenager, there is a high concern for toxic shock syndrome. A complete physical exam including a vaginal exam should be completed to ensure there are no retained foreign bodies. Classically, toxic shock syndrome has been associated with highly absorbent tampons, but the removal of these tampon types from the market has greatly decreased its incidence. It is
Q1525 | STEM: A 17-year-old female presents to the emergency department with left knee and right wrist pain as well as subjective fever for the previous 2 days. The physical exam is remarkable for swollen, erythematous, and warm joints with significant effusions. The skin exam is shown below. Which of the following is the likely underlying cause of this patient’s illness? | ANSWER: Disseminated gonorrhea | EXPLANATION: Explanation
A. Dermatomyositis
Dermatomyositis shows Gottron's papules, which are errythematous lesions over

the metacarpal-phalangeal and interphalangeal joints. The image in this question shows a hemorrhagic pustule pathognomonic for another disease.
B. Disseminated gonorrhea
Disseminated gonococcal infection (DGI) usually is one of two major clinical syndromes: arthritis-dermatitis syndrome and localized purulent arthritis without skin lesions. The patient presented is manifesting the arthritis-dermatitis syndrome which has the triad of tenosynovitis, dermatitis, and polyarthralgia. The skin lesions are pustular or vesicular , and hemorrhagic pustules are thought to be pathognomonic for gonorrhea. Ceftriaxone given intravenously or intramuscularly is the preferred initial antibiotic of choice.
Further information: https://www.ncbi.nlm.nih.gov/books/NBK470439/
C. Juvenile idiopathic arthritis (JIA)
Juvenile idiopathic arthritis (JIA) manifests with an erythematous macular rash. The image in this question shows a hemorrhagic pustule pathognomonic for another disease.
D. Systemic

7. Arabic rapid-review summary

تُراجع هذه المفاهيم ضمن سياق التقييم الإسعافي، التشخيص التفريقي، العلاج الأولي، وإعادة التقييم، مع الانتباه إلى التريكات والأخطاء الشائعة.

8. Source-question coverage map

This chapter consolidates all source questions assigned to Systemic Infectious Disorders.

REFERENCES AND GUIDANCE PORTALS

[1] Royal College of Emergency Medicine (RCEM), Clinical Standards and Guidance: https://rcem.ac.uk/clinical-standards-and-guidance/
[2] European Resuscitation Council (ERC), Guidelines 2025 English: https://www.erc.edu/science-research/guidelines/guidelines-2025/guidelines-2025-english/
[3] NICE, Emergency and acute medical care in over 16s (NG94): https://www.nice.org.uk/guidance/NG94
[4] Resuscitation Council UK: https://www.resus.org.uk/

[5] American College of Emergency Physicians, Clinical Policies: https://www.acep.org/patient-care/clinical-policies
[6] International Liaison Committee on Resuscitation: https://www.ilcor.org/
[7] British Thoracic Society: https://www.brit-thoracic.org.uk/
[8] National Poisons Information Service / TOXBASE: https://www.toxbase.org/